

Intraday Market Return Predictability Culled from the Factor Zoo^{*}

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Abstract

We provide strong empirical evidence for time-series predictability of the intraday return on the aggregate market portfolio by exploiting lagged high-frequency cross-sectional returns on the factor zoo. Our results rely on the use of modern Machine Learning techniques to regularize the predictive regressions and help tame the signals stemming from the zoo, together with techniques from financial econometrics to differentiate between continuous and theoretically non-predictable discontinuous high-frequency price increments. Using the predictions from the model estimated for the aggregate market portfolio in the formulation of simple intraday trading strategies for a set of highly liquid ETFs results in sizeable out-of-sample Sharpe ratios and alphas after accounting for transaction costs. Further dissecting the abnormal intraday returns, we find that most of the superior performance may be traced to periods of high economic uncertainty and a few key factors related to tail risk and liquidity, pointing to slow-moving capital and the gradual incorporation of new information as the underlying mechanisms at work.

Keywords: High-frequency data, market return predictability, factor zoo, machine learning, market timing, market frictions, slow-moving capital.

JEL classification: G12, G14, G17, C45, C55.

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1 Introduction

Is the return on the aggregate stock market predictable, and if so is it possible to generate abnormal returns by trading on said predictions? These dual questions lie at the heart of the efficient market hypothesis (Fama, 1970) and easily ranks among the most studied questions in finance. The vast majority of the existing literature has focused on return predictability over daily, or longer monthly and quarterly return horizons, utilizing the information in lagged market returns, or a relatively small number of predictor variables, including valuation ratios, dividend yields, interest rates, and volatility-type measures, reaching mixed conclusions (see, e.g., Pesaran & Timmermann, 1995; Rapach, Strauss, & Zhou, 2010; Welch & Goyal, 2008). Meanwhile, a smaller strand of the literature has sought to explain higher-frequency intraday return predictability patterns by various imperfections or market frictions (see, e.g., Chordia, Roll, & Subrahmanyam, 2005; Heston, Korajczyk, & Sadka, 2010).

Set against this background, armed with high-frequency intraday returns on an expansive set of more than two hundred cross-sectional risk factors (Aleti, 2023), together with estimation techniques adopted from machine learning to succinctly distill the pertinent information, we provide strong empirical evidence that the intraday return on the aggregate market portfolio is indeed predictable. Using the predictions from our main forecasting model in the formulation of a simple trading strategy for the SPY ETF results in an annualized intraday transaction-cost-adjusted Sharpe ratio of 1.37, along with an annualized Fama-French six-factor risk-adjusted alpha of 20.83%. By comparison, the intraday Sharpe ratio for the SPY over the same out-of-sample forecast period is just 0.09.¹ Consistent with the idea that the predictability exploited by our strategy may

¹ The Sharpe ratio for a traditional buy-and-hold strategy for the SPY that also includes the overnight

be attributed to the gradual incorporation of new information and slow-moving capital, we trace the accrual of most of the abnormal intraday returns to periods of high overall economic uncertainty and the outsized importance of a few key high-frequency factors related to liquidity and tail risk.

Our work may be seen as part of the recent literature that seeks to exploit the idea that the useful information in most high-dimensional datasets effectively lies in a lower-dimensional space.² In that regard, we rely on various machine learning models as a way to uncover intraday time-series return predictability for the aggregate market portfolio. Importantly, however, rather than searching for predictive signals within the set of individual stock returns, as traditionally done in the literature, we search for predictive signals in the returns on the myriad of characteristic-sorted portfolios established in the asset pricing literature. Hypothesizing that the predictive signals lie in this alternate subspace, we in turn rely on an array of machine learning techniques, ranging from ridge, LASSO and elastic net to neural nets, random forests, and gradient boosted regression trees, to regularize and estimate our predictive models.

In addition, motivated by the fact that in an arbitrage-free, continuous-time setting price jumps are inherently unpredictable (see, e.g., Fontana, Pelger, & Platen, 2019), we deliberately filter out excessively large returns before the estimation of our prediction models, thereby allowing us to more keenly focus on the continuous, and potentially theoretically predictable, smooth high-frequency returns. To render such a decomposition meaningful, we purposely restrict our forecasting models to the active portion of the trading day for which reliable high-frequency data are available. Meanwhile, it is well-

return is 0.49.

²This idea is closely related to the so-called “Manifold Hypothesis” in the machine learning literature. It is, of course, also directly related to the ubiquitous use of factor models in asset pricing as a way to summarize systematic risks in terms of a few common factors.

established that most of the return on the market actually manifests during the overnight period (see, e.g., Hendershott, Livdan, & Rösch, 2020), thus in effect leaving us with a more substantial challenge when it comes to generating economic gains from trading on the intraday return predictions only.³

Our basic idea of melding the literature on cross-sectional return predictability with that of time-series return predictability is akin to Dong et al. (2022), who similarly exploit a large set of asset pricing factors for uncovering market return predictability at the monthly level.⁴ However, in contrast to their study, which rationalizes the apparent predictability by time-varying risk premia, the intraday return predictability documented here suggests a very different economic explanation rooted in market frictions and the gradual incorporation of new information. In this regard, our analysis and empirical findings are perhaps most closely related to Huddleston, Liu, and Stentoft (2023), who similarly exploit a variety of machine learning algorithms together with a large set of individual stock returns in their search for intraday market return predictability, and Chinco, Clark-Joseph, and Ye (2019) who rely on LASSO for estimating intraday prediction models for individual stock returns based on the full cross-section of returns.

By comparison, our reliance on well-defined asset pricing factors as predictors, together with ETF-based trading strategies, allows us to more directly pinpoint what drives the transaction-cost-adjusted abnormal returns. In particular, relying on so-called SHAP values (Lundberg & Lee, 2017) for additively decomposing the superior performance unequivocally points to factors based on various measures of turnover and liquidity as the

³Lou, Polk, and Skouras (2019) attribute the large difference between average intraday and overnight returns to a “tug of war” between differently informed groups of investors; see also Bogousslavsky (2021) and Akbas et al. (2022). Relatedly, Gao et al. (2018) and Li, Diks, and Pachenko (2020) find that the overnight return help forecast intraday momentum-type effects.

⁴By comparison, Engelberg et al. (2023) find little evidence that cross-sectional firm-level characteristics have out-of-sample time series predictability for monthly market returns.

key drivers of our superior Sharpe ratios and alphas. This, of course, is in line with the classic literature on cross-autocorrelations and the finding that the returns on larger firms tend to lead the returns on smaller firms (see, e.g., Chordia, Sarkar, & Subrahmanyam, 2007; Hou, 2007; Lo & MacKinlay, 1990). It also echoes the past work by Chordia and Swaminathan (2002) and the finding that stocks with higher turnover tend to lead those with lower turnover, and the more recent work by Caporin, Kolokolov, and Renò (2017) and the finding that more actively traded stocks generally respond more quickly to important news.⁵ In contrast to all of these earlier studies, however, which seek to explain lower-frequency lead-lag return relations between different groups of stocks by group-specific measures of trading activity, our prediction models for the intraday return on the market portfolio derive their explanatory power from aggregate measures of turnover, as embodied in the returns on the high-frequency systematic risk factors.

The finding that our trading strategies tend to perform the best when economic uncertainty and financial market volatility are high is consistent with market frictions being more acute in such periods thereby limiting mispricing correction (see, e.g., Gromb & Vayanos, 2010; Shleifer & Vishny, 1997).⁶ Relatedly, the recent study by Avramova, Cheng, and Metzker (2023) finds that the anomalous returns for individual stocks tend to concentrate in more difficult-to-arbitrage stocks and periods of high overall market volatility and low liquidity. It also echoes Farmer, Schmidt, and Timmermann (2023), and their findings that return predictability over longer return horizons tends to be confined to relatively short calendar time spans and pockets-of-predictability.

⁵In a similar vein, Huang et al. (2022) document that the lead-lag relationship observed empirically between economically-linked firms appears to be mainly driven by “slow-moving” news.

⁶The theoretical models of Gârleanu and Pedersen (2013, 2016) also further explain how limited risk-bearing capacity and transaction cost may naturally curtail mispricing correction. Similar arguments based on limits to arbitrage have also been invoked to explain a number of cross-sectional pricing anomalies, including, e.g., the idiosyncratic volatility puzzle (Stambaugh, Yu, & Yuan, 2015) and the differential pricing of up- and downside beta risks (Bollerslev, Patton, & Quaedvlieg, 2022).

At a more technical level, our finding of intraday market return predictability also connects to a series of recent studies emphasizing possible short-lived violations of the benchmark Itô semimartingale model used for continuous-time modeling of returns and corresponding arbitrage opportunities associated with drift bursts (Christensen, Oomen, & Renò, 2022), pockets of extreme predictability (Andersen et al., 2023), or intermittent trading (Bandi et al., 2023). These studies, however, primarily focus on violations that occur at even higher intraday sampling frequencies than the 15-minute frequency on which our forecasting models and practical trading strategies are based. Similarly, the recent study by Aït-Sahalia et al. (2022) on the predictability of individual stock returns primarily emphasize the effect of market microstructure frictions and order imbalances over ultra-short millisecond time scales. That study also finds that the return predictability uncovered by their LASSO-based estimation procedures essentially vanishes over longer 5-minute return intervals.

At a broader level, our paper is also part of the rapidly growing literature on the use of tools from machine learning to address a long list of empirical questions in finance, including the estimation and forecasting of risk premia. As a case in point, Gu, Kelly, and Xiu (2020) find that the use of machine learning algorithms results in substantial performance gains over traditional regression-based approaches (see also Bianchi, Büchner, & Tamoni, 2021; Fan et al., 2022; Guijarro-Ordóñez, Pelger, & Zanolini, 2022; Jiang, Kelly, & Xiu, 2023; Ke, Kelly, & Xiu, 2021; Leippold, Wang, & Zhou, 2022). In a similar vein, several studies have exploited machine learning techniques to help better explain the cross-sectional variation in returns through the construction of improved factor models, or the direct estimation of a more accurate stochastic discount factor (SDF) (see, e.g., Aletti & Bollerslev, 2024; Bryzgalova, Pelger, & Zhu, 2024; Chen, Pelger, & Zhu, 2023;

Giglio & Xiu, 2021; Korsaye, Quaini, & Trojani, 2019; Lettau & Pelger, 2020). Our paper aligns more closely with the former literature, while focusing primarily on predictably induced by frictions rather than time-varying risk premia.

The remainder of the paper is organized as follows. Section 2 begins by summarizing the data underlying our empirical analyses. Section 3 discusses the formulation of our forecasting models and our implementation of the different machine learning algorithms. Section 4 presents the resulting out-of-sample high-frequency forecasts for the market portfolio, while Section 5 discusses our main empirical findings pertaining to the practical implementation of the new forecasting models for trading the SPY ETF. Section 6 further shows that most of the predictability underlying the superior performance may be attributed to a few liquidity-related factors. Section 7 concludes. Additional theoretical justification and empirical analyses are provided in an Online Supplemental Appendix.

2 Data

Our primary dataset consists of high-frequency returns for the 272 portfolios constructed in Aleti (2023), spanning January 2, 1996 to December 31, 2020. The 272 portfolios comprise 218 characteristic-sorted portfolios capturing the many factors proposed in the asset pricing literature, 48 industry portfolios summarizing sector-specific movements in the stock market, and 6 additional factor portfolios consisting of the Fama-French five plus momentum.⁷ To alleviate any concerns about market microstructure noise, we follow Aleti (2023) and sample the returns at a “coarse” 15-minute frequency, for a total of 169,965 intraday return observations on each of the factors.

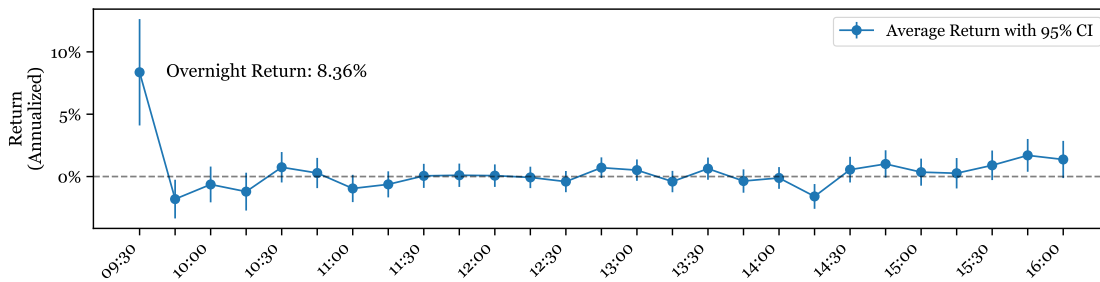
Table 1 reports various descriptive statistics for different subsets of the resulting in-

⁷The 218 characteristic-based factors collectively capture all the factors studied in Jensen, Kelly, and Pedersen (2023) and Chen and Zimmermann (2022).

Table 1: Descriptive Portfolio Statistics

Averages (%)	Portfolios			
	Market	FF6	Industry 48	Factors
Return	1.07	3.74	1.32	2.93
Realized Volatility	14.42	8.59	19.62	8.87
Relative Jump Measure	2.76	4.76	5.21	4.69
Autocorrelation	3.56	6.11	3.28	4.00
Correlation with Market	100.00	0.68	67.32	-9.28

Note: The table reports descriptive statistics for different subsets of the portfolios used in our empirical analyses. Market refers to the high-frequency Fama-French market portfolio. FF6 refers to the five workhorse factors from Fama and French (2015), plus the momentum factor. Industry 48 refers to the 48 industry portfolios, while Factors refers to the 218 characteristic-sorted portfolios. The numbers are calculated as the cross-sectional average of the time-series averages for each of the different subsets. The sample period is from 1996-2020. All of the measures are reported in annualized percentage form.

Figure 1: Intraday Fama-French Market Portfolio Returns

Note: The figure reports the average intradaily 15-minute returns on the Fama-French market portfolio over 1996 to 2020. The first return, labeled 09:30, corresponds to the average close-to-open return. The confidence intervals are computed using the corresponding realized volatility estimates.

tradaily portfolio returns. The first column pertains to the Fama-French market portfolio, constructed as the value-weighted average of every common stock. It shows that the average intraday return on the market was indeed quite small, equalling just 1.07% per year. This, of course, is consistent with the above-mentioned extant literature, and past work showing that most of the equity risk premium appears to be earned during the overnight period (see, e.g., Bogousslavsky, 2021; Bondarenko & Muravyev, 2023; Cliff, Cooper, & Gulen, 2008).

Further illustrating this point, Figure 1 plots the average market return earned across the day. As the figure shows, a disproportionate fraction of the market return is earned

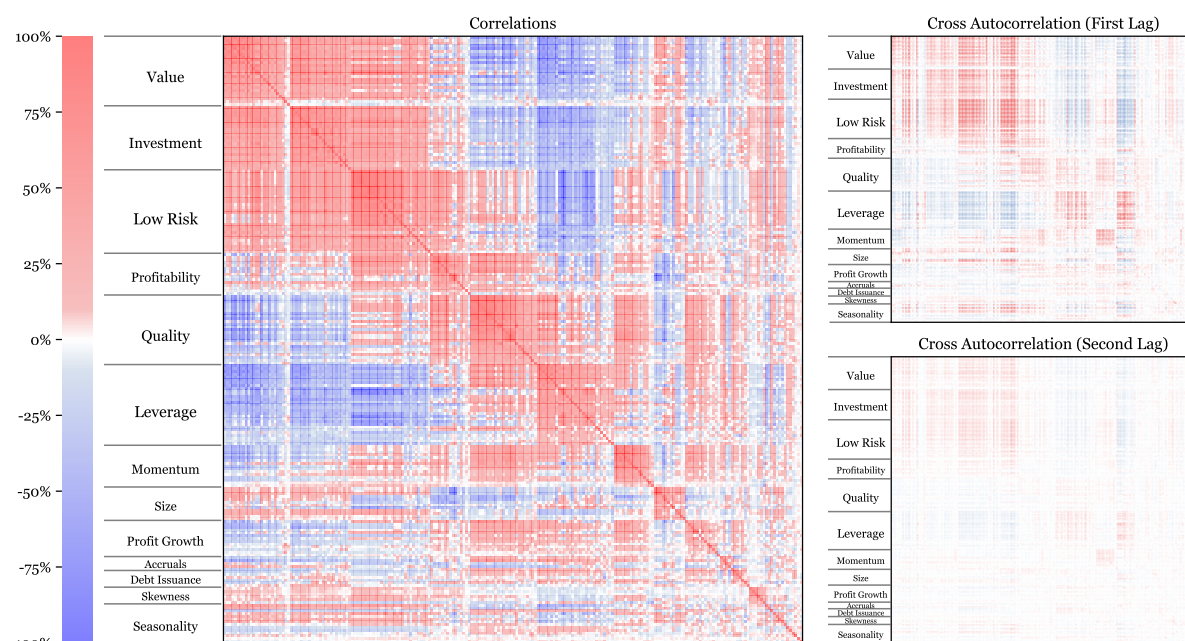
overnight, while the intraday returns are almost “flat” and close to zero throughout the active part of the trading day. Further underscoring the difficulty of predicting and earning a positive return during the day, the intraday market returns are also approximately serially uncorrelated, as evidenced by the average first-order autocorrelation coefficient of just 3.56% reported in [Table 1](#).

The remaining summary statistics for the different subsets portfolios, reported in columns two through four, tell a similar story. Not surprisingly, the 48 industry portfolios are most closely correlated with the market portfolio. Meanwhile, all of the high-frequency portfolio returns appear approximately serially uncorrelated. A cursory look at the lagged cross-autocorrelations with the return on the market portfolio also suggests little predictive information. In particular, looking at the 218 characteristic-sorted portfolios, the lagged first, second, and third-order cross-autocorrelations with the return on the market portfolio, corresponding to 15-, 30-, and 45-minute shifts in the factor returns, equal -0.75%, -0.43%, and -0.35%, respectively. Similar results obtain for the Fama-French workhorse factors and the 48 industry portfolios. Of course, as further explored in the Online Appendix, these averages may mask stronger correlations for a few of the individual portfolios, although the ex-ante identification of such predictors remains a challenge.

The strong correlation between many of the factors in the zoo further complicates this. To illustrate, [Figure 2](#) plots the full sample high-frequency correlations between the 218 characteristic-sorted portfolios, together with their lag-one and lag-two cross-autocorrelations.⁸ Echoing earlier work based on more coarsely sampled daily and monthly factor returns, the figure evidences clear clusters within the factor zoo. In other words, the

⁸The factors are ordered based on the classifications in Jensen, Kelly, and Pedersen (2023) and Aletti (2023).

Figure 2: High-Frequency Factor Correlations



Note: The leftmost larger heatmap shows the contemporaneous full-sample pairwise cross-correlations between the 15-minute returns on each of the 218 characteristic-sorted portfolios. The upper and lower right-hand-side heatmaps show the corresponding lag-one (15-minute) and lag-two (30-minute) cross-autocorrelations, respectively.

relevant information in the high-dimensional factor zoo seemingly lies in a much lower dimensional space, hinting at the value of using modern machine learning techniques in trying to identify the most useful predictors (see also the earlier related analysis in Lewellen, Nagel, & Shanken, 2010). We turn next to a discussion of the specific prediction models and related estimation procedures that we rely on for doing so.

3 Methodology

We begin with a brief discussion of the formal setting that motivates our approach and the separation of the returns into jump and continuous components. We then describe the various forecasting specifications and machine learning models that we rely on.

3.1 Continuous and Jump Price Dynamics

Let $P = (P_t)_{t \geq 0}$ denote the multivariate continuous-time log-price process representing both the market and our factor portfolios. To fix ideas, assume that the dynamics of the prices may be described by a standard jump-diffusion process, as commonly employed in the high-frequency financial econometrics literature (see, e.g., Aït-Sahalia & Jacod, 2014),

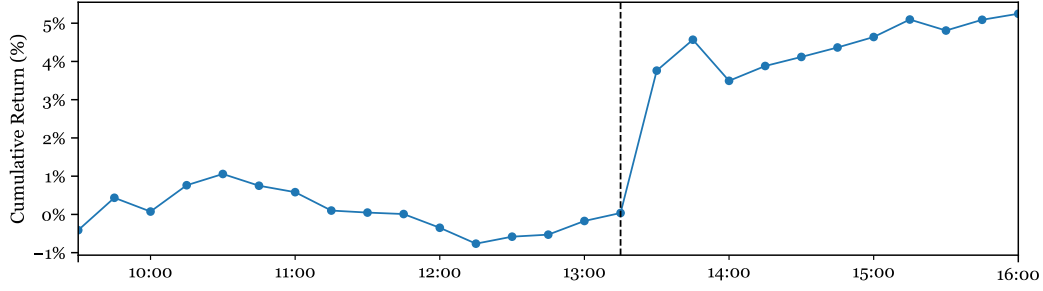
$$P_t = P_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s + \sum_{s=1}^{N_t} J_s, \quad t \geq 0, \quad (1)$$

where $\mu = (\mu_t)_{t \geq 0}$ denotes the drift, $\sigma = (\sigma_t)_{t \geq 0}$ is a stochastic volatility process, $W = (W_t)_{t \geq 0}$ is a standard Brownian motion, and $J = (J_s)_{s=1, \dots, N_t}$ denotes a sequence of non-zero random variables of jump sizes with corresponding jump times $\tau = (\tau_s)_{s=1, \dots, N_t}$. In the absence of arbitrage, the discontinuous jump component is inherently unpredictable.⁹ By contrast, the drift and diffusive components, which capture the risk premia and the potential integration of slow-moving information, are both continuous and possibly jointly predictable. Hence, this naturally suggests treating the diffusive and jump components differently in the formulation of the forecasting models for future market returns.

In practice, of course, we do not observe the prices on a true continuous timescale, rendering the exact identification of the separate components impossible. The actually observed prices are also invariably contaminated by market microstructure noise stemming from bid-ask spreads, price discreteness, and other complications. As previously noted, to help ameliorate the impact of such noise, we follow the common practice in the high-frequency literature of deliberately sampling the prices at a coarse 15-minute

⁹ This holds even without a semimartingale restriction as shown by Fontana, Pelger, and Platen (2019).

Figure 3: Market Portfolio Return on January 3, 2001



Note: The figure plots the cumulative 15-minute returns on the high-frequency Fama-French market portfolio on January 3, 2001. The vertical dashed line marks the time of the surprise FOMC rate cut announcement.

frequency. To fix notation, let

$$r_{t,i} \equiv P_{t+i\Delta_n} - P_{t+(i-1)\Delta_n}, \quad (2)$$

denote the corresponding 15-minute returns, where $i \in \{1, \dots, n\}$, $\Delta_n \equiv 1/n$, and $n = 26$ refers to the number of intraday returns per day. Armed with these returns, we rely on a standard truncation-based procedure, as further detailed in Appendix A, to disentangle the continuous and jump components of the factors.

To help further intuit the idea and motivate the separate consideration of jumps in the formulation of our forecasting models, Figure 3 plots the cumulative 15-minute returns on the Fama-French market portfolio on January 3, 2001. On this day, the FOMC issued a surprise rate cut at exactly 13:15, evidently causing a sharp rise in the price, or a jump, at that exact time. As the figure further shows, there also appears to be a slight momentum in the return after the announcement. By contrast, neither the occurrence of the jump, nor its size and direction, appears to be predictable based on the earlier returns over the day. Underscoring this visual impression, the estimation of a simple AR(1) model based on the 15-minute intraday returns for the whole day results in an

R^2 of just 0.69%. On the other hand, estimating an AR(1) model with the jump return from 13:15-13:30 omitted produces an R^2 of 8.96%. Interestingly, relying on the estimates from the former AR(1) model to predict the non-jump returns results in a negative R^2 of -0.19%. As such, this indirectly suggests that by including the jump return in the regression, we effectively “overfit” the unpredictable variation associated with the jump, thereby completely obscuring the smaller, but potentially predictable return variation over the rest of the day. Therefore, in the results that follow, we will consider alternative forecasting specifications that exclude market jumps from the regressand.

3.2 Forecasting Model Specifications

Each of our specifications are based on autoregressive-type structures and the use of the lagged factor returns for predicting the future aggregate market portfolio return. Inspired by the Heterogeneous AutoRegressive (HAR) specification originally proposed by Corsi (2009) for modeling and forecasting time-varying volatility, we similarly rely on a cascade-type structure as a convenient way to impose a layer of sparsity in the dynamic dependencies.

Our baseline specification, which we will refer to as *All*, *All* in the sequel, uses the 15-minute market return $r_{t,i+1}^{mkt}$ as the regressand and all the lagged factor returns $X_{t,i} \equiv \bigcup_{j \in factors} X_{t,i,j}$ as the regressors. Owing to the HAR structure, each factor j results in three separate regressors in the form of the lagged 15-minute, hourly, and daily portfolio returns, $X_{t,i,j} \equiv [r_{t,i}^j, r_{t,i|i-3}^j, r_{t-1}^j]$. Accordingly, our baseline specification that includes all of the 272 factor portfolios entails a $3 \times 272 = 816$ dimensional “ β ” parameter to be estimated.

We further augment this already high-dimensional model by separating the portfolio

returns into their continuous and jump components. The resulting *All*, *All+Jmp* specification in turn includes $X_{t,i} \cup X_{t,i}^{jmp}$ as regressors, where $X_{t,i}^{jmp} \equiv [r_{t,i}^{j,jmp}, r_{t,i|i-3}^{j,jmp}, r_{t-1}^{j,jmp}]$ is defined analogously to $X_{t,i}$ with the estimated jump returns for each of the portfolios in place of their total returns. It obviously encompasses the previous *All*, *All* specification, but further allows for the possibility that the predictive content of the lagged jump factor returns may differ from that of the lagged continuous factor returns.

In addition, building on the idea that the true latent market jump returns should be inherently unpredictable, we consider two additional specifications in which we explicitly target only the continuous return on the market rather than the entire market return, by replacing the market return regressand in the above two specifications with $r_{t,i+1}^{mkt,cts}$. If the market jump returns are indeed unpredictable, explicitly excluding the jump component from the regressands would naturally enhance the accuracy of the resulting *Cts*, *All* and *Cts*, *All+Jmp* based predictions.¹⁰ Additionally, as illustrated by the discussion of the single-day market returns depicted in Figure 3 and the estimated intraday AR(1) models, omitting the large jump returns from the regressands also helps prevent potential “overfitting” of a few especially large returns.

Altogether, our two most general *All*, *All+Jmp* and *Cts*, *All+Jmp* specifications both involve a total of $3 \times 272 \times 2 = 1,632$ parameters to be estimated. Even though we have a large number of high-frequency returns at our disposal for doing so, the unrestricted estimation by standard OLS of such high-dimensional models is simply not practically feasible. Instead, we rely on various machine learning approaches to discipline the estimation and construct effective forecasts.

In addition to the factor-based model forecasts, we also consider a series of simple

¹⁰ These specifications can also be viewed as weighted regressions, where the weight on the regressand is an indicator variable pertaining to whether or not the market return is flagged as continuous.

benchmarks in the form of AR(1), AR(4), and AR(26) models for the market portfolio return, thus incorporating the past 15-minute, 1-hour, and 1-day worth of market information, respectively. In parallel to the more general specifications discussed above, we consider four different versions of each of the AR models based on the decomposition of the market return into its continuous and jump components. Since none of these models exploit the information in the factor zoo, they serve as natural benchmarks for what can be learned from the market portfolio itself, implicitly speaking to a narrower form of weak form market efficiency. As a final benchmark, we also consider the sample mean return. In so doing, we deliberately rely on the ex-post average return on the market to provide a more challenging, practically infeasible, competitor.

3.3 Machine Learning Algorithms

Following the rapidly growing financial machine learning (ML) literature (see, e.g., Avramova, Cheng, and Metzker, 2023; Coulombe et al., 2023; Dong et al., 2022; Fan et al., 2022; Gu, Kelly, and Xiu, 2020; Guijarro-Ordóñez, Pelger, and Zanotti, 2022, along with the many additional references therein), we employ on an array of ML algorithms to construct our forecasts. All of the different procedures that we use are now considered fairly standard in the finance and machine learning literature.

More precisely, we employ the following linear procedures: OLS, Ridge, Lasso, Elastic Net (ENet), Adaptive Lasso (AdaLasso), Principal Components Regression (PCR), and Partial Least Squares (PLS). The non-linear ML algorithms we consider are: Feedforward Neural Networks (FNN), Random Forests (RF), and Gradient Boosted Regression Trees (GBRT). Following common practice, we also consider an equally-weighted average of all of the above models sans OLS, thus collapsing all the forecasts into a single so-called

ensemble forecast. In order to conserve space, we defer more detailed descriptions of each of the ML algorithms and their practical implementation, including the choice of hyperparameters, to the Online Appendix. For an extensive review of the same models and various applications thereof in the finance literature, we refer to the recent review by Kelly and Xiu (2023).

3.4 Practical Implementation

Each machine learning model is defined by a set of hyperparameters and their main parameters. The hyperparameters, such as the L1 and L2 penalty pair for the elastic net, must be chosen before estimating the model itself to obtain the main parameters.

To this end, we rely on a commonly used data-driven approach from the ML literature known as a training/validation/testing split. We begin by splitting a contiguous region of the data into three subsets: a train, valid, and test period. Keeping the splits contiguous ensures that we do not violate the time-series structure of the data. We then start by using the data from the training period to fit the parameters for a given ML algorithm over a grid of hyperparameter sets. Using each of these fitted models, we next construct pseudo out-of-sample forecasts over the validation period for each hyperparameter set. We then rank the resulting forecasts by their MSE and define the hyperparameter set corresponding to the forecast with the lowest MSE as the “optimal” hyperparameter set for the model. We then reuse the fitted parameters for the ML algorithm previously estimated with these optimal hyperparameters to construct out-of-sample forecasts over the test period. To summarize, the fitted parameters are derived from the training period, the optimal hyperparameters are derived from the validation period, with the forecasts constructed using these hyperparameters and the parameters from the testing period,

thereby ensuring that our forecasts are genuinely out-of-sample.

This rolling train/valid/test sampling-splitting procedure requires its own set of tuning parameters. Following common practice in the ML literature, we rely on an expanding window, using the first 80% of the window for training and the latter 20% for validation. Next, we fix the size of our out-of-sample window, or testing period, to one calendar year, refitting our models on an annual basis.¹¹ Lastly, we begin our out-of-sample analysis starting in 2004, producing new forecasts until the end of 2020, for a total out-of-sample evaluation period of 17 years, or 111,280 15-minute intraday observations. Incidentally, this out-of-sample period also closely corresponds to the “late post-decimalization period” (Huddleston, Liu, & Stentoft, 2023), commensurate with a substantial increase in market liquidity thereby allowing for more meaningful high-frequency-based analyses. A series of robustness checks concerning the tuning parameters used in our estimation and other normalizations are detailed in the Online Appendix.

4 Forecasting the Market Portfolio

We begin our discussion of the empirical results by considering the statistical accuracy of the forecasts for the market portfolio. Following common practice in the literature, we rely on the out-of-sample R^2

$$R^2 = 1 - \frac{\sum_{(t,i) \in \text{OOS}} (r_{t,i}^{mkt} - \hat{r}_{t,i}^{mkt})^2}{\sum_{(t,i) \in \text{OOS}} (r_{t,i}^{mkt})^2}, \quad (3)$$

for doing so, where $\hat{r}_{t,i}^{mkt}$ denotes the forecasts produced by a specific model. We purposely report the uncentered rather than the centered R^2 to allow for a direct comparison

¹¹ For the non-linear FNN, RF, and GBRT models, we only refit every five years to reduce the computational burden.

with other recent studies (e.g., Huddleston, Liu, & Stentoft, 2023). The uncentered R^2 is also especially adept in the high-frequency context, where the drift that drives expected returns is generally negligible in magnitude compared to that of the martingale component.

To establish a benchmark for the R^2 to expect with 15-minute returns, consider the predictability bound of $R^2 \leq \gamma^2 \text{Var}(r_{t,i}^{mkt})$ derived by Ross (2004), where γ refers to the risk aversion coefficient for the maximally risk-averse CRRA investor in the economy. Using the realized variance of our market portfolio returns as an estimate of the variance results in an upper bound on the 15-minute R^2 of just 0.00777% for a conservative choice of $\gamma = 5$. As such, this clearly shows that we should not expect R^2 's around say 1%-5% as sometimes reported in the literature focused on predictability over longer monthly horizons.¹² Alternatively, suppose that the continuous-time process for the market portfolio is known and may be described by a simple model with constant drift and volatility. As shown in the Online Appendix, in this situation even perfect prediction of the drift would only result in a 15-minute $R^2 = 0.00008\%$.

Turning to the actual forecast results obtained for each of our different specifications and ML tools, many of the 15-minute out-of-sample R^2 's reported in Table 2 are substantially larger than the benchmarks discussed immediately above. In particular, the overall highest R^2 of 0.212% is obtained by the Ensemble *All, All* specification. Using split regressors and/or targeting the continuous component of the market portfolio returns only seemingly results in marginally lower Ensemble-based R^2 's. Putting the results further into perspective of the numerical bounds discussed above also suggests that the fore-

¹²By comparison, evaluating the same bound with monthly returns results in 4.72%, which incidentally is also closely in line with the highest out-of-sample R^2 's obtained by Gu, Kelly, and Xiu (2020) through the use of ML tools for uncovering monthly market return predictability.

Table 2: Out-of-Sample R^2 s

Regressors	All		All+Jmp	
	All	Cts	All	Cts
AR1	0.036	0.051	0.051	0.097
AR4	0.024	0.051	-0.120	-0.063
AR26	-0.015	0.017	-0.410	-0.382
Linear	-5.123	-5.012	-12.408	-11.868
Ridge	0.172	0.152	0.126	0.148
Lasso	0.151	0.119	0.165	0.113
ENet	0.153	0.127	0.164	0.119
AdaLasso	0.137	0.137	0.099	0.107
PCR	0.172	0.177	0.053	0.069
PLS	0.148	0.106	0.100	0.122
FNN	0.111	0.131	-0.044	0.002
RF	0.118	0.007	0.052	0.090
GBRT	-0.161	0.116	0.128	-0.225
Ensemble	0.212	0.205	0.199	0.178

Note: The table reports the out-of-sample R^2 s for each of our forecasts for the 15-minute return on the Fama-French market portfolio from 2004 to 2020. Each column corresponds to one of our four specifications, while each row corresponds to a particular machine learning algorithm. All of the values are reported in percentages.

casting models are likely not predicting temporal variation in the equity risk premium.¹³

Instead, the predictability seems more likely to arise from short-lived violations of the basic semimartingale assumption, possibly associated with the “slow” incorporation of new information.

To more formally assess the statistical significance of the reported differences in R^2 's, we also calculate Diebold and Mariano (1995) (DM) tests based on the out-of-sample MSEs. To conserve space, Table 3 only reports the t -statistics for comparing the four benchmark models to the different Ensemble specifications. As evidenced by the negative DM t -statistics, all four Ensemble specifications offer statistically significant improvements over the benchmarks. Meanwhile, testing the four different Ensemble specifications against one another, none of the corresponding DM t -statistics are significant at conventional levels. In short, the Ensemble forecasts beat all the benchmarks, while the equivalent statistical comparisons of the four different Ensemble specifications do not

¹³ Generating a Ross-bound of 0.2%, commensurate with the R^2 obtained for the *All*, *All* Ensemble specification, would require an implausible large risk aversion coefficient of $\gamma = 25.3$.

Table 3: Diebold-Mariano Tests on MSE for the Ensemble Model

Regressors	All		All+Jmp	
	All	Cts	All	Cts
SampleMean	-4.315	-5.019	-4.275	-4.576
AR1	-2.544	-2.417	-2.264	-1.204
AR4	-2.398	-2.125	-3.586	-2.723
AR26	-2.409	-2.101	-4.550	-4.163

Note: The table reports the Diebold-Mariano test-statistics for the Ensemble model versus the four different benchmark models discussed in the text for forecasting the 15-minute return on the Fama-French market portfolio. Each column corresponds to one of our four specifications, while each row corresponds to a particular baseline model. Negative values indicate that the baseline model performs worse than the Ensemble-based forecast.

point to a clear significant winner.

In the Online Appendix, we also explore the sign classification accuracy of each of the models by comparing the signs of the forecasts to the signs of the realized returns. In so doing, we find that the specifications that only target the continuous component of the market returns perform statistically significantly better than those targeting the total returns. In other words, foreshadowing the trading performance results discussed in the next section, the directional forecasting ability of our models is better for the *Cts*, *All* and *Cts, All+Jmp* specifications.

5 Trading ETF Market Proxies

As emphasized by Campbell and Thompson (2008), even small and seemingly insignificant improvements in R^2 's can result in large improvements in Sharpe ratios and other performance measures. Kelly, Malamud, and Zhou (2024) similarly cautions against evaluating high-dimensional prediction models developed by techniques from machine learning based on their R^2 's. Hence, we now turn to a more direct economic comparison based on the transactions-cost-adjusted performance of various trading strategies that

directly utilize the forecasts from our different specifications.

5.1 Practical Implementation

The Fama-French market portfolio isn't directly tradeable. Instead, we rely on various market proxies in the form of different ETFs. To conserve space, for our main empirical analyses discussed below, we focus on the highly liquid SPY ETF designed to track the S&P 500. The Online Appendix reports additional complementary results for the QQQ (Nasdaq-100), the IVV (S&P 500), and the IWM (Russell 2000), which are also highly liquid with low transaction costs and may similarly serve as market proxies.¹⁴

We deliberately rely on the forecasts for the Fama-French market portfolio for defining our trading signals, rather than re-estimating new forecasts for the ETF returns. Our motivation for doing so is driven by the fact that the different ETF returns are contaminated by contract-specific microstructure noise and tracking error. Consequently, high-frequency forecasts constructed directly for specific ETF returns may end up targeting economically irrelevant variation, thereby resulting in inferior trading performance (see also the discussion in Cui, Hu, & Wang, 2024). Indeed, as we show in the Online Appendix, trading on the direct forecasts for the ETFs typically generates returns and Sharpe ratios that are markedly worse than the corresponding numbers obtained by trading on the forecasts for the Fama-French market portfolio. This approach, of course, also keeps the trading results tightly linked with the statistical forecasting results discussed in the previous section and substantially reduces the overall computational burden.

¹⁴ As a point of reference, the average time-weighted percentage spreads for the SPY, QQQ, IVV, and IWM over the 2004-2020 period were 0.0067%, 0.0167%, 0.0156%, and 0.008%, respectively.

5.2 Trading Strategies

To keep our analysis concise and straightforward, we consider a limited set of four trading strategies. As a benchmark, we also consider the trivial strategy of simply buying and holding the market.

Our first strategy seeks to time the market based solely on the signs of the forecasts. That is, the strategy responds to a positive (negative) forecast by going long (short) in the market. Let $w_{t,i}$ denote the weight for the market portfolio on day t and intraday time-interval i . The *Sign* strategy is then simply defined by:

$$[\textit{Sign}] \quad w_{t,i+1} = \begin{cases} 1 & \text{if } \hat{r}_{t,i+1}^{mkt} \geq 0, \\ -1 & \text{if } \hat{r}_{t,i+1}^{mkt} < 0, \end{cases} \quad (4)$$

where $\hat{r}_{t,i+1}^{mkt}$ denotes the forecast of the market return by any one of the forecasting models.

The *Sign* strategy effectively treats short and long positions symmetrically. However, short-selling is generally more costly to implement. Moreover, legal constraints and charters also impede many institutional investors from short-selling; see, e.g., the discussion in Pontiff (1996) and Shleifer and Vishny (1997). Accordingly, our *Positive* strategy modifies the sign-based strategy by simply zeroing out the market exposure whenever the forecasts are negative:

$$[\textit{Positive}] \quad w_{t,i+1} = \begin{cases} 1 & \text{if } \hat{r}_{t,i+1}^{mkt} \geq 0, \\ 0 & \text{if } \hat{r}_{t,i+1}^{mkt} < 0. \end{cases} \quad (5)$$

Although this strategy helps circumvent the higher costs typically associated with short-

selling, it still rebalances the entire portfolio whenever the sign of $\hat{r}_{t,i+1}^{mkt}$ changes. As such, it can obviously be quite costly to implement.

Several procedures have been proposed in the literature to help mitigate trading costs. However, these procedures are typically geared toward the specific setting and strategy at hand and they can be difficult to realistically implement in other situation. Instead, motivated by Gârleanu and Pedersen (2013, 2016) and the idea of only trading partially “towards the target,” our third and fourth strategies only trigger rebalance in response to sufficiently strong predictive signals, as measured by the lagged half-spread,

$$[S\text{-}Sign] \quad w_{t,i+1} = \begin{cases} 1 & \text{if } \hat{r}_{t,i+1}^{mkt} > Spread_{t,i}/2, \\ -1 & \text{if } \hat{r}_{t,i+1}^{mkt} < Spread_{t,i}/2, \\ w_{t,i} & \text{otherwise} \end{cases} \quad (6)$$

and

$$[S\text{-}Positive] \quad w_{t,i+1} = \begin{cases} 1 & \text{if } \hat{r}_{t,i+1}^{mkt} > Spread_{t,i}/2, \\ 0 & \text{if } \hat{r}_{t,i+1}^{mkt} < Spread_{t,i}/2, \\ w_{t,i} & \text{otherwise.} \end{cases} \quad (7)$$

As discussed in more detail in the Online Appendix, the *S-Sign* strategy represents the optimal solution to a static optimization problem, where an agent aims to maximize expected returns while facing transaction costs from paying the half spread on each trade. This same argument holds true for the *S-Positive* strategy if we further assume that the trader is constrained to long-only positions only. Both of these strategies are also closely in line with the strategies previously employed by Huddleston, Liu, and Stentoft, 2023

and Chincó, Clark-Joseph, and Ye (2019), although in contrast to these earlier studies we deliberately choose to maintain the previously held position if the signal isn't "strong" enough, rather than putting the weight to zero.

Lastly, whenever the *Positive* and *S-Positive* strategies do imply a zero weight for the market portfolio, we simply associate a zero return with that time interval. Of course, allowing for alternative cash investments like Treasury Bills would enhance the calculated returns.

5.3 Economic Performance Measures

Having defined the portfolio weights, we assess the trading performance by computing the resulting average intraday returns and Sharpe ratios over our out-of-sample period. That is

$$\text{AvgRet}^{F,m,k} = \frac{1}{T_{OOS}} \sum_{t,i} w_{t,i}^k r_{t,i+1}^F, \quad (8)$$

and

$$\text{Sharpe}^{F,m,k} = \underbrace{\left(\frac{1}{T_{OOS}} \sum_{t,i} (w_{t,i}^k r_{t,i+1}^F - r_{t,i+1}^f) \right)}_{\text{AvgExcRet}^{F,m,k}} / \underbrace{\left(\sqrt{\frac{1}{T_{OOS}} \left(\sum_{t,i} (w_{t,i}^k r_{t,i+1}^F)^2 \right)} \right)}_{\text{AvgVol}^{F,m,k}}, \quad (9)$$

where T_{OOS} denotes the number of years in the out-of-sample period, m refers to a particular model (specification + machine learning algorithm), k refers to a strategy, and F refers to the return on the specific market proxy. We proxy for the intraday risk-free rate $r_{t,i+1}^f$ using the daily Treasury bill returns obtained from Ken French's website, proportionally distributing the daily returns over the intraday and overnight intervals. In

line with the underlying forecasting models, the above-defined average return and Sharpe ratio formulas are both based exclusively on intraday returns. However, to allow for more a direct comparison with existing work, we also subsequently consider overnight-inclusive versions of both measures.

To account for the transaction costs associated with the practical implementation, we begin by defining

$$\text{Turn}_{t,i}^{m,k} \equiv |w_{t,i}^{m,k} - w_{t,i-1}^{m,k}|. \quad (10)$$

We then calculate the (transaction cost) adjusted returns as:

$$\text{AdjAvgRet}_{t,i}^{F,m,k} \equiv r_{t,i}^{F,m,k} - \underbrace{\text{Turn}_{t,i}^{m,k} \times \frac{\tau_{t,i}^F}{2}}_{\text{trading cost}}, - \underbrace{\mathbb{1}[w_{t,i}^{m,k} < 0] \times v_{t,i}^F}_{\text{shorting cost}} \quad (11)$$

where $\tau_{t,i}^F$ denotes the corresponding spread and $v_{t,i}^F$ is the borrow fee for shorting. Correspondingly, our adjusted Sharpe ratio is defined by

$$\text{AdjSharpe}^{F,m,k} \equiv \text{AvgExcAdjRet}^{F,m,k} / \text{AvgVol}^{F,m,k}, \quad (12)$$

where the average excess adjusted return is computed by subtracting the risk-free rate from $\text{AdjAvgRet}_{t,i}^{F,m,k}$, analogous to equation (9).

In the calculations reported below, we measure $\tau_{t,i}^F$ by the “Time-Weighted Quoted Spread” provided by the NYSE TAQ database. This implicitly assumes that rebalancing always incurs half the spread in transaction costs. Lower costs may likely be attained by some investors, thus further enhancing the performance of our strategies. Additionally, we rely on a very conservative fixed annualized value of 2% for the borrow rate, or shorting cost $v_{t,i}^F$.¹⁵

¹⁵ As a reference, InteractiveBrokers (<https://www.interactivebrokers.com/en/pricing/short-sale-cost>).

Although our trading strategies, and our assessment of their performance, are exclusively based on the intraday returns, the above transaction cost calculations invariably depend on the positions over the first and last 15-minute intervals of the trading day, and in turn the treatment of the overnight period. Motivated by the finding that much of the market return has historically been earned overnight, in the calculations reported on below, we fix the weight for the overnight period for all of our strategies to be one.¹⁶ Accordingly, the weight at the start of each day is always set to unity, or $w_{-1,t}^{m,k} \equiv 1$.

5.4 General Results

Motivated by the forecast results for the market portfolio discussed in Section 4, we begin our discussion of the ETF-based trading strategies with a broad overview of the performance measures achieved by the forecasts from the various Ensemble specifications to trade the SPY. In addition to the above-defined performance measures, Table 4 also reports the Fama-French six-factor alphas, computed from regressing the high-frequency returns for each of the strategies on the high-frequency factor returns.¹⁷ For completeness, we report separate results based exclusively on the intraday returns (left part of the table) and the results that obtain by also incorporating the overnight returns (right part of the table).

php) charges a 0.25% fee for shorting highly liquid securities, so that our use of 2% is eight times higher than the cost faced by many investors.

¹⁶The additional analysis in Online Appendix shows that the intraday performance measures discussed below are essentially unaltered when closing out the market position at the end of the day, or when maintaining the weight defined by the intraday strategy for the 15:45-16:00 time interval during the overnight period. It is, of course, possible that by actively trading the overnight returns even better strategies may be obtained. This, however, would also entail the development of very different forecasting models from the high-frequency models underlying our current intraday forecasts, and we will not pursue this any further here.

¹⁷To allow the factor loadings to vary over time, we estimate the regressions on a rolling annual basis. Of course, since all the strategies only invest in the market portfolio, fixing the factor loadings for the market factor to be equal to the high-frequency market weights, all of the alphas would be identically equal to zero. Instead, the reported alphas should be interpreted as traditional measures of risk-adjusted performance “as if” the strategies were investments into some unknown assets or actively traded funds.

Table 4: Trading Performance of Ensemble Model

Adjusted Intraday Performance					Adjusted Overnight-Inclusive Performance				
Regressand	Regressors				Regressand	Regressors			
	All		All+Jmp			All		All+Jmp	
	All	Cts	All	Cts		All	Cts	All	Cts
Panel A1: Return (%)					Panel A2: Return (%)				
Positive	7.24	8.67	6.12	6.90	Positive	14.52	15.97	13.42	14.22
Sign	12.09	15.02	9.88	11.50	Sign	18.94	21.93	16.78	18.45
S-Positive	9.65	11.13	9.50	8.96	S-Positive	16.89	18.40	16.77	16.24
S-Sign	16.86	19.89	16.61	15.56	S-Sign	23.64	26.73	23.45	22.43
Panel B1: Sharpe					Panel B2: Sharpe				
Positive	0.64	0.75	0.54	0.58	Positive	0.89	0.97	0.83	0.86
Sign	0.83	1.03	0.67	0.78	Sign	1.02	1.18	0.90	0.99
S-Positive	0.89	0.99	0.89	0.79	S-Positive	1.06	1.14	1.06	1.00
S-Sign	1.16	1.37	1.14	1.07	S-Sign	1.28	1.44	1.26	1.20
Panel C1: Alpha (%)					Panel C2: Alpha (%)				
Positive	6.83	8.07	5.88	6.42	Positive	9.48	10.48	8.28	8.67
Sign	13.19	15.74	11.32	12.46	Sign	18.32	20.40	15.95	16.79
S-Positive	9.29	10.64	9.26	8.53	S-Positive	12.04	13.19	11.72	10.74
S-Sign	18.06	20.83	18.04	16.62	S-Sign	23.38	25.76	22.80	20.88
Panel D1: Alpha t-statistic					Panel D2: Alpha t-statistic				
Positive	3.63	4.32	3.09	3.41	Positive	4.57	5.13	3.93	4.22
Sign	3.56	4.28	3.01	3.37	Sign	4.54	5.15	3.89	4.22
S-Positive	4.96	5.70	4.86	4.55	S-Positive	5.80	6.43	5.54	5.22
S-Sign	4.88	5.68	4.78	4.49	S-Sign	5.78	6.48	5.51	5.21

Note: The table reports the out-of-sample, adjusted for transactions costs annualized returns, Sharpe ratios, and alphas obtained by trading the SPY ETF based on our Ensemble model forecasts. Panels D1 and D2 report the t -statistics for the adjusted alphas. The sample period spans 2004-2020. The intraday performance measures reported in the left panels, labelled 1, exclude the 16:00-9:30 overnight returns, while the overnight-inclusive performance measures reported in the right panels, labelled 2, include the overnight returns. The alphas are estimated using yearly regressions against the FF6 factors.

Looking at the results, the *Cts*, *All* specification, which uses the continuous component of the returns as the regressand and the total lagged returns as the regressors, seemingly outperforms all the other approaches in terms of the average adjusted returns, the adjusted Sharpe ratios, and the FF6 alphas, regardless of the specific strategy used to translate the forecasts into trades. Moreover, for both the *All* and *All+Jmp* regressor specifications, using the continuous returns as the regressand results in the best performance according to all the different measures. As such, the results in Table 4 directly corroborate our earlier conjecture that only targeting the potentially predictable return variation does indeed result in more economically meaningful out-of-sample forecasts.

Table 5: Intraday Adjusted Sharpe Ratios for (Cts, All) Specifications

Model	Strategy			
	Positive	Sign	S-Positive	S-Sign
AR1	-1.42	-2.03	0.18	0.16
AR4	-0.18	-0.33	0.44	0.51
AR26	-0.41	-0.63	0.38	0.44
Linear	0.06	-0.00	0.12	0.09
Ridge	0.49	0.62	0.61	0.79
Lasso	0.44	0.54	0.68	0.88
ENet	0.41	0.50	0.67	0.86
AdaLasso	0.44	0.52	0.56	0.68
PCR	0.47	0.58	0.75	0.95
PLS	0.44	0.54	0.93	1.21
FNN	0.77	1.06	0.90	1.25
RF	0.70	0.99	0.42	0.48
GBRT	0.66	0.92	0.99	1.44
Ensemble	0.79	1.09	0.98	1.36

Note: The table reports the 2004-2020 out-of-sample adjusted Sharpe ratios for trading the SPY ETF based on the forecasts for each of our machine learning algorithms under the *Cts, All* specification. All of the values are annualized.

The table also reveals that the *S-Positive* and *S-Sign* strategies, designed to help mitigate transaction costs, generally perform better than their simpler counterparts. Further underscoring the ability of these two strategies to better manage the cost of trading, the relative performance of the same two strategies when ignoring transaction costs is markedly worse than their simpler counterparts. It is also noteworthy that allowing for short positions results in substantial economic gains, as manifest in the intraday adjusted Sharpe ratio for the *Cts, All* specification rising from 0.99 to 1.37 when shifting from the *S-Positive* to the *S-Sign* strategy and the corresponding alphas more than doubling. Allowing for short positions also nearly doubles the returns, suggesting that the increase in alpha is indeed driven by genuine predictability rather than simply reducing the factor exposures through negative weights. Consistent with the idea that much of the return on the market is earned overnight, comparing the results in the left and right panels, the inclusion of the overnight returns naturally also increases all of the different performance measures.

Next, to better understand the performance of our suite of machine learning models,

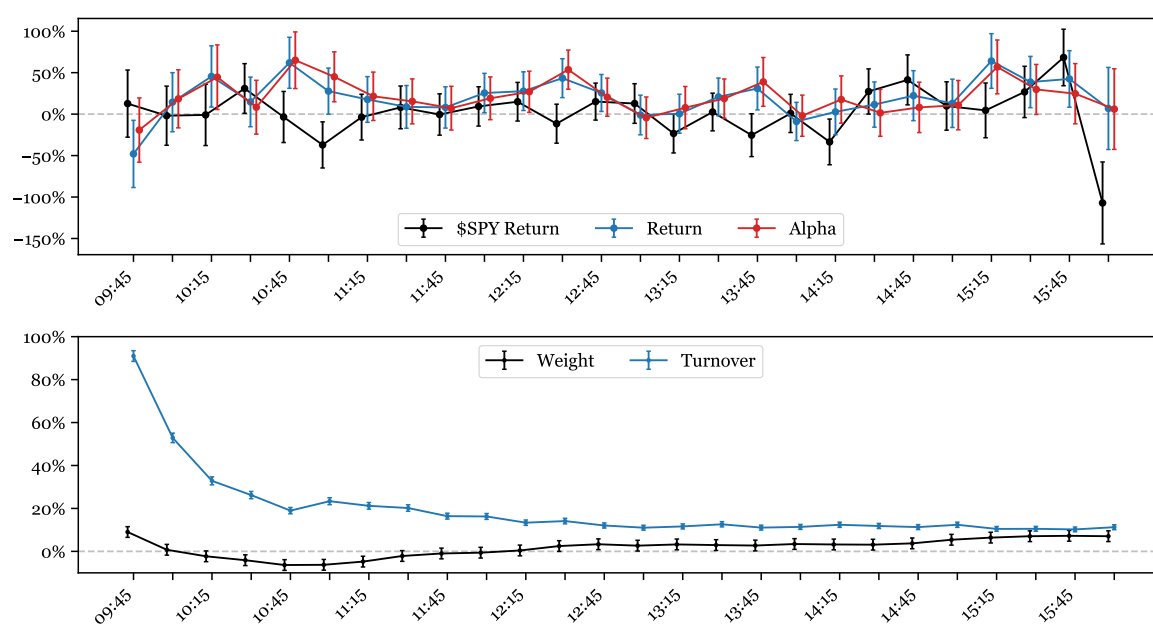
Table 5 reports the intraday adjusted Sharpe ratios, along with the Sharpe ratios for our simple benchmark models. As the table shows, the benchmark models, together with traditional linear OLS-based estimation of the different specifications, all perform quite poorly under the *Positive* and *Sign* trading strategies. The use of the *S-Positive* and *S-Sign* strategies to translate the forecasts into trades improves the performance. However, the results for the benchmark models are markedly worse than the results for all the machine learning models. The forecasts from Partial Least Squares (PLS), Feedforward Neural Networks (FFN), and Gradient Boosted Regression Trees (GRBT), in particular, when combined with the *S-Sign* strategy all achieve Sharpe ratios in excess of 1.20. Interestingly, the Ensemble average forecasts attain the overall highest Sharpe ratios under both the *Positive* and *Sign* strategies, and very close to the highest ratios under the *S-Positive* and *S-Sign* strategies, thus effectively obviating the difficult task of trying to identify a single best model.¹⁸ We turn next to a more detailed analysis of when these superior returns are earned.

5.5 Intraday and Interday Patterns

Motivated by the results in the previous section, to streamline the presentation, we will focus our more in-depth analyses on the Ensemble model and the *Cts*, *All* specification, along with the *S-Sign* strategy for trading the SPY. We begin by characterizing the intradaily variation in the performance, followed by a look at the longer-run interday temporal variation and the relation to underlying economic conditions.

¹⁸ Further along these lines, the Online Appendix reports SHAP values for each of the individual forecasting models and their contributions to both the Sharpe ratio and the alpha for the (*Cts*, *All*)-(*Ensemble*)-*S-Sign* model; SHAP values are defined more precisely in Section 6 below. The patterns observed for these contributions generally resemble the patterns observed for the individual forecasts themselves. Interestingly, the linear and the non-linear models both appear important in terms of driving the superior performance of the Ensemble average.

Figure 4: Average Intraday Returns and Turnover



Note: The first subplot shows the average annualized returns over each 15-minute interval within the day for the SPY ETF, the *(Cts, All)-(Ensemble)-(S-Sign)* strategy for trading the SPY, and the corresponding FF6 alphas. The bars refer to 90% confidence intervals. The bottom subplot shows the average portfolio turnover and weight across the day. Both plots are based on returns and portfolio weights for the 2004-2020 out-of-sample period. The markers are staggered for visibility.

The top panel in Figure 4 reports the average returns over each of the 15-minute intervals of the trading day, together with the average returns and FF6 alphas earned from our trading strategy together with the corresponding 90% confidence intervals. The average returns for the SPY, not surprisingly, closely mirror the on-average close to zero and flat average intraday returns for the Fama-French market portfolio previously shown in Figure 1. The most notable systematic differences from zero seemingly occur around 14:15-14:45, probably as a result of FOMC announcements at that time, and toward the end of the trading day, likely due to increased selling pressure at that time.¹⁹ The average intraday returns for our main trading strategy in contrast show some notable variation, especially around 10:45 coincident with the time at which many macroeconomic data

¹⁹ The intraday pattern for the Fama-French market portfolio, previously depicted in Figure 1 averaged over the full 1996-2020 sample period, also evidences a negative, albeit not significantly so, average return over the last fifteen-minute of the active portion of the trading day when restricting the sample to the 2004-2020 out-of-sample period.

are announced (see, e.g., Andersen et al., 2003), and around the 14:15-14:45 FOMC announcement time. The alphas display a similar pattern, suggesting that our strategy may be in part exploited a delayed reaction to economic information. We also find slightly larger returns during the middle of the day, mirroring Huddleston, Liu, and Stentoft (2023) who find an inverse-U pattern in the degree of predictability across the day, and their trading strategies correspondingly earning the highest abnormal returns in the middle of the day and performing worse than the SPY at the end of the day.

Looking at the bottom panel in Figure 4 reveals that the proportion invested in the SPY remains fairly stable across the day with a slight dip around 10:45. This again is most likely explained by the systematic release of macroeconomic news at that time. Meanwhile, turnover peaks at the beginning of the day and proceeds to decline throughout the rest of the day.²⁰ The sharp decline observed at the start of the day is readily explained by the fact that our overnight weight is fixed at 100%, thus resulting in high turnover as the strategy readjust to the on-average much lower optimal intraday weights. More broadly, the overall patterns in both the weights and the turnover also mimic the well-documented U-shaped pattern in intraday volatility (see, e.g., Andersen & Bollerslev, 1997), suggesting that there is more predictable variation in the high-frequency returns to be exploited when volatility is high.

Turning to the longer-run interday dynamic dependencies, the top panel in Figure 5 shows the cumulative daily returns over the full 2004-2020 out-of-sample period. The cumulative return on the SPY at the end of the sample is less than 50%, consistent with the on-average very low intraday return on the market, and the SPY, discussed earlier. By comparison, our main strategy delivers a cumulative return of more than 300% by

²⁰ Recall that the turnover defined in (10) is computed as the absolute difference in the weights. Hence, a strategy that switches between being fully invested in the SPY and cash over each successive 15-minute interval would have a turnover of 100%.

Figure 5: Cumulative Interday Returns and Turnover – (Cts, All)-(Ensemble)-(S-Sign)



Note: The first subplot reports the cumulative intradaily returns for the SPY ETF, the (Cts, All)-(Ensemble)-(S-Sign) strategy for trading the SPY, and the corresponding FF6 alphas. The trading strategy returns and alphas are both adjusted for transaction costs. The second subplot shows the average monthly weights for the SPY and turnover for our main trading strategy. The average weights and turnover are computed on a daily basis and further smoothed with a 21-trading-day EWMA. Shaded regions mark NBER-defined recessions.

the end of the sample. Interestingly, it appears that the largest gains manifest during recessions, exactly when the market tends to perform poorly. However, the returns are not simply coming from exiting the market during bad times. In fact, using the long-only *S-Positive* strategy instead of the *S-Sign* strategy still delivers markedly higher returns than the SPY itself over the same recession periods. Put differently, our trading strategy does not simply exit the market during bad economic times, but instead actively trades to exploit market dislocations that occur at such times. The large returns and alphas observed at the beginning of 2020 and the time of the COVID-induced recession also support this conjecture.

Further elaborating on this theme, the bottom panel in Figure 5 shows the rolling monthly weight on the SPY and the turnover of the main trading strategy. The average turnover hovers around 20% for most of the sample, suggesting a moderate, but not

Table 6: Trading Performance versus Monthly Uncertainty Measures

	SPY	Return	Alpha	SPY	Return	Alpha	SPY	Return	Alpha
Intercept	1.63 (1.18)	-7.98*** (-2.82)	-9.77*** (-2.80)	1.47 (0.94)	-7.99** (-2.30)	-9.61** (-2.22)	0.60 (0.60)	-5.06* (-1.66)	-5.92* (-1.68)
Financial Uncertainty	-0.28 (-1.00)	1.81*** (3.28)	2.16*** (3.14)						
Macro Uncertainty				-0.22 (-0.80)	1.55*** (2.61)	1.82** (2.46)			
Real Uncertainty							-0.08 (-0.44)	1.09** (1.99)	1.25* (1.95)
R^2	0.011	0.209	0.239	0.008	0.171	0.193	0.001	0.110	0.114

Note: The table reports the results from regressions of the monthly returns of the SPY ETF, the *(Cts, All)-(Ensemble)-(S-Sign)* strategy for trading the SPY, and its corresponding FF6 alphas. All of the regressions are based on 204 monthly observations. The three uncertainty measures are obtained from Jurado, Ludvigson, and Ng (2015) and are normalized by their sample standard deviations. Robust *t*-statistics are reported in parentheses, with *, **, and *** indicating significance at the 90%, 95%, and 99% levels, respectively.

excessive amount of portfolio rebalancing. On the other hand, the average weight tends to fluctuate quite substantially throughout the sample, again underscoring the previous point that the strategy actively seeks to find alpha via trading at all times.

5.6 Performance versus Economic Conditions

To help further assess the relationship between overall economic conditions and the performance of our main strategy, we consider its' relation to the monthly uncertainty measures proposed by Jurado, Ludvigson, and Ng (2015). Lagging the measures by a month to avoid any look-ahead-bias, we regress the total monthly intraday returns, and the monthly FF6 alphas, on each of the different measures. For ease of interpretation, we further normalize each of the uncertainty measures by their sample standard deviation. As a reference, we also include the same regressions involving the monthly intraday returns for the SPY. All of the regressions span the full 2004-2020 out-of-sample period, for a total of 204 monthly observations.

The resulting estimates, reported in Table 6, show that the performance of the strategy is indeed strongly positively related to all three uncertainty measures. A one stan-

Table 7: Trading Performance versus Daily Volatility Measures

	SPY	Return	Alpha	SPY	Return	Alpha
Intercept	-0.00 (-0.13)	-0.07*** (-2.60)	-0.08** (-2.54)	-0.10** (-2.30)	-0.24*** (-3.58)	-0.22*** (-3.11)
SPY Realized Volatility	1.10 (0.26)	17.33*** (4.69)	18.38*** (4.50)	-8.55 (-1.04)	0.96 (0.15)	4.29 (0.66)
VIX				15.51* (1.86)	26.33*** (2.76)	22.66** (2.27)
R^2	0.000	0.024	0.023	0.003	0.034	0.030

Note: The table reports the results from regressions of the daily returns of the SPY ETF, the *(Cts, All)-(Ensemble)-(S-Sign)* strategy for trading the SPY, and its corresponding FF6 alphas. All of the regressions are based on 4,282 daily observations. The realized variance (RV) of the SPY is computed using 15-minute intraday and overnight returns. The VIX index is obtained from the CBOE website and is rescaled to match the daily realized volatility estimates. Both regressors are lagged by one day. Robust t -statistics are reported in parentheses, with *, **, and *** indicating significance at the 90%, 95%, and 99% levels, respectively.

dard deviation increase in “Financial Uncertainty” or “Macro Uncertainty” increases the monthly return by one to two percent. The “Financial Uncertainty” and “Macro Uncertainty” measures also both do a remarkable job in terms of explaining the monthly cumulative alphas, with R^2 s of 23.9% and 19.3%, respectively. As such, this again supports the idea that our trading strategy is able to exploit dislocations that tend to occur during highly uncertain, or bad economic, times. By contrast, all of the estimated slope coefficients for the SPY returns are insignificant and the R^2 ’s close to zero. This lack of significance for the SPY, of course, is also in line with the vast market predictability literature referenced earlier, emphasizing the difficulty of reliably predicting aggregate stock market returns; see, e.g., Welch and Goyal (2008).

To provide an even more detailed look at when the strategy performs well, Table 7 reports the results from finer daily regressions against the lagged CBOE VIX index and the lagged realized volatility of the SPY. To be comparable with the VIX, the realized volatility of the SPY is computed using the summation of the squared 15-minute intraday and overnight returns, with both measures rescaled to a daily magnitude. Looking at the results reported in the second column, a 1pp increase in the daily realized volatility

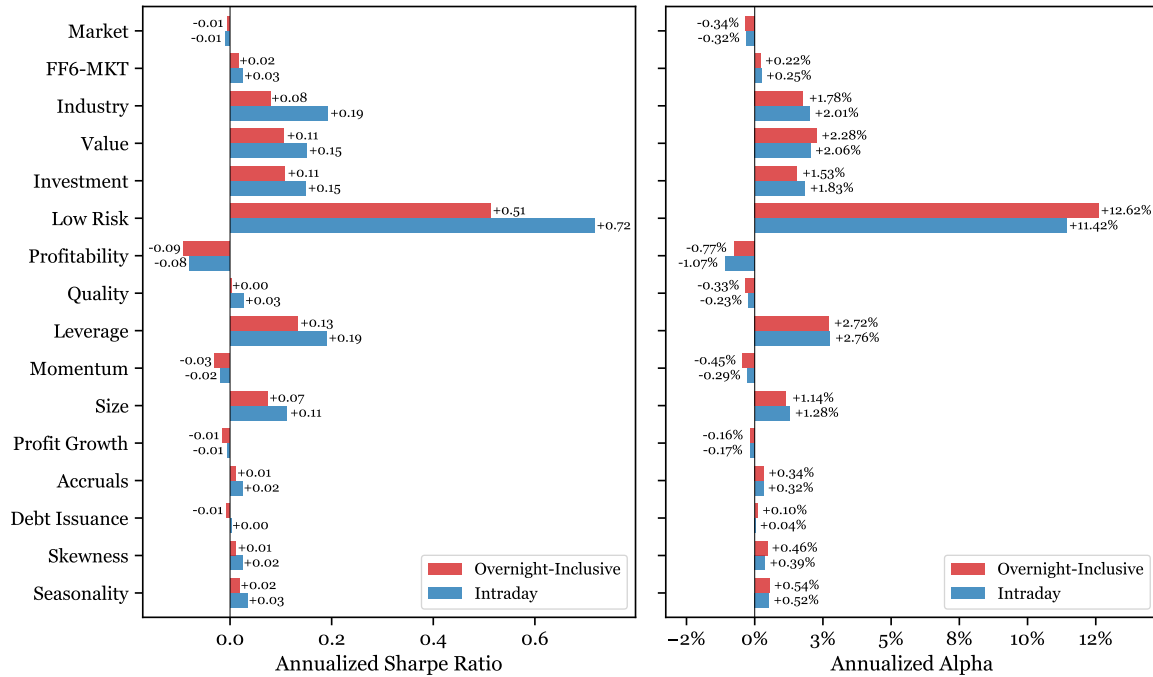
is associated with an impressive 17.33pp increase in the annualized intraday return for our main strategy. The results obtained for the alphas are similar, with the equivalent annualized increase equal to 18.38pp. The corresponding daily regression R^2 's of 2.4% and 2.3%, respectively, are also quite high. By comparison, the regression coefficient for the SPY returns, reported in the first column, is insignificant, and the R^2 is also effectively zero.

Looking at the second set of regressions, the results for the SPY appear reminiscent of Bollerslev, Tauchen, and Zhou (2009), and the finding that the variance risk premium, defined as the difference between the two volatility measures, is able to predict the return on the market. However, the estimated coefficients are still insignificant and they also have the “wrong” sign vis-a-vis the results in the extant literature based on the returns over longer monthly and quarterly horizons that include the overnight returns. Meanwhile, the regressions involving the intraday returns on our *(Cts, All)-(Ensemble)-(Sign)* trading strategy and its FF6 alphas again point to significantly better performance when the VIX is high, suggesting that implied volatility, or “investor fears” (Bollerslev & Todorov, 2011), rather than realized volatility is most closely associated with its superior performance.

6 Feature Importance and Model Interpretability

To help clarify which of the factors drive this performance, it is instructive to compute so-called SHAP (SHapley Additive exPlanations) values (Lundberg & Lee, 2017). SHAP values are now routinely used in the machine learning literature as a way to succinctly quantify variable importance in high-dimensional settings. They have also recently been used in the context of finance applications (see, e.g. Coulombe et al., 2023; Erel et al.,

Figure 6: Factor Cluster SHAP Contributions



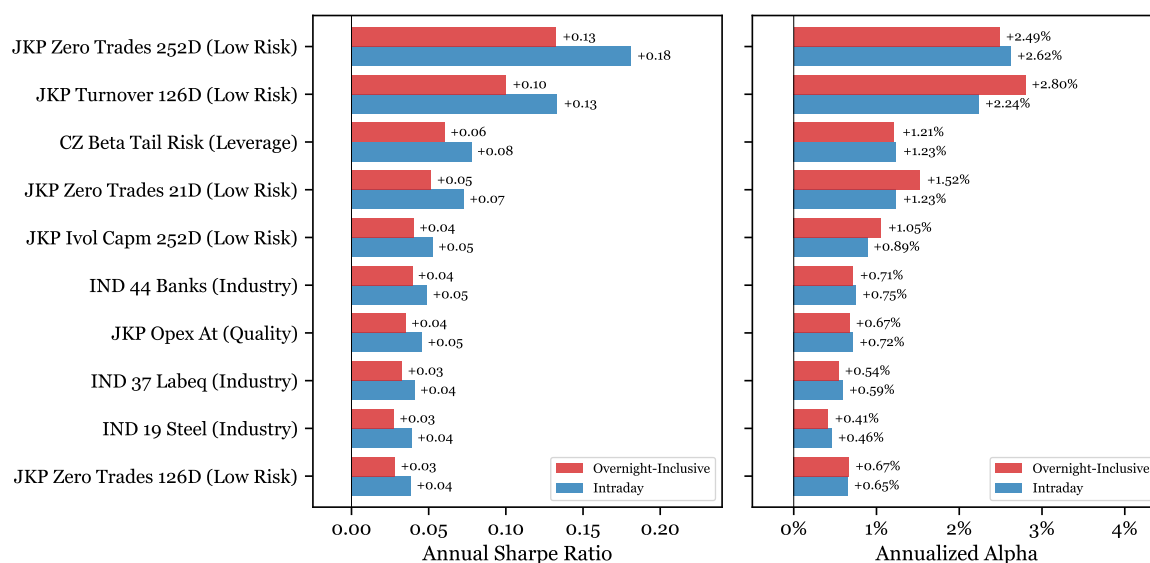
Note: The figure reports the SHAP values for the different factor clusters, the market portfolio (MKT), the Fama-French six portfolios except for the market (FF6-MKT), and the 48 industry portfolios (Industry), pertaining to the intraday and overnight-inclusive Sharpe ratios and FF6 alphas from trading the SPY ETF based on the *(Cts, All)-(Ensemble)-(S-Sign)* strategy. The Sharpe ratios and alphas are both reported in annualized units.

2021; Moehle, Boyd, & Ang, 2021) and more general economic time-series forecasting applications (see, e.g. Borup et al., 2022). Put simply, SHAP values split a specific performance measure for a given model into the marginal contributions stemming from each of the variables included in said model; unlike certain other variable importance measures, the SHAP values are additive – the values for each regressor variable ultimately sum to the performance of the baseline model. Precise details concerning our actual calculation of the SHAP values are provided in Appendix B.

6.1 SHAP Values

We begin with a birds-eye view of the variable importance for the different factor clusters previously illustrated in Figure 2 and their contributions to the Sharpe ratios and FF6

Figure 7: Individual Factor SHAP Contributions



Note: The figure reports the SHAP values for each of the individual factors pertaining to the intraday and overnight-inclusive Sharpe ratios and FF6 alphas from trading the SPY ETF based on the *(Cts, All)-(Ensemble)-(S-Sign)* strategy. The Sharpe ratios and alphas are both reported in annualized units. For brevity, the figure only reports the factors with the top ten most positive SHAP contributions to the intraday Sharpe ratio. The specific clusters associated with each of the factors are reported in parentheses.

alphas. In addition to the fourteen different factor clusters, we also include the Fama-French market portfolio, the six Fama-French portfolios excluding the market portfolio, and the 48 industry portfolios, as additional sets of clusters. For completeness, we report the results for both the intraday exclusive and the overnight inclusive Sharpe ratios and FF6 alphas.

Looking at the results in Figure 6, quite remarkably the vast majority of the superior performance, both in terms of Sharpe ratios and alphas, may be attributed to a single cluster, namely the Low Risk cluster. Much smaller gains also arise from the Industry, Value, Investment, and Leverage clusters, while all of the other clusters, as well as the market and the Fama-French factors, appear to contribute very little, if anything, to the performance.

To further help identify the specific factors within the different clusters that drive the performance, Figure 7 reports the SHAP values for the top ten factors by their

intraday Sharpe ratios.²¹ The names of the factors are adopted directly from Chen and Zimmermann (2022) and Jensen, Kelly, and Pedersen (2023), as indicated by the “CZ” and “JKP” labeling in the figure. Consistent with the previous Figure 6, four out of the top-five factors come from the Low Risk cluster, with the remaining factor stemming from the Leverage cluster. Interestingly, the SHAP values appear to be fairly evenly spread across several factors. This finding aligns with the use of models such as Ridge and PLS, which implicitly assume that the predictive signals are relatively evenly distributed among many predictor variables. By contrast, sparse models like Lasso naturally result in sparser SHAP values. The results depicted in Figure 7 pertaining to the Ensemble model naturally exhibit both features.

Digging further into the specific factors that matter the most, the “JKP Zero Trades 252D” and “JKP Zero Trades 21D” factors are both based on sorting stocks by the number of days with zero trades over the past 252 and 21 days, respectively (Liu, 2006), while the “Turnover 126D” factor is defined by sorting stocks based on shares traded divided by shares outstanding averaged over the previous 126 trading days (Chordia, Subrahmanyam, & Anshuman, 2001). The “Beta Tail Risk” factor sorts stocks based on a nuanced measure of their tail risk (Kelly & Jiang, 2014), while the “IVol CAPM 252D” factor sorts stocks based on their rolling CAPM idiosyncratic volatility (Ali, Hwang, & Trombley, 2003). Echoing the discussion in the previous section, the tail risk factor is naturally associated with time-varying economic and financial uncertainty. Meanwhile, the four Low Risk factors, which account for most of the superior performance, all speak to notions of market liquidity. It is natural to hypothesize that the corresponding long-short portfolios essentially capture the return differentials between stocks that respond

²¹ Complementary analysis in the Online Appendix further reveals that the SHAP values are generally concentrated in a few factors within each cluster.

more quickly to new information versus stocks that respond more slowly to news, thereby allowing for the identification of periods with more marked cross-sectional differences in the rate of news absorption and poorer overall market quality. The analysis in the next section further strengthens this argument and more directly illustrates how the Low Risk factors translate into useful trading signals.

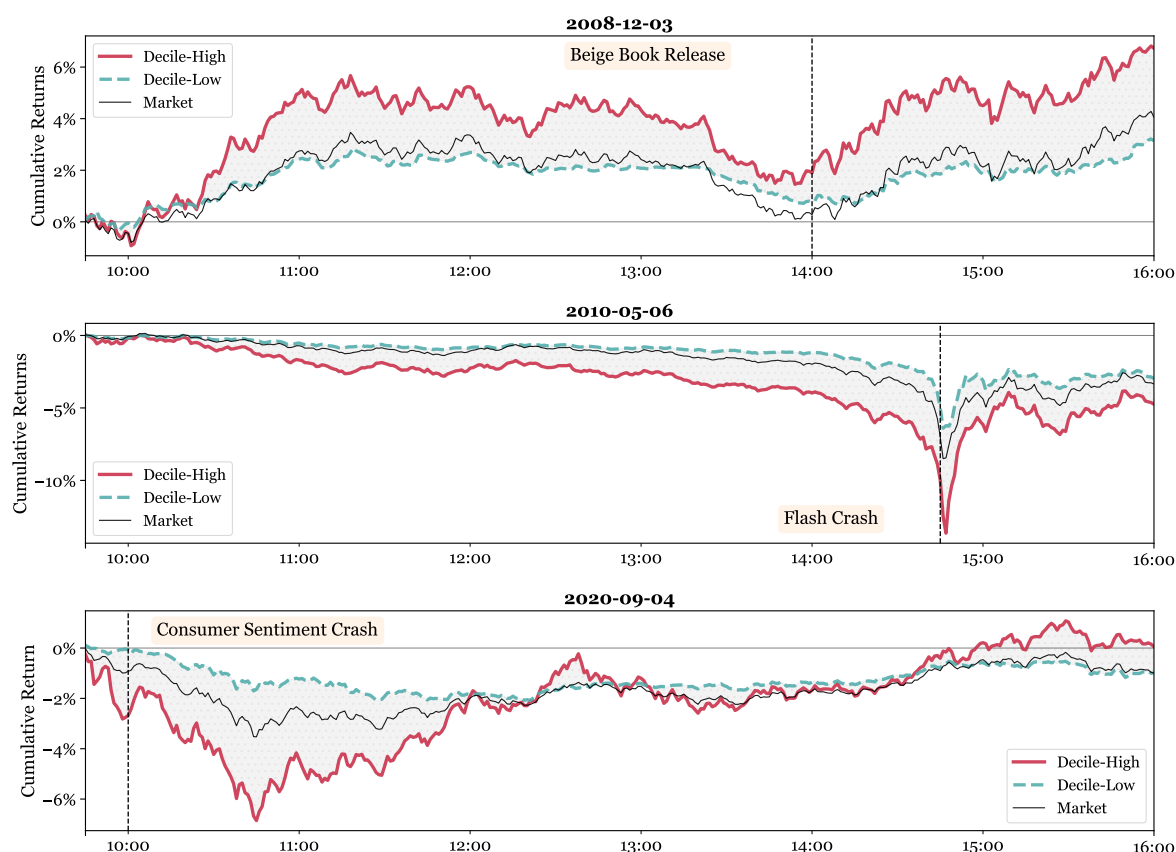
6.2 Liquidity Factors and Predictability

Although each of the four factors from the Low Risk cluster identified above contribute substantially to the total trading performance, they are all designed to capture similar aggregate liquidity-type effects. Hence, not surprisingly, as previously illustrated in [Figure 2](#), the four factors are also strongly correlated, with the full-sample realized correlations between each of the pairs lying around 90-95%.²² In other words, the predictive content of the factors is likely quite similar. Thus, for brevity in the analysis that follows, we will focus exclusively on the “turnover” factor, but very similar numerical results and pictures to the ones discussed below obtain for the three other Low Risk factors.

To help better illuminate which leg of the turnover factor primarily drives the performance, we begin by constructing value-weighted decile-sorted turnover portfolios at an even finer one-minute frequency, following the exact same approach used in the construction of the original “turnover” factor by Chordia, Subrahmanyam, and Anshuman (2001). [Figure 8](#) plots the resulting cumulative returns on the highest and lowest decile portfolios for three select days with various important economic news: the Beige Book release at 14:00 on March 12, 2008; the Flash Crash at around 2:45 on May 6, 2010; and the release of the Consumer Sentiment index at 10:00 on April 9, 2020. As the figure shows, the high

²² By comparison, the average 15-minute return correlations between each of the Low Risk factors and the tail risk factor from the Leverage cluster is only around 66%.

Figure 8: Turnover Portfolio Returns



Note: The figure reports the cumulative intradaily returns on the high and low decile turnover portfolios, together with the cumulative returns on the Fama-French market portfolio, for three select days. The return differentials between the high and low decile portfolios are shaded in gray. The (approximate) times of the labeled news events are indicated by dashed vertical lines.

decile portfolio, comprised of stocks that tend to trade more frequently, clearly responds more quickly to the specific news. Interestingly, the gap between the cumulative returns on the two decile portfolios, which essentially corresponds to the cumulative return on the lower-frequency “turnover” portfolio used in our forecasting models, also appears to foreshadow the subsequent direction of the market. These general features are perhaps most clearly evident in the second subplot and the day of the Flash Crash in 2010.²³ In fact, it appears as if the high decile portfolio anticipates the crash. Correspondingly, there is a steadily increasing gap between the returns on the high and low decile portfo-

²³The Flash Crash has, of course, been extensively studied by market regulators and elsewhere in the academic literature; see, e.g., Andersen and Bondarenko (2014) and the many additional references therein.

lios leading up to the crash. Our trading strategy takes advantage of this and effectively uses the high-minus-low portfolio return as a leading indicator for exiting the market in advance of the crash. Likewise, the widening negative gap observed in the bottom panel following the release of the Consumer Sentiment index signals that it is a good time to exit the market, while the converse widening positive gap observed in the top panel following the Beige Book release suggests that it is a good time to enter the market. Put differently, relevant new information appears to be absorbed more rapidly into the high turnover portfolio, with the high-low return differential foreseeing subsequent aggregate market moves.

As discussed earlier, the basic premise that high-turnover portfolios tend to lead low-turnover portfolios is hardly new to the literature. Chordia and Swaminathan (2002), for example, has previously shown that portfolios comprised of high-volume stocks appear to lead those of low-volume stocks, in turn inducing significant cross-autocorrelations in the returns over short horizons. Existing more theoretically oriented work, including Chan (1993) and Bernhardt and Mahani (2007), have also demonstrated how these differences in the absorption rates of market-wide information, or speeds of adjustments, may naturally arise from informational and trading frictions, with Mech (1993) positing that differential costs of trading provide the most likely explanation for the observed lead-lag relationships. However, we still find a non-trivial, and highly statistically significant, amount of alpha even after accounting for transaction costs from trading on the innovations of the high turnover stocks through our turnover-sorted factor portfolio.

To further develop the empirical support for this thesis, we consider two additional simple trading strategies based on either the long or the short leg of the high-minus-low turnover portfolio. Importantly, to eliminate irrelevant common variation, we reconstruct

these high and low turnover portfolios as the 15-minute return on each of the underlying legs minus the 15-minute return on the market portfolio, following the same approach for market-neutralization as in Dong et al. (2022). We then use the lagged 15-minute returns on the resulting High-Minus-Mkt and Mkt-Minus-Low portfolios to define a simple trading strategy where we go long/short based on whether this variable is above/below a one-standard deviation threshold.²⁴

Consistent with the idea that the superior performance attributed to the turnover factor primarily stems from more frequently traded stocks that exhibit a faster speed of adjustment, this simple High-Minus-Mkt strategy attains an intraday Sharpe ratio of 0.50 from trading the SPY ETF. By comparison, the Sharpe ratio for the Mkt-Minus-Low strategy only equals -0.13. The transaction-cost adjusted intraday FF6 alphas for the same two strategies equal 4.90% ($t = 2.03$) and -0.44% ($t = -0.17$), respectively. In short, the economically relevant predictive content from our High-Minus-Low turnover factor lies in the “high” leg. Meanwhile, the intraday Sharpe ratio and alpha for the simple High-Minus-Mkt strategy are still markedly lower than the 1.37 Sharpe ratio and 20.83% alpha obtained for the *(Cts, All)-(Ensemble)-(S-Sign)* strategy reported in Table 4. Thus, even though this analysis points to some performance from merely exploiting the long leg of the turnover factor, much of the superior performance of our main strategy still stems from the use of modern machine learning techniques in conjunction with the full factor zoo. Of course, the above analysis and discussion are also inherently ex-post in nature, relying on the estimation results for the *(Cts, All)* Ensemble model together with the SHAP values to actually identify the most important factors, and the turnover factor in particular.

²⁴ Intuitively, the strategy only trades if the forecast is sufficiently large/small. The standard deviation that we use for the threshold is computed on a rolling basis with a 3-month window, corresponding to $26 * 21 * 3 = 1,638$ high-frequency observations.

7 Conclusion

We document significant intraday market return predictability from the use of high-frequency intraday factors combined with modern machine learning techniques. Utilizing the ensemble forecasts from our prediction models for the market portfolio in the construction of simple trading strategies for a series liquid ETFs including the SPY results in large economic gains. The trading models perform particularly well during periods of increased economic uncertainty and heightened financial market volatility, generally thought to be associated with greater market dislocations. Consistent with that idea, we further trace most of the superior performance to a few liquidity-related factors that effectively capture differences in the rate of news absorption across stocks.

The factors that form the foundation for our empirical analyses have chiefly been proposed to explain cross-sectional, as opposed to time-series, variation in equity returns. As such, it is possible that similar techniques to the ones developed here may give rise to even better prediction models and trading strategies when applied to individual stocks. Similarly, while our use of an ensemble approach involving a simple average of an array of different machine-learning-based forecasts delivers superior performance, it is possible that even better results may be obtained through the use of dynamic forecast combinations (see, e.g. Diebold & Shin, 2019). Further gains may also be available through the use of more advanced dynamic trading strategies designed to more effectively control transaction costs compared to the simple S-Sign strategy that we primarily rely on here. We leave further work on all of these issues for future research.

Appendix

A Jump Detection

Our jump identification scheme relies on the now commonly-used truncation procedure originally proposed Mancini (2001). Intuitively, the procedure classifies a high-frequency return as a jump if the return is deemed “too large” to be compatible with the realization from a normal distribution with the locally estimated variance. More formally, we identify the return $r_{t,i}$ on day t and intraday time-interval i as a jump if:

$$|r_{t,i}| > \alpha \hat{\sigma}_{t,i} \Delta_n^{\varpi}, \quad (\text{A.13})$$

where α and ϖ are tuning parameters, $\hat{\sigma}_{t,i}$ denotes an estimate of the spot volatility at the time, and n refers to the number of high-frequency observations per day. Following the literature, we set α and ϖ to 3.5 and 0.49, respectively (Bollerslev & Todorov, 2011). Unlike the existing literature, however, which generally estimates the spot volatility using the ex-post volatility for the full day t , we instead rely on rolling ex-ante volatility forecasts to ensure that our jump detection scheme is implementable in real-time, while still incorporating the most recent information.

To do so, we employ a simple exponentially weighted moving average (EWMA) of the rolling product of adjacent absolute returns:

$$\hat{\sigma}_{t,i}^2 \equiv \frac{\pi}{2} \cdot n \cdot \text{EWMA}(|r_{t,i-1} \cdot r_{t,i-2}|, \text{half-life} = n) \cdot \tau_{t,i}, \quad (\text{A.14})$$

$$\tau_{t,i} = \left(\sum_{s=1}^k r_{t-s,i}^2 \right) / \left(\frac{1}{n} \sum_i \sum_{s=1}^k r_{t-s,i}^2 \right), \quad (\text{A.15})$$

where $\tau_{t,i}$ is a time-of-day adjustment factor. This construction is directly motivated by the bipower variation measure traditionally used in the literature, $BV_t \equiv \frac{\pi}{2} \cdot \frac{n}{n-1} \cdot \sum_i |r_{t,i} \cdot r_{t,i-1}|$ (Barndorff-Nielsen & Shephard, 2004). However, unlike the bipower variation, which uses the ex-post returns over the full day t , our EWMA-based estimator avoids look-ahead-bias by only using intraday realized returns preceding time (t, i) . Additional details, including comparisons with other procedures, are provided in the Online Appendix.

B Shapley Values Computation

The following algorithm provides a step-by-step guide to our computation of SHAP values.

1. Let π_p represent a permutation of all the regressors possibly included in the model.

We will write $\pi_p(j)$ to refer to the j 'th element of the permutation of $\{1, \dots, J\}$ defined by π_p . Correspondingly, the permuted set of regressors may be written as $\{X_{t,i}^{\pi_p(1)}, \dots, X_{t,i}^{\pi_p(J)}\}_{t,i}$, where $X_{t,i}^j$ refers to the value of the j 'th regressor on day t and intraday time-interval i .

2. Using this sequence of regressors, the predictions $\{\hat{r}_{t,i}^{\pi_p,0}, \hat{r}_{t,i}^{\pi_p,1}, \dots, \hat{r}_{t,i}^{\pi_p,J}\}_{t,i}$ based on the following expression:

$$\hat{r}_{t,i}^{\pi_p,j} = \sum_{j'=1}^j \hat{\beta}_{\pi_p(j')} X_{t,i}^{\pi_p(j')} + \sum_{j''=j+1}^J \hat{\beta}_{\pi_p(j'')} \bar{X}^{\pi_p(j'')},$$

where $\bar{X}^j$ refers to the sample average of the j 'th regressor, and $\hat{\beta}_j$ refers to the estimated coefficient for that same regressor.

3. Produce the first set of return predictions $\{\hat{r}_{t,i}^{\pi_p,0}\}_{t,i}$ by replacing all of the regressors with their sample averages. Then, incrementally replace each regressor's mean

with its actual values to produce the subsequent set of return predictions, where the order by which the regressors are replaced is determined by the permutation π_p . The resulting sequence of sets of predictions, $\{\hat{r}_{t,i}^{\pi_p,j}\}$, incrementally incorporate the information from each regressor.

4. Using each set of predictions $\{\hat{r}_{t,i}^{\pi_p,j}\}_{t,i}$ for $j \in \{1, \dots, J\}$, compute the returns from the strategy and the performance measure of interest, say $\{\xi_{\pi_p}^0, \dots, \xi_{\pi_p}^J\}$.
5. Compute the marginal contribution of regressor $\pi_p(j)$ as the marginal effect of adding that regressor to the model that only includes regressors $\{\pi_p(1), \dots, \pi_p(j-1)\}$ and replaces the remaining regressors with their sample averages as $\xi_{\pi_p}^j - \xi_{\pi_p}^{j-1}$.
6. Repeat all of the steps above for each permutation of the J regressors. The SHAP value for the j 'th regressor is then defined as $\frac{1}{J!} \sum_p \xi_{\pi_p}^{\pi_p^{-1}(j)} - \xi_{\pi_p}^{\pi_p^{-1}(j)-1}$, where π_p^{-1} inverts the permutation π_p .

The additive nature of SHAP values, whereby the marginal contributions for each of the regressors $\pi_p(j)$ defined by $\xi_{\pi_p}^j - \xi_{\pi_p}^{j-1}$ adds up to the performance of the full model, $\xi_{\pi_p}^J$, makes them particularly attractive and easy-to-interpret.²⁵ In particular, as discussed in the main text, the additive nature of SHAP values allows us to aggregate our different performance measures by individual factors and factor clusters.

At the same time, the computation of SHAP values for large-dimensional models, including some of the ones used here, can be quite challenging from a computational perspective. Hence, rather than relying on the full set of $J!$ possible permutations, we instead randomly sample a smaller set of permutations and then take the sample average

²⁵ For brevity, we do not report the SHAP value associated with the null model, $\xi_{\pi_p}^0$, in the various figures and tables of our paper. Note that when studying intraday trading performance, this value ends up approximately corresponding to the performance of simply holding cash, since the ML model forecasts when the regressors are zero are similarly close to zero. Thus, the SHAP value for the null model will be close to zero anyways, whether we are studying Sharpe ratios or alphas.

of the resulting permutation-specific estimates. This approach, dubbed “Monte Carlo Permutation Sampling” by Rozemberczki et al. (2022), was originally proposed by Castro et al. (2017), and it is now fairly standard in the machine learning literature. For our main results, we use a total of 400 such samples.²⁶

We also further employ antithetical sampling to help reduce the standard error of the estimates. This is a common technique in Monte Carlo simulations, where samples are deliberately chosen to induce a negative correlation between draws. Following Lomeli et al. (2019) the specific Monte Carlo sampling scheme that we rely on, consists of drawing a random permutation and then reversing that permutation for the subsequent trial. In other words, the “even” Monte Carlo trial iterations consist of random permutations, while the “odd” trials simply use the reversed permutation underlying the preceding even trial, thereby substantially reducing the standard errors of the resulting SHAP estimates (see also the discussion in Mitchell et al., 2022).

²⁶In terms of computational time, each Monte Carlo trial takes about 17 hours, so this still amounts to 6,800 hours of processing.

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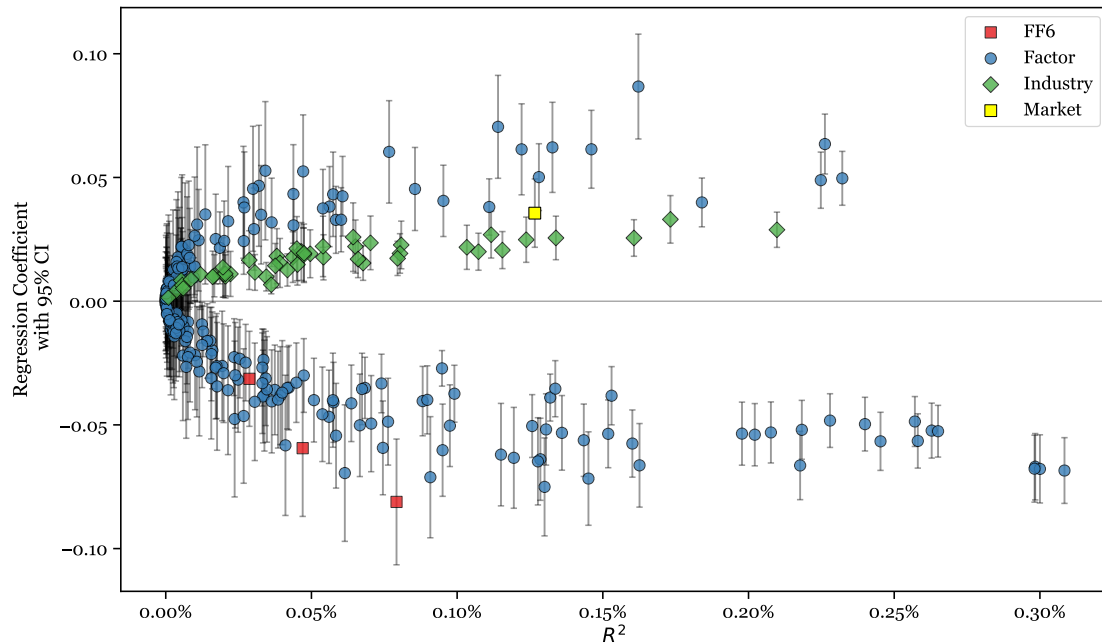
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Online Supplemental Appendix to:
“Intraday Market Return Predictability Culled from
the Factor Zoo”

Saketh Aleti, Tim Bollerslev, & Mathias Siggaard

Figure OA.1: UNIVARIATE PREDICTIVE MARKET RETURN REGRESSIONS



Note: The figure plots the estimated regression coefficients against the corresponding R^2 s from univariate regressions of the 15-minute return on the Fama-French market portfolio on the lagged 15-minute returns on each of our 272 portfolios. The error bars show the corresponding 95% confidence intervals, computed using heteroskedasticity robust standard errors.

OA.1 Univariate Regressions

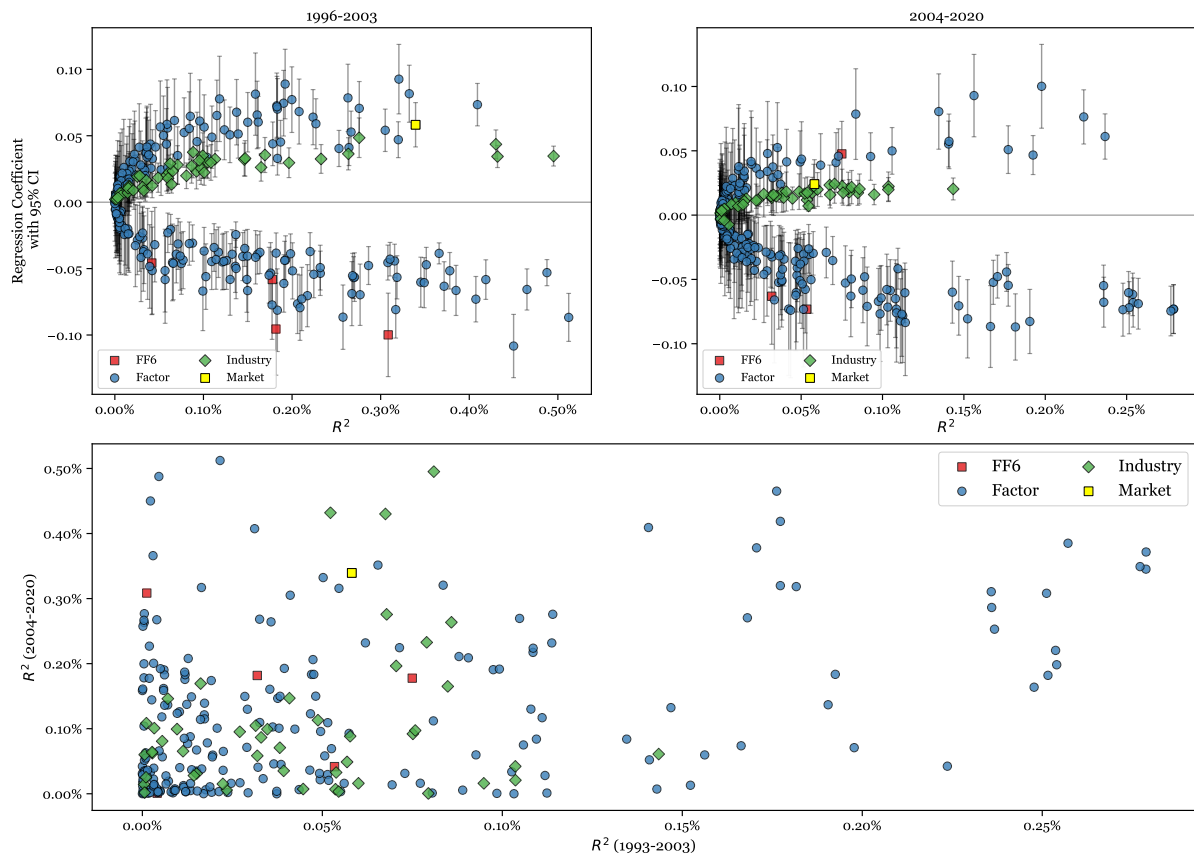
In the main text, we have argued that, despite weak average correlations between the broader zoo and the market portfolio, there may yet be some portfolios in the zoo that are useful in a predictive sense. To explore this, [Figure OA.1](#) plots the estimated regression coefficients, together with their 95% confidence intervals against the corresponding regression R^2 s from the 272 simple univariate regressions of the 15-minute market return on each of the lagged 15-minute portfolio returns. Perhaps surprisingly, some of the individual portfolios do appear to provide significant predictive power, with the R^2 s as high as 0.30%.¹ At the same time, however, the vast majority of the portfolios result in R^2 s less than 0.05%. By comparison, the first-order autocorrelation for the market portfolio translates into an R^2 of 0.12%.

Nevertheless, the regressions depicted in [Figure OA.1](#) are all ex-post, in the sense that they utilize the returns over the full sample. As such, it is unlikely that the specific portfolios with the highest R^2 s could actually have been identified ex-ante. Indeed, as shown in [Figure OA.2](#), repeating the same time-series regressions over the separate in-sample (1996-2003) and out-of-sample (2004-2020) periods underlying our subsequent

¹ Our discussion of the R^2 's in the main text puts the magnitudes of these seemingly very low 15-minute R^2 s into a proper perspective.

empirical analyses does not identify the same portfolios as the ones with the highest predictive R^2 s. In fact, the correlation between the two sub-sample R^2 's across the cross-section of univariate regressions is 37.7%. Put differently, the ex-ante identification of the most useful predictors necessitates the use of more sophisticated real-time variable selection procedures. Moreover, it is also very likely that predictive regressions involving not just one, but multiple portfolios as regressors, may result in more accurate predictions.

Figure OA.2: UNIVARIATE SUBSAMPLE REGRESSIONS



Note: Paralleling the estimates for the full 1996-2020 sample reported in Figure OA.1 in the main text, the figure reports the analogous univariate regression results obtained for the two different subsamples spanning 1996-2003 and 2004-2020, respectively. The lower scatter plot compares the R^2 s obtained for the two different subsamples.

OA.2 Jump Detection

As detailed in our main text, we rely on the usual thresholding approach (Mancini, 2001) to identify jump returns. This approach classifies returns as jumps if their absolute value exceeds the threshold $\alpha\sqrt{\sigma_{t,i}^2}\Delta_n^\varpi$ where $\sigma_{t,i}$ is the continuous component of quadratic variation for day t and interval i , Δ_n is the sampling interval length, and (α, ϖ) are tuning parameters. Intuitively, this approach classifies a return as a jump if it is seemingly too large to be driven by the Brownian motion component of the log price process; in a finite

sample setting, the $\sqrt{\sigma_{t,i}^2} \Delta_n^\varpi$ term can be interpreted as approximately the t-statistic for the return itself; the classification procedure then amounts to rejecting the “null” of a continuous return if the statistic exceeds α . In order to implement this procedure in practice, we must estimate the term $\sigma_{t,i}^2$ which on the price process of interest. Our approach for doing so is detailed in [Section OA.2.1](#). Once we have formed estimates for each asset, we can obtain the jump classifications. We describe the classified jumps in [Section OA.2.2](#).

OA.2.1 Volatility Estimation

We begin by detailing our approach for the $\sigma_{t,i}^2$ term. The literature traditionally estimates this term as $BV_t \cdot \tau_i$ where BV_t is the bipower variation (Barndorff-Nielsen & Shephard, 2004) computed over the trading day t and $\tau_{t,i}$ is a time-of-day correction for interval i of the same day. The advantage of this traditional approach is that the bipower variation is known to converge to precisely $\sigma_t^2 \equiv \int_t^{t+1} \sigma_s^2 ds$ in the infill asymptotic limit while the time-of-day correction effectively adjusts for the well known intraday u-shape pattern in volatility; combining these two terms, we may obtain fairly precise estimates of $\sigma_{t,i}^2$. Although this approach is likely the best and most standard, it also uses data from the end of the trading day therefore creating look-ahead bias in any downstream application of the jump classifications.

Thus, we deliberately take an alternative approach and instead estimate $\sigma_{t,i}^2$ using forecasts thereof to ultimately obtain ex ante jump classifications. But, this approach has its own drawback – using ex-ante forecasts rather than the ex-post realized measures introduces additional noise – forecast error – into the estimates of the threshold thereby affecting the precision of the jump detection. This loss of precision motivates us to consider and compare a few different approaches for forecasting $\sigma_{t,i}^2$.

OA.2.1.1 Approaches

Approach 1 (CHAR): The first approach we use is to directly form forecasts of the future volatility σ_t^2 and then follow up with a time-of-day adjustment to obtain $\sigma_{t,i}^2$. For volatility forecasting, we rely on the so-called CHAR model (Andersen, Bollerslev, & Diebold, 2007) which essentially consists of the usual HAR setup from Corsi (2009) – a constrained AR(21) model consisting of lagged daily, weekly, and monthly averages of past volatility – but instead uses bipower variation as the regressand and regressors.

Formally, we have

$$\text{(Def.)} \quad BV_t \equiv \frac{\pi}{2} \cdot \frac{n}{n-1} \cdot \sum_i |r_{t,i} \cdot r_{t,i-1}|$$

$$\text{(Model)} \quad BV_t \sim \beta_0 + \beta_{(d)} BV_{t-1} + \beta_{(w)} \left(\frac{1}{5} \sum_{s=1}^5 BV_{t-s} \right) + \beta_{(m)} \left(\frac{1}{21} \sum_{s=1}^{21} BV_{t-s} \right)$$

where the forecasted BV_t on the left-hand-side is used as an ex ante proxy for σ_t^2 .

In regards to the implementation, the forecasts from this regression are formed using a rolling train/test split where the train split is used to get ex ante estimates of the parameters and the test split is used to construct the forecasts. We use 1000 trading days for the train split and one trading day for the test split, updating the parameters each day following Bollerslev, Patton, and Quaedvlieg (2016). Overall, the implementation is similar to the train/valid/test split used for the return forecasts except that validation is not necessary here due to the fact that the CHAR model does not have any hyperparameters.

Next, a time-of-day adjustment τ_i is constructed by computing

$$\tau_{t,i} = \left(\sum_{s=1}^k r_{t-s,i}^2 \right) / \left(\frac{1}{n} \sum_i \sum_{s=1}^k r_{t-s,i}^2 \right)$$

where k is some window length and $r_{t,i}$ is the return on the asset on day t at time i . The time-of-day adjustment is thus a simple ex ante measure of the fraction of daily realized variation that arrives at each time-of-day, and is computed on a rolling basis ending on date $t-1$. The estimate for $\sigma_{t,i}^2$ is ultimately given by $\hat{\sigma}_{t,i}^2 \equiv \widehat{BV}_t \tau_{t,i}$. In sum, this approach – using a standard volatility forecasting model plus a time-of-day adjustment – is maybe the most obvious first choice, but still lacks in one critical way: it misses developing information from within the intraday as the forecasts themselves are updated on daily basis, using data from the previous trading day, rather than on an intraday basis.

Approach 2 (Rolling): In contrast, our second approach does incorporate intraday information and is actually simpler. This approach simply consists of estimating the usual bipower variation measure but on an intraday rolling basis. That is, $\sigma_{t,i}^2$ is estimated using

$$\begin{aligned} \hat{\sigma}_{t,i,A2}^2 &\equiv \frac{\pi}{2} \cdot \text{RollingSum}(|r_{t,i-1} \cdot r_{t,i-2}|, n) \cdot \tau_{t,i} \\ &= \frac{\pi}{2} \cdot \left(\sum_{s=1}^n |r_{t,i-s} \cdot r_{t,i-s-1}| \right) \cdot \tau_{t,i}, \end{aligned}$$

where $n = \lfloor 1/\Delta_n \rfloor$ is the number of observations per day, the $\pi/2$ term is an adjustment for the fact that “ $\mathbb{E}[|\mathcal{N}(0, \sigma)|]$ ” = $\sigma\sqrt{2/\pi}$, and $\tau_{t,i}$ is the same time-of-day correction from before. Because this matches the usual bipower variation formula, the volatility estimate for the first interval of each day is simply the bipower variation of the previous

day: $\hat{\sigma}_{t,0,A2}^2 = BV_{t-1}\tau_{t,i}$. Moreover, each proceeding estimate rolls the returns forward, incorporating the latest intraday information unlike Approach 1.

Approach 3 (EWMA): Our third approach is quite similar to our second, but simply replaces the rolling average with an EWMA:

$$\hat{\sigma}_{t,i,A3}^2 \equiv \frac{\pi}{2} \cdot n \cdot \text{EWMA}(|r_{t,i-1} \cdot r_{t,i-2}|, \text{half-life} = n) \cdot \tau_{t,i}.$$

Since we are again using the product of absolute returns, the $\pi/2$ adjustment factor is needed. Additionally, to convert the EWMA term into a sum that estimates the volatility for the full day, we multiply by n . Thus, this setup is analogous to that used in Approach 2 but with the benefit of putting a heavier weight on more recent information, potentially improving the precision of the jump detection when the true volatility is strongly time-varying. The choice of using a half-life of one trading day worth of observations, n , is somewhat arbitrary but we find that it works well in practice compared to using a half-life of 1 (one observation) or $5 \cdot n$ (one trading week worth of observations).

Benchmark: Our final approach is an ex post benchmark. This approach is exactly that which was discussed earlier – using the ex post computed bipower variation multiplied by a time-of-day correction to estimate the volatility term $\sigma_{t,i}^2$. More formally, this is

$$\hat{\sigma}_{t,i,\text{benchmark}}^2 \equiv BV_t \cdot \tau_{t,i}.$$

where BV_t and $\tau_{t,i}$ are the same as defined earlier. While this approach is itself returns an estimate of the volatility rather than the truth, it is also the standard approach in the literature and thus represents a feasible albeit ex post benchmark.

OA.2.1.2 Results

In order to determine the accuracy of our volatility estimation approaches, we rely on a number of different metrics with each comparing against the benchmark specification.

We begin by computing panel R^2 's for the variance estimates $\hat{\sigma}_{t,i}^2$, as well as the corresponding threshold estimates $\alpha\sqrt{\hat{\sigma}_{t,i}^2}\Delta_n^\varpi$ with $(\alpha, \varpi) = (3.5, 0.49)$ as in the main text. The “truth” is defined through out ex post Benchmark defined above. In terms of the data, we focus on our set of 272 portfolios over the sample period 2004-2020, the same as our trading implementation window. We report the panel R^2 measures in [Table OA.1](#). Our results show that the first approach of using daily-updated volatility forecasts quite noticeably underperforms the latter two approaches which update on an intraday basis. The performance differential is larger for the variance than the threshold. Additionally, the performance of the latter two approaches is fairly similar.

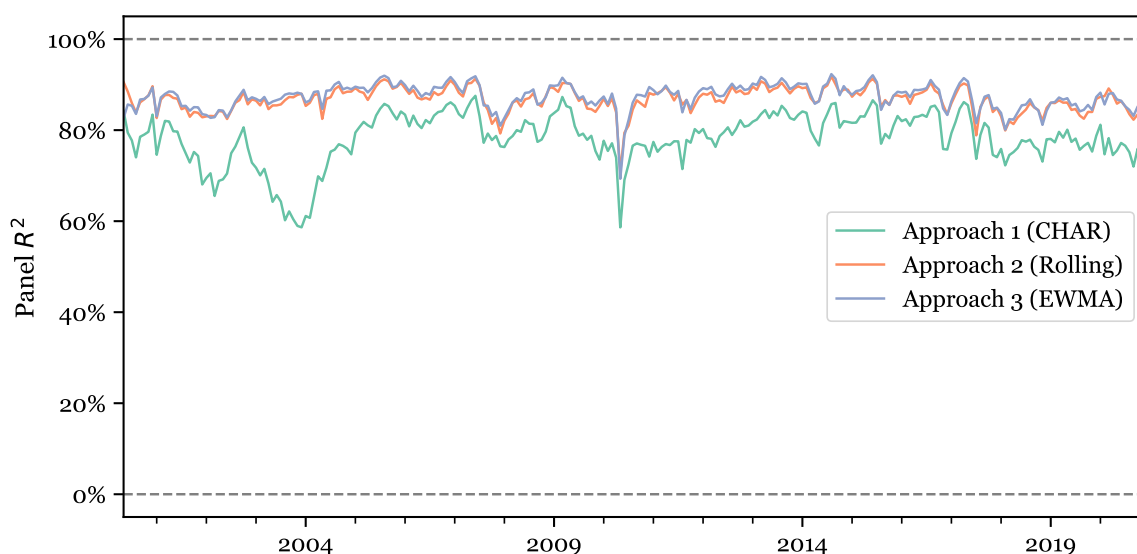
To study whether the performance of the approaches is consistent across time, we

Table OA.1: PERFORMANCE OF VOLATILITY FORECASTS – PANEL R^2 'S FOR VARIANCE AND THRESHOLD

Model	Target	
	Variance	Threshold
Benchmark	100.00%	100.00%
Approach 1 (CHAR)	51.79%	78.42%
Approach 2 (Rolling)	75.96%	86.48%
Approach 3 (EWMA)	74.89%	86.92%

Note: The table reports the panel R^2 's of our volatility forecasts computed on our set of 272 portfolios for the period 2004-2020. The variance column reports this metric as computed on the target $\sigma_{t,i}^2$ for each asset; the truth is defined using the ex post estimate labelled Benchmark. The threshold column is similar but the R^2 is instead on the target $\alpha\sqrt{\hat{\sigma}_{t,i}^2}\Delta_n^\varpi$ where $(\alpha, \varpi) = (3.5, 0.49)$.

Figure OA.3: PERFORMANCE OF VOLATILITY FORECASTS – TIME-VARIATION IN PANEL R^2 FOR THRESHOLD



Note: The figure reports the panel R^2 's of our volatility forecasts computed on our set of 272 portfolios for the period 2004-2020. The panel R^2 is computed on a transformation of the raw variance forecasts: the threshold $\alpha\sqrt{\hat{\sigma}_{t,i}^2}\Delta_n^\varpi$ where $(\alpha, \varpi) = (3.5, 0.49)$. This metric is recomputed on a monthly basis and passed through an EWMA with a half-life of one-month.

further plot the same panel R^2 's for the threshold, computed on a monthly basis and smoothed with a EWMA with a one-month half-life, in Figure OA.3. Again, we find that the first approach quite underperforms the latter two approaches but further note that this underperformance is consistent across the sample. Interestingly, all three approaches see a dip in 2010, which may be related to the heightened volatility around that period. In any case, this figure does not reveal a clear distinction in the performance of the *Rolling*

Table OA.2: PERFORMANCE OF VOLATILITY FORECASTS – JUMP CLASSIFICATION METRICS

Model	Classification Metrics				
	Type I Error	Type II Error	Precision	Recall	Accuracy
Benchmark	0.00%	0.00%	100.00%	100.00%	100.00%
Approach 1 (CHAR)	0.39%	61.62%	25.07%	38.38%	99.40%
Approach 2 (Rolling)	0.57%	26.10%	30.53%	73.90%	99.34%
Approach 3 (EWMA)	0.39%	40.35%	34.47%	59.65%	99.48%

Note: The table reports the various classification metrics for our jump detections. The classifications of the benchmark approach are treated as the ground truth. The underlying panel data consists of detected jumps for our 272 portfolios from 2004-2020. Let TP, FP, FN, and TP represent the number of true positives, false positives, false negatives, and true positives. The Type I Error is defined as $(FP / (FP+TN))$, the Type II error is $(FN / (FN+TP))$, Precision is $(TP / (TP+FP))$, Recall is $(TP / (TP+FN))$, and Accuracy is $(TP + TN) / (TP + TN + FP + FN)$.

and *EWMA* approaches, so we study additional metrics to better identify the superior approach.

To this end, we now study the accuracy of the jump detection afforded by each of the three approaches, thus focusing on a more relevant metric for the downstream application. We will again treat the Benchmark approach as the truth, motivated by the fact that this approach does indeed consistently identify the jump returns in the infill asymptotic limit. We begin by computing some simple statistics on the counts of the identified jumps. More precisely, we compute the average number of jumps per year for each of our 272 portfolios and then compute the cross-sectional average of this number. We find that, on average across assets, the ex-post Benchmark approach finds 18.26 jumps per year, while Approach 1 (CHAR) finds 25.2, Approach 2 (Rolling) finds 42.52, and Approach 3 (EWMA) finds 27.12. In other words, Approach 1 and 3 slightly over-detect jumps while Approach 2 greatly over-detects jumps.

Next, we study the time-variation in the jump counts to determine whether they reasonably match the benchmark. To do so, we first compute the cross-sectional averages of the monthly jump counts under each of our approaches and then compute the correlation matrix of these time series. We find that Approach 1 has a correlation of 38.0% with respect to the Benchmark, while the equivalent statistics for Approach 2 and 3 are 96.3% and 92.5%. In other words, while Approach 1 produces a reasonable level of jump counts, it struggles to produce time-variation in the jump counts themselves that is similar to that of the Benchmark. On the other hand, both Approach 2 and 3 quite closely match the benchmark. In sum, the results thus far provide additional evidence against using Approach 1 and also suggest that Approach 2 is somewhat problematic since it vastly over-detects jumps.

Lastly, we compute directly the accuracy of the jump classifications as measured by

standard classification metrics employed in the machine learning literature. We report our results in Table OA.2. Consistent with the results arguing over-detection above, Approach 2 shows relatively higher Type I error than the other two metrics but relatively less Type II error and higher Recall. Despite the improvements in the latter two metrics, Approach 2 obtains the smallest overall accuracy simply because there are only a relatively small number of jumps overall in the data – in other words, its over-detection of jumps is problematic. In contrast, Approach 3 offers relatively more balanced performance, being an intermediate of Approach 1 and 3 in most metrics but ultimately obtaining the highest overall Accuracy.

To conclude, we find that Approach 1 and Approach 2 both have sharp drawbacks that led us to opt for Approach 3. Approach 1 seems to be fairly imprecise, producing a threshold that markedly deviates from the Benchmark threshold as measured by the Panel R^2 's; the time-variation in the jump detections of Approach 1 also do not line up well with that of the Benchmark. Approach 2 is more precise but still greatly over-detects jumps. In fact, for the Fama-French market portfolio, Approach 2 finds 52 jumps per year compared to the Benchmark measure of 21 and the other approaches which find around 35. Therefore, we settle on using Approach 3 which appears to be comparatively acceptable on all fronts.

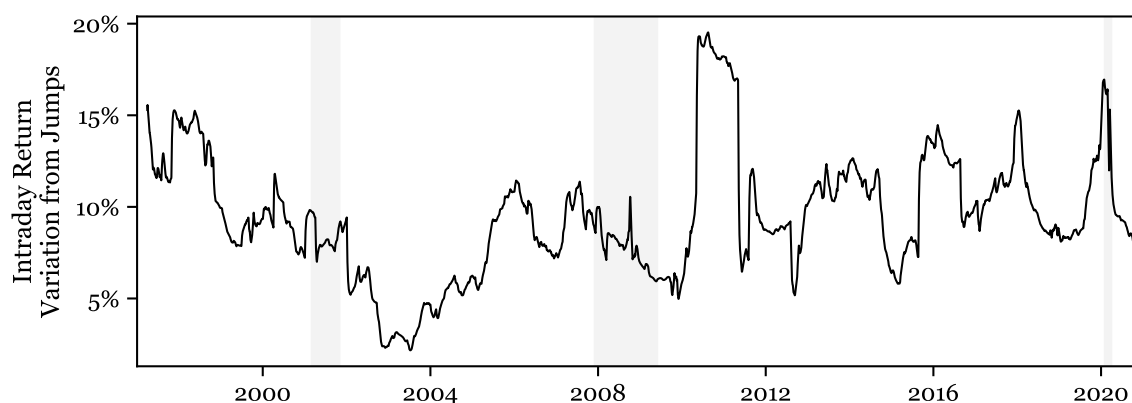
OA.2.2 Classifications

Having decided to use Approach 3 (*EWMA*) for the volatility estimation, we now study the jump classifications in greater detail, performing a case study on the market portfolio as well as the broader examination of the full set of 272 portfolios. To show as much information as possible, we focus on the period to 1997 to 2020; the first year is discarded due to lack of data for the backward-looking estimation of the volatility forecasts and time-of-day correction.

OA.2.2.1 Case Study: The Fama-French Market Portfolio

We begin by computing the jump variance contributions using our ex ante detected jumps for the Fama-French market portfolio. This proportion is computed by dividing the sum of squared jump returns by the total sum of squared returns and, in a theoretical sense, converges to the proportion of quadratic variation coming from the discontinuous component of the log price process. When computed across our sample for the intraday, we find that the contribution of jump returns to the overall variance is 8.44%, in line with past work by Huang and Tauchen (2005). This finding suggests that a fairly non-trivial proportion of the variation in the market return is coming from these unpredictable returns. Correspondingly, dropping said returns from forecasting regressions that minimize mean squared error may be helpful towards improving their out-of-sample performance

Figure OA.4: YEARLY JUMP VARIANCE CONTRIBUTIONS – FAMA-FRENCH MARKET PORTFOLIO



Note: The figure reports the jump variation contributions, defined as the sum of squared jump returns divided by the sum of squared total returns, for the Fama-French market portfolio as computed on 15-minute sampled returns. The contributions are computed on a rolling basis with a window of 365 calendar days and updated each day. The series is further smoothed with an EWMA with a half-life of seven calendar days. The gray areas highlight NBER-defined recessions.

and especially useful guiding the forecasts towards economically relevant/exploitable variation.

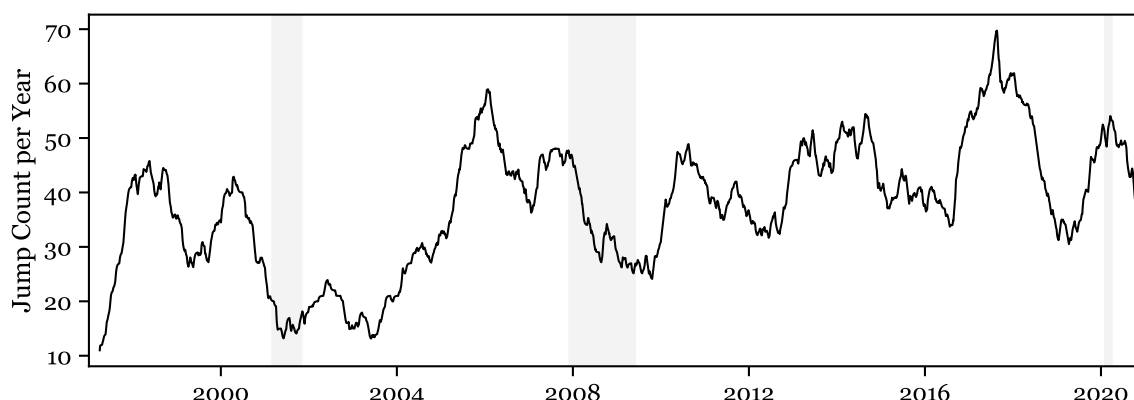
Additionally, to further illustrate the time-variation in the jump contributions, we recompute the variations on a daily basis using a yearly window, reporting our results in Figure OA.4. We find that this statistic shows substantial time-variation, peaking around 2011 in the wake of the European Sovereign Debt Crisis (coincident with a global stock market crash in August 2011) and peaking again in 2020 during the COVID pandemic. Figure OA.5, which shows the average yearly jump counts over time, shows similarly pronounced time-variation, generally matching the pattern in the variance contributions. It is also notable that the jump variation and variation do not noticeably peak in the first two recessions; this may simply be driven by the fact that average volatility was higher during those periods, making it more difficult to separate the continuous and jump components of the return.²

OA.2.2.2 All Portfolios

We now turn to statistics for the full set of 272 portfolios. We summarize our results broadly in Figure OA.6 and Figure OA.7 which, respectively, report the jump variance contributions and average jump counts for each of the portfolios. As with the market, the variance contributions are non-trivial, averaging 7.78% with a standard deviation of 1.76% across our set of portfolios. The counts are similarly non-trivial with an average of 27.2

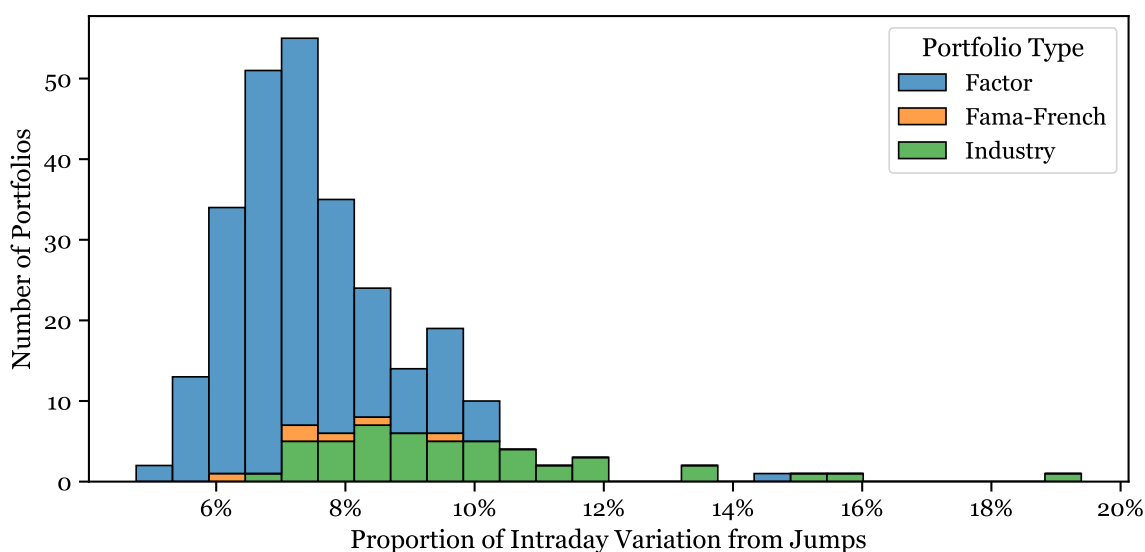
² The same pattern is present if jumps were detected using the usual Benchmark approach, so this is not an artifact of our ex ante focused jump detection scheme.

Figure OA.5: AVERAGE NUMBER OF JUMPS PER YEAR – FAMA-FRENCH MARKET PORTFOLIO



Note: The figure reports the jump counts, defined by simply counting the number of jumps, for the Fama-French market portfolio. The counts are computed on a rolling basis with a window of 365 calendar days and updated each day. The series is further smoothed with an EWMA with a half-life of seven calendar days. The values can be interpreted as the conditional mean of the number of jumps in the market portfolio. The gray areas highlight NBER-defined recessions.

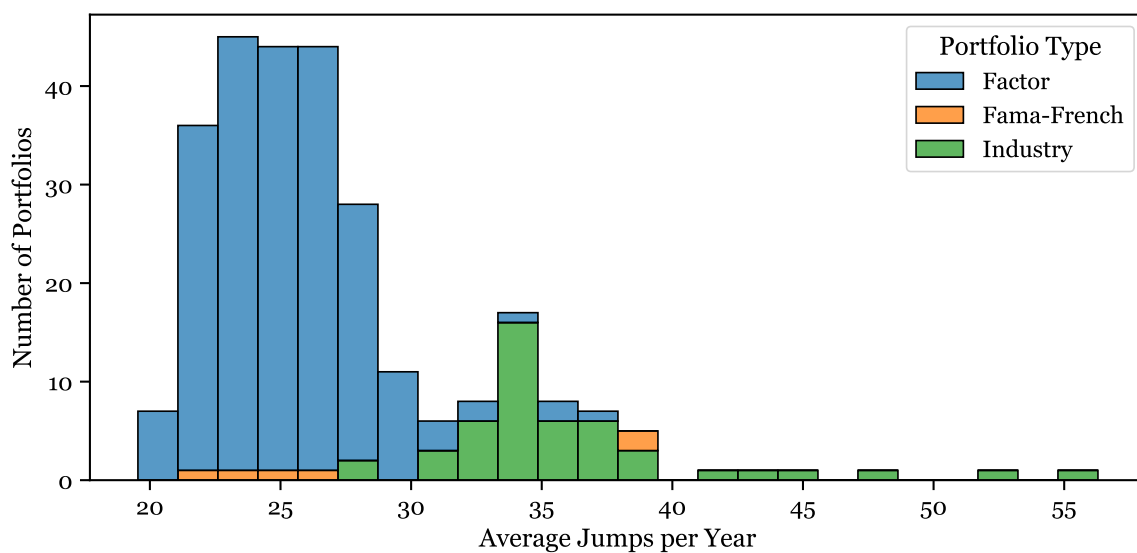
Figure OA.6: JUMP VARIANCE CONTRIBUTIONS – ALL PORTFOLIOS



Note: The figure reports the jump variation contributions, defined as the sum of squared jump returns divided by the sum of squared total returns, for all 272 portfolios as computed on 15-minute sampled returns. The bars are colored by the portfolio type with Factor referring to the 218 JKP and CZ portfolios, Fama-French referring to the usual six factors, and Industry referring to the 48 industry portfolios.

jumps per year (stdev of 5.46) across the portfolios. In general, the industry portfolios tend to have slightly more jump variation and counts than the factor portfolios, since the latter are much less exposed to market jump risk. That is, non-market jumps tend to be lower in frequency and size than market jumps. In any case, all the portfolios exhibit jumps and non-trivial jump variation.

Figure OA.7: AVERAGE NUMBER OF JUMPS – ALL PORTFOLIOS



Note: The figure reports the average number of jumps per year for all 272 portfolios. The bars are colored by the portfolio type with Factor referring to the 218 JKP and CZ portfolios, Fama-French referring to the usual six factors, and Industry referring to the 48 industry portfolios.

OA.3 Machine Learning Algorithms

In this section, we provide additional details regarding the implementation of the machine learning algorithms defined in the main text. We then report the validated hyperparameters obtained under *Cts*, *All* specification for each of these models.

OA.3.1 Descriptions

We begin by detailing and motivating each of the ML algorithms we consider.

Linear: This algorithm is simply conventional OLS. In a sense, it serves as a benchmark for what can be achieved without relying on anything more than the classical econometric toolkit. However, given the high-dimensional nature of the regressors and the low signal-to-noise ratio of financial market data, it is unlikely that the out-of-sample forecasts obtained from OLS will perform well. For this reason, we explore penalized linear specifications as well.

Ridge: The ridge regression is a penalized linear regression via the L2 norm (Hoerl & Kennard, 1970). In general, shrinkage on linear models allows for a more reasonable bias-variance trade-off especially in high-dimensional settings such as ours. This regression also has a Bayesian interpretation as a standard linear model with a zero-mean Gaussian prior on the coefficients; accordingly, ridge creates attenuation in the estimated coefficients, implicitly betting that the predictive content in the dataset is dispersed relatively evenly across the regressors.

Lasso: The lasso is another penalized linear regression but via the L1 norm (Tibshirani, 1996). Like the ridge regression, the Lasso is particularly helpful in high-dimensional settings and has a Bayesian interpretation as a linear model with a prior on the coefficients. However, unlike the ridge, the prior is a Laplace distribution, thereby promoting solutions with a concentration of coefficients at precisely zero which is particularly helpful when the predictive content of the dataset is sparse, i.e., concentrated in a small number of the regressors. Given our data, lasso allows us to explore the idea that only a small subset of the portfolios in the factor zoo are relevant for forecasting, analogous to how the cross-sectional asset pricing literature has assumed sparsity.

ENet: The elastic-net regression combines the L1 and L2 penalties of lasso and ridge, allowing for both variable selection and regularization (Zou & Hastie, 2005). The use of an L2 penalty is particularly helpful when variables are strongly correlated, while the use of an L1 penalty is what allows for variable selection. Since the ratio of the L1 and L2 penalties may lie anywhere in $[0, 1]$, elastic net nests both ridge and lasso.

AdaLasso: Adaptive lasso is similar to lasso but employs a stochastic L1 penalty (Zou, 2006). The motivation behind this penalty is based on a key issue with the conventional lasso regression – in an asymptotic setting, the rate at which the L1 penalty shrinks

can guarantee either consistent variable selection or asymptotically normally-distributed estimates but not both. Adaptive lasso solves this by taking a two-step approach. In the first step, an initial $\sqrt{n}$ -consistent regression (e.g., lasso) is run to estimate the coefficients on each regressor; in the second step, another lasso regression is run with L1 penalties that are inversely proportional to the first-step coefficient estimates. These adaptive penalties allow for consistent variable selection as well as asymptotically normally-distributed estimates for the selected coefficients. We refer to Horowitz (2015) for additional information and Freyberger, Neuhierl, and Weber (2020) for an application in finance.

PCR: The principal component regression is a two-step estimator which consists of first extracting a number of principal components from the regressors and performing the remainder of the regression using said components (Jolliffe, 1982; Massy, 1965). The benefit of this approach is that it substantially shrinks the dimensionality of the downstream linear regression while maintaining most of the information embedded in the original regressors. A key drawback, however, is that the principal components capture the statistical factors that explain the majority of the variation in the original regressors and not what underlying factors may provide the best explanatory power for the regressand.

PLS: The partial least squares regression solves the issue of PCR by finding lower-dimensional, latent factors in the regressors that maximally covary with the regressand (Abdi, 2003; Geladi & Kowalski, 1986; Helland, 1990). In doing so, the discovered statistical factors are thus relevant for prediction, rather than being optimal for a pure dimensionality reduction of the regressors as in PCR. In any case, like PCR, PLS is also a linear model.

FNN: We also consider feedforward neural networks for prediction.³ An FNN consists of an input layer, a sequence of hidden layers, and an output layer. The input layer simply passes the regressors into the hidden layers. Each hidden layer consists of “neurons” that perform a linear transformation of their inputs, pass this linear transform through a non-linear “activation” function, and finally pass the output to the next hidden layer. This process repeats until the final hidden layer passes to the output layer, which performs one last linear transformation plus non-linear activation to arrive at the prediction. The use of long sequences of hidden layers allow for repeated linear and non-linear transformations giving rise to greater representational flexibility – the application thereof is often referred to as “deep learning” (Goodfellow, Bengio, & Courville, 2016; LeCun, Bengio, & Hinton, 2015). The success of neural networks in practice lies in the fact that they can allow for a high degree of non-linearity in this manner while still being relatively less prone to overfitting than more naive non-linear methods such as a polynomial regression. Our implementation further applies dropout (Srivastava et al., 2014), early

³To ensure that the structure of the regressors for the neural networks is consistent with the other models, we leave an exploration of recurrent neural networks, which implicitly construct features from time-series data, for future work.

stopping (Prechelt, 2002), batch normalization (Ioffe & Szegedy, 2015), and ensembling as further regularization.

RF: Random forests are ensembles of decision trees (Breiman, 2001). Each decision tree forms relatively simple predictions by partitioning the features and associating each resulting group with a fixed estimate; for instance, we may predict a value of a if $((X_1 > 0) \cap (X_2 < 0))$ and a value of b otherwise. The number of features used in each tree determines the depth – greater depth allows for more representational ability (less bias) but also leading to potential overfitting (more variance). Random forests themselves are constructed by sub-sampling the data, or so-called “bagging,” and building shallow decision trees on each subsample; the collection of “trees” forms the “forest” with the average output of the underlying trees serving as the final, ensemble prediction.

GBRT: In addition to bagging, ensembles of decision trees can also be implemented via “boosting,” which consists of sequentially applying decision trees to fit the prediction residuals of the previous tree (Friedman, 2001). Within each of these sequential steps, the forecasts from each following tree are shrunk by a constant factor to prevent overfitting. This process is repeated for a prespecified number of steps; the use of decision trees at each step gives rise to the name Gradient Boosted Regression Trees (GBRTs). Like bagging, the key idea behind boosting is to allow ensembles of “weak learners,” or shallow decision trees in this context, to form a “strong learner” with relatively greater predictive power than its constituents and relatively less overfitting than fitting a singular complex tree.

Ensemble: Our final model is a simple average of the preceding models excluding *Linear*. The *Ensemble* model can thus be interpreted as a bet on the non-trivial machine learning algorithms with a fully uncertain prior over their performance. Relatedly, it is possible that there exist more optimal weights when combining forecasts. However, in practice, the estimation of such weights is quite difficult and tends to produce results that often perform worse than a simple average – this is sometimes referred to as the “forecast combination puzzle” (Claeskens et al., 2016). For this reason, we focus on only a simple average for our ensemble approach.

OA.3.2 Implementation Details

OA.3.2.1 Preprocessing

Each of our models are implemented via the train/valid/test split defined in the main text. These splits break the data into chunks with the train split being used for fitting the parameters, the valid split being used to determine the optimal hyperparameters, and the test split ultimately representing the data used for the out-of-sample forecasts themselves. Before passing the data to the models for training/validation/forecasting, each subset of the data undergoes a normalization. The normalization consists and demeaning the data

and standardizing the columns to ensure the resulting preprocessed regressors are mean zero and variance one. In order to ensure the normalization is ex ante, only the train data is used for estimating the means and variances of the standardization; these fitted standardization parameters are then used to normalize the data in all three subsets. This procedure ensures that there is no look-ahead bias in the normalization.

OA.3.2.2 Model Information

Once the regressors have been preprocessed, we then train each model under a specific set of hyperparameters drawn from our manually specified hyperparameter grid. Below, we list the hyperparameter grids and estimation procedures for each of our models. We also list the Python (3.10.13) packages used for each model.

Linear: The linear model does not require any hyperparameters and is run as a standard “ridgeless” regression where the pseudo-inverse is used to estimate the precision matrix. This inverse is computed using the `pinv` command in `numpy 1.23.5`. Note that, given the density of our high-frequency data, there is never a case in our out-of-sample period from 2004 onwards where the number of regressors exceeds the sample size; thus, co-linear regressors are the main challenge faced by the linear model, not solely a lack of sample size.

Ridge: The ridge regression contains a singular hyperparameter, the L2 penalty. For the log of the L2 penalty, $\log(\lambda_2)$, we use a grid of 100 equally spaced points between -1 and 8. The penalty is also implicitly normalized by the sample size. The regression itself is fit in the usual way, as $(X^T X/n + \lambda_2 I_p)^{-1}(X^T Y/n)$, where n is the number of samples, p is the number of regressors, and λ_2 is the penalty. Note that this corresponds to the objective

$$(n)^{-1} \cdot \|Y - X\beta\|_2^2 + \lambda_2 \|\beta\|_2^2,$$

which is the usual ridge setup with a normalization of n .

Lasso: The lasso regression also contains a singular hyperparameter, the L1 penalty. For the log of the L1 penalty, $\log(\lambda_1)$, we use a grid of 100 equally spaced points between -6 and -1. During validation, the regression is fit using the `sklearn.linear_model.Lasso_path` function with 5000 maximum iterations and a tolerance of $1e - 4$; the scikit-learn version is 1.3.2. After validation, the regression is refit using the `sklearn.linear_model.Lasso` with 10000 maximum iterations. The refit post-validation after choosing the optimal hyperparameters helps improve the numerical precision of the estimates while respecting computational constraints. In both stages, the optimization objective is

$$(2n)^{-1} \cdot \|Y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1$$

and the underlying numerical algorithm is coordinate descent.

ENet: The elastic net involves two hyperparameters, the L1 penalty and the L2 penalty. We use the `sklearn` implementation and therefore instead define the hyperparameters as the L1 ratio γ and the penalty scale α . The objective function in this case is

$$(2n)^{-1} \cdot \|Y - X\beta\|_2^2 + \gamma\alpha\|\beta\|_1 + \alpha(1 - \gamma)(1/2)\|\beta\|_2^2$$

where the L1 and L2 penalties are $\gamma\alpha$ and $0.5\alpha(1 - \gamma)$. The log penalty scale $\log(\alpha)$ is chosen from 50 equally spaced points from -6 to 1 while the L1 ratio γ is chosen from 50 equally spaced points from 0.05 to 0.95, where exactly zero and one are discluded for numerical stability reasons. During validation, the model is fit with `sklearn.linear_model.enet_path` with 5000 maximum iterations and a tolerance of $1e - 4$; after validation, the regression is refit with `sklearn.linear_model.ElasticNet` with 10000 maximum iterations. As with Lasso, the underlying numerical algorithm is coordinate descent.

AdaLasso: Adaptive lasso is a two-step estimator requiring $\sqrt{n}$ -consistent estimates of the linear model parameters in the first stage followed by a weighted lasso fit in the second stage. Accordingly, in the first stage, we obtain pilot estimates for the parameters using a standard linear regression following Zou (2006).⁴ Then, in the second stage, we solve the optimization problem

$$(2n)^{-1} \cdot \|Y - X\beta\|_2^2 + \lambda_1 \sum_{j=1}^p \hat{w}_j |\beta_j|$$

where $\hat{w}_j \equiv (\hat{\beta}_j^{(1)})^{-\gamma}$ with $\hat{\beta}^{(1)}$ being the first-stage estimate and γ being a hyperparameter. For the log penalty $\log(\lambda_1)$, we use 50 equally spaced points from -25 to -5 and, for γ , we use the set $\{0.5, 1, 1.5\}$. In order to implement this weighted lasso estimation in practice, we simply rescale each X_j by $\hat{w}_j^{-1}$ and then re-use the `sklearn`-based implementation of Lasso discussed earlier. Additionally, we do a simple rescaling of the first-stage estimates by their standard deviation for numerical stability; this has no effect on the latter stages since the scale of the final L1 penalty is handled by the λ_1 term. This approach allows us to implicitly induce coefficient-specific L1 penalties without having to redefine the numerical procedure for Lasso from scratch.

PCR: For the principal components regression, we first run PCA on the features to obtain the set of principal components (PCs). These are obtained using the `sklearn.decomposition.PCA` class. We then use these PCs in a simple linear regression to form our forecasts. The main hyperparameter here is the number of PCs; we choose from the

⁴ We have also experimented with using Lasso in the first stage (combined with AIC for penalty selection) and found essentially the same forecast performance as quantified by the Sharpe of the corresponding forecast-based trading strategy. It should also be noted that, in all periods, we have greater samples than features, so the OLS solution is always well-defined.

set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150\}$. After fitting the parameters, we transform the regression betas on the PCs back into betas for the original features. These transformed betas are used for constructing the out-of-sample forecasts; in other words, PCA is only run once on the training data.

PLS: The partial least squares regression is run in the same way as the principal components regression. The only hyperparameter is the number of factors; we choose from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. The factors themselves are estimated using the `sklearn.cross_decomposition.PLSRegression` class.

FNN: The feedforward neural networks are implemented using `tensorflow 2.15.0` and `keras 2.15.0`. There are an extensive number of hyperparameters involved, so we bullet them below.

- *Hidden layers depth:* This determines the number of (dense) hidden layers used in the FNN. We choose from the set $\{1, 2, 3\}$; given that financial market data tends to have a low signal-to-noise ratio, only a small number of hidden layers are likely needed.
- *Hidden layers width:* This determines the width of the layers via the geometric pyramid rule (Masters, 1993). So, for instance, if this hyperparameter is set to 32 and the depth is set to 3, then the width of the layers will be $[32, 16, 8]$. In other words, the width is halved as we iterate through each hidden layer. For this hyperparameter, we pick from the grid $\{8, 16, 32\}$.
- *Intermediate activation:* Each of the hidden layers consists of a linear transformation of the input followed by a non-linear transformation called the activation. We fix the activation for our neural nets to the so-called ReLU function (Nair & Hinton, 2010), which is fairly popular in the finance and deep learning literature (Goodfellow, Bengio, & Courville, 2016; LeCun, Bengio, & Hinton, 2015).
- *Output activation:* The final layer, or output layer, of the FNN is given an activation as well. Here, we simply use a linear output, so the range of the output may be any value in $\mathbb{R}$.
- *Batch Normalization:* Batch normalization (Ioffe & Szegedy, 2015) is a method that normalizes the output of each neuron to zero mean and unit variance. The purpose of the procedure is to reduce so-called “internal covariate shift” which refers to a shift in the parameters of the hidden layers that occurs when the distribution of the inputs change; this phenomenon supposedly reduces convergence and accuracy (Bjorck et al., 2018). The normalization is generally implemented as an auxillary layer that follows each of the standard, dense hidden layers of the FNN. Additionally, the estimation of the means and variance of the neurons used for the

normalization – done using a moving average of the neuron outputs requires its own parameters. Here, we rely on the `keras.layers.BatchNormalization` class with the default arguments.

- *Dropout Fraction:* Dropout (Srivastava et al., 2014) is a technique where neurons are randomly zeroed out in terms of output during the training process. This approach essentially regularizes the FNN as to avoid relying on any particular neuron too much (Wager, Wang, & Liang, 2013). As with batch normalization, the auxiliary dropout layers used to apply this method require a singular tuning parameter: the fraction of neurons dropped from the output. We follow the original paper, we fix the dropout fraction to 50%, avoiding expanding our hyperparameter grid and therefore reducing the computational burden. We apply a dropout layer after each batch normalization layer following the recommendation of Li et al. (2019). The layer itself is implemented using the `keras.layers.Dropout` class.
- *Solver:* Neural networks are generally optimized using gradient descent, but the exact optimizer may vary. We rely on the well-known “Adam” optimizer proposed by Kingma and Ba (2014). We set the learning rate on this optimizer to 0.001 and use the `keras.optimizers.Adam` class defaults for the remaining optimizer arguments.
- *Optimization Iterations:* In order to train the network, we use the usual batch gradient descent. We fix the batch size to 10000 as in Gu, Kelly, and Xiu (2020) and then consider a range of epochs: {128, 256, 512, 1024}. Note that a batch refers to the size of the dataset sampled when running a gradient descent step while an epoch consist of passing the entire dataset through gradient descent once; so, for instance, if the sample size is 30000, running one epoch would take us three batches. The purpose of using less than an arbitrarily large number of epochs is primarily to further regularize the network (Prechelt, 2002).
- *Ensemble Size:* To further improve the performance of our neural network, we rely on a standard ensemble strategy. That is, we fit several FNNs independently with different seeds and then define our ultimate prediction as the average prediction. Using a large number of FNNs in the ensemble would improves the ensemble performance by way of variance reduction, but would also increase the training time substantially. Thus, we fix the ensemble size to five FNNs to balance this trade-off.

RF: Random Forests, which we implement using the `sklearn.ensemble.RandomForestRegressor` class, similarly require multiple hyperparameters detailed below.

- *Number of Trees*: Random forests themselves are ensemble estimators that fit a large number of decision trees to the underlying data and then take an ensemble to form the final prediction. The number of trees within this “forest” is itself a hyperparameter. While further increasing the number of trees produces a greater variance reduction via the averaging effect, there are also diminishing returns and linearly increasing computational costs. For this reason, we simply fix the number of trees to a large number following the suggestion of Probst and Boulesteix (2018).
- *Number of Features*: Each time an underlying decision tree is constructed, a subset of the features available are considered when looking for the best split. The number of features in this subset is a hyperparameter. We decide to validate on this hyperparameter, following Gu, Kelly, and Xiu (2020), and pick this value from the set $\{3, 5, 10, 20, 30, 50\}$.
- *Max Depth*: Each decision tree consists of at least one split/decision. The number of splits/decisions defines the depth of the tree. The maximum depth is thus another hyperparameter to tune. We pick this value from the set $\{1, 2, 3\}$.
- *Remaining hyperparameters*: There are a number of other hyperparameters for the Random forest such as the split criterion, the minimum samples per split, and the minimum samples per leaf; we leave these hyperparameters and others at their default value in our implementation. We refer to Breiman (2001) for more details about these terms.

GBRT: We implement gradient boosted regression trees using the XGBoost algorithm (xgboost 2.0.3), which was originally proposed by (Chen & Guestrin, 2016). This implementation, well known for its exceptional performance in data science competitions, requires a number of hyperparameters. To better understand the hyperparameters, recall that GBRT consists of repeatedly fitting decision trees such that each decision tree is trained on the residuals of the previous tree, a process known as boosting. Each tree, as in the Random Forest, consists of nodes that conditionally split the data into increasingly finer partitions; each of these partitions is associated with a prediction. Because the trees are built up from a “root node,” there is a sequential ordering to the nodes; nodes that precede others are “parents” (closer to the root) and those that follow are called “children.” The final nodes of the tree are called “leaves” and are not split any further. With these terms defined, we detail the hyperparameters below. In general, we validate along the same hyperparameters that Gu, Kelly, and Xiu (2020) consider.

- *Number of Trees*: First is the number of estimators which we pick from the set $\{50, 100, 150, 200, 250, 300, 350\}$. Since GBRTs rely on boosting rather than bagging, increasing the number of estimators decreases the bias at the cost of higher

variance, so the estimator count parameter is related to regularization. It is for this reason, we do not simply fix the number of trees as with the Random Forest.

- *Learning Rate*: Each time a new tree is added to the model, a shrinkage factor is applied to the tree new predictions to prevent overfitting on the present residuals. This factor is also referred to as “eta” or the learning rate for Gradient Boosting. We set the grid for this hyperparameter to $\{0.01, 0.05, 0.1\}$.
- *Min Child Weight*: Each time a tree is partitioned, there are an increasingly smaller number of relevant samples within each child. For instance, a tree might be defined as $\hat{y}_i = 0 + 5 \cdot \mathbb{1}[X_i > 0]$ where the children are defined implicitly by whether X_i is greater or less than zero. If, in the data, there is only a singular sample within one of the children, the tree may simply be overfitting on that sample by devoting an entire child to explain just that observation. The min child weight, in the context of a regression, solves this issue by only allowing the construction of nodes with a minimum number of samples. For this hyperparameter, we use the grid $\{1, 2\}$ where higher values correspond to greater regularization.
- *Max Depth*: This hyperparameter is defined in the same way as in Random Forest. We pick this value from the set $\{4, 5, 6\}$.
- *Remaining hyperparameters*: As with Random Forest, there are a number of other hyperparameters that we do not explore. We refer to the original paper (Chen & Guestrin, 2016) for a more detailed description of these hyperparameters, which we leave at their default values in our Python-based implementation.

After each model is trained and the forecasts are formed using the standardized regressors, we reverse the normalization to obtain the forecasts for the raw market returns. An important side note is that our non-linear models take an extensive amount of computational resources to fit, even with the use of GPUs. For this reason, we refit these models every five years instead of every one year. Otherwise, the forecasting pipeline for these models is the same as the others.

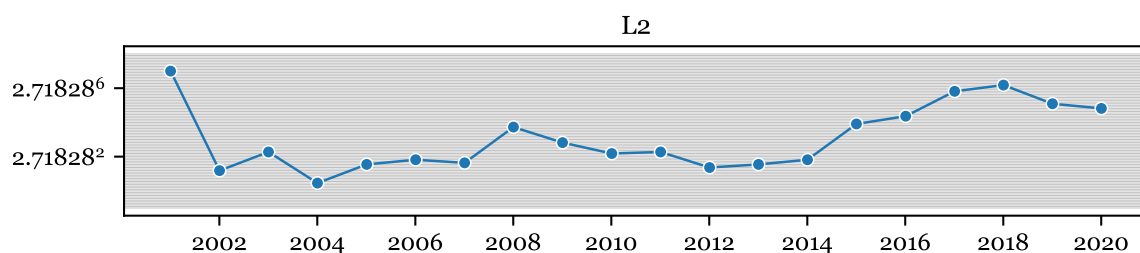
OA.3.3 Validated Hyperparameters

We now report the validated hyperparameters for each of our models. For brevity, we only report the validated hyperparameter choices for the *Cts*, *All* specification. We plot our results in a series of figures below; each figure shows the hyperparameter for the out-of-sample period starting with the marked date with separate subplots for each of the model-specific hyperparameters discussed above in Section OA.3.2.2. Note that our forecasting data begins in 2000, so the plotted validated hyperparameter results start a

few years before our primary out-of-sample analysis period, 2004 to 2020. Additionally, the non-linear models have fewer markers since their out-of-sample window is five years instead of one, as stated above. In any case, for every single model, there are well-defined hyperparameter choices for the full 2004-2020 period.

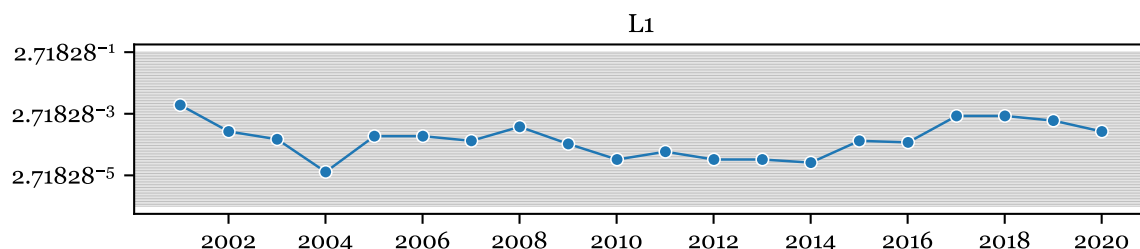
One important takeaway from the figures is how the validated hyperparameter choices relate to the underlying, pre-specified hyperparameter grids, which are marked in each plot with horizontal lines. In general, the chosen grids for most of the models do an effective job of encompassing the validated hyperparameter choices, suggesting that our results are not particularly dependent on the grids. More importantly, this encompassing suggests that most of the models have not had their hyperparameter choices “cut off” by the grid choice. This is always the case for the linear models but not for the non-linear models. For these later models which tend to have a larger number of hyperparameters, we have deliberately limited the grid sizes for computational reasons; this choice is driven by the fact that the number of grid points is increasing multiplicatively and the dimension of the grid is already large, so even moderately increasing the grid for any of the hyperparameters would drastically increase the search space.

Figure OA.8: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(RIDGE)



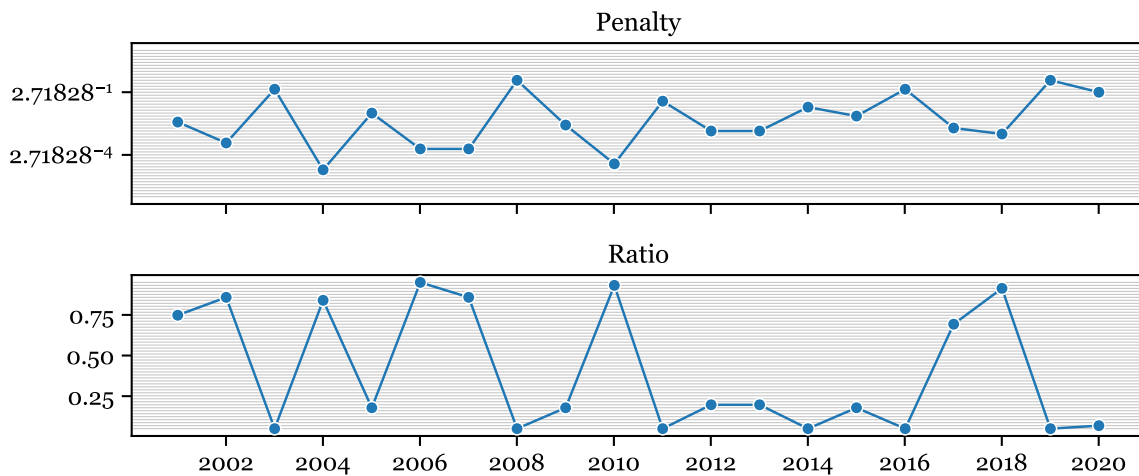
Note: The figure reports validated hyperparameters for the ridge regression. The only hyperparameter is the L2 penalty which is reported on a log scale. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.9: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(LASSO)



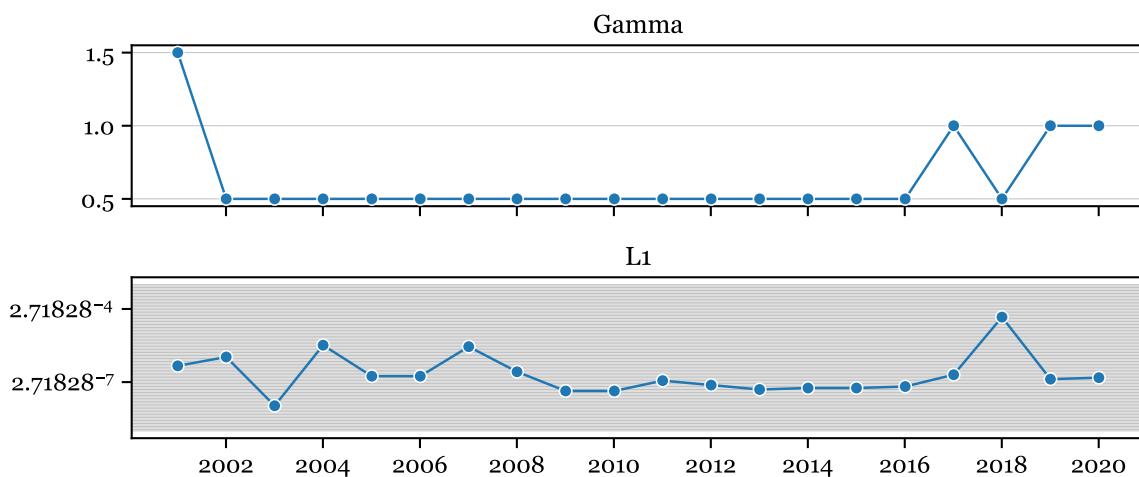
Note: The figure reports validated hyperparameters for the lasso regression. The only hyperparameter is the L1 penalty which is reported on a log scale. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.10: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(ENet)



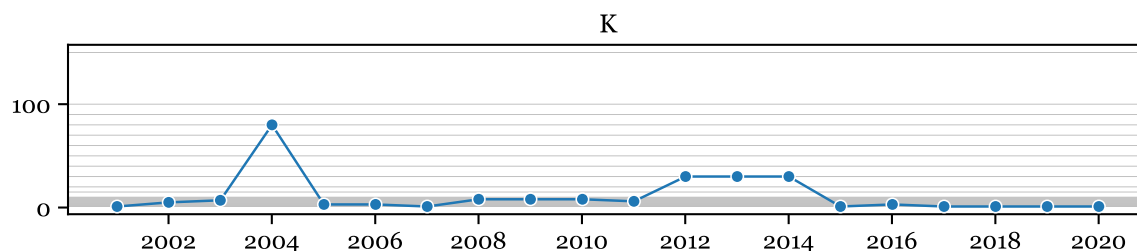
Note: The figure reports validated hyperparameters for the elastic net regression. The two hyperparameters are the L1 penalty ratio and the penalty scale; the latter is reported on a log scale. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.11: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(ADALASSO)



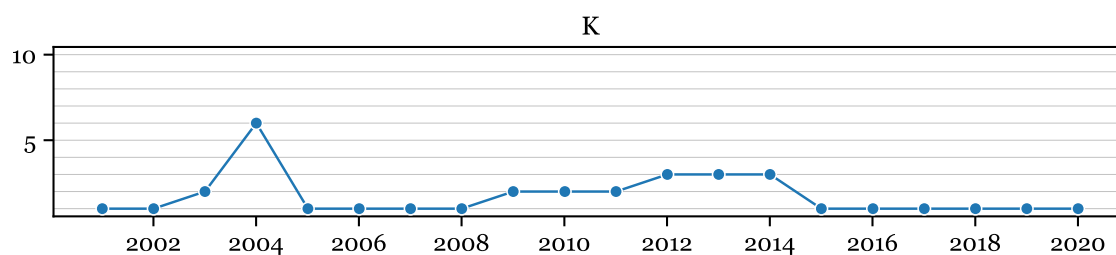
Note: The figure reports validated hyperparameters for the adaptive lasso regression. The two hyperparameters are the gamma parameter, controlling the coefficient on the weights, and the L1 penalty scale; the latter is reported on a log scale. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.12: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(PCR)



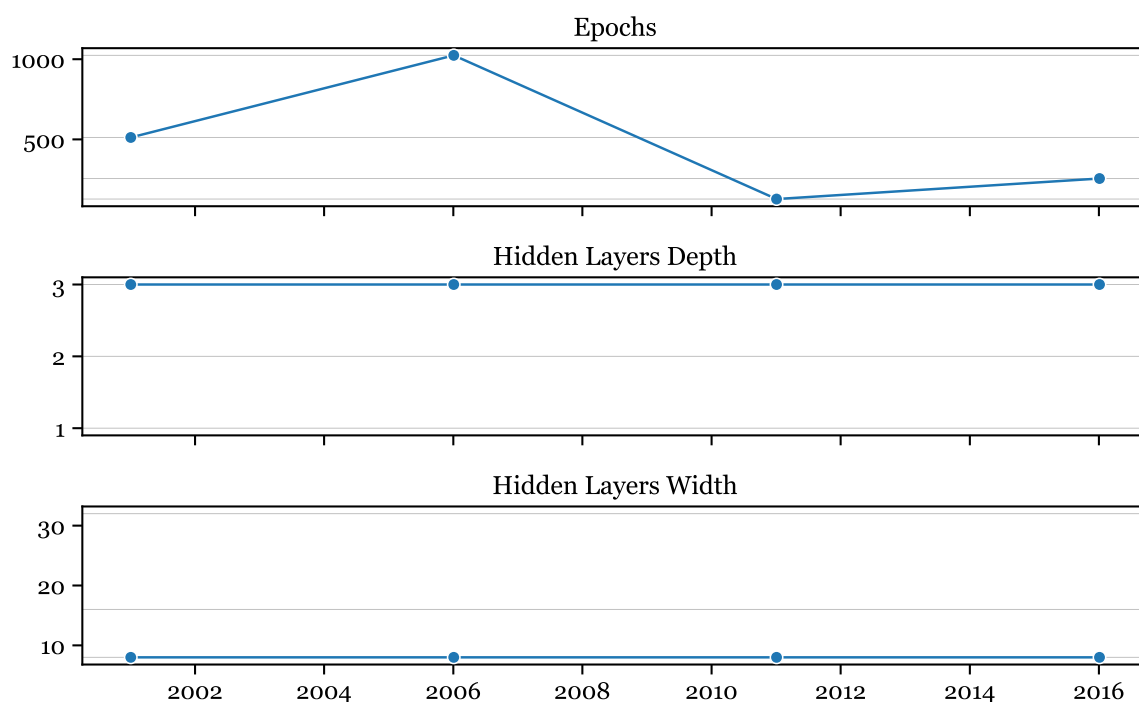
Note: The figure reports validated hyperparameters for the principal components regression. The only hyperparameter is the number of principal components used which is reported a log scale. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.13: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(PLS)



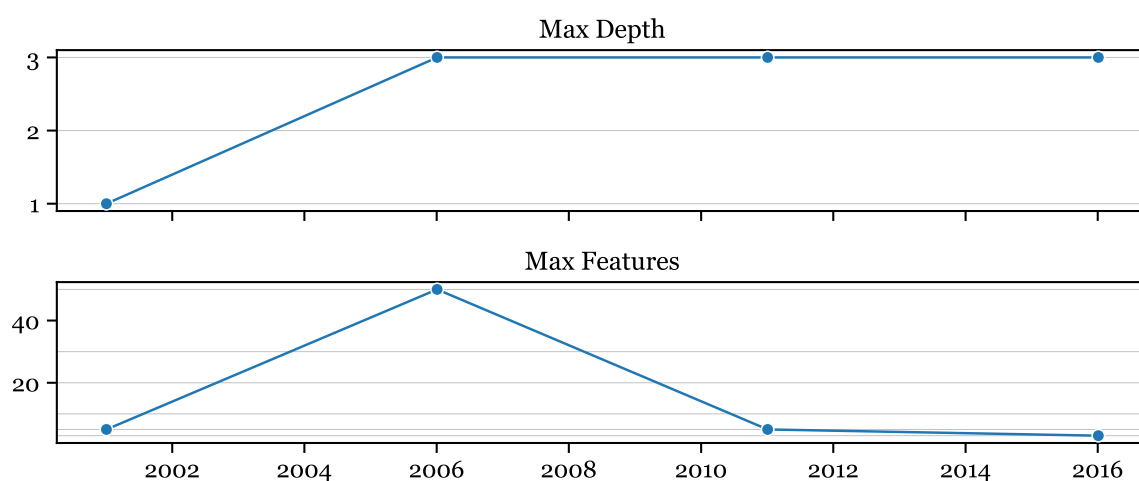
Note: The figure reports validated hyperparameters for the partial least squares regression. The only hyperparameter is the number of factors used which is reported a log scale. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.14: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(FNN)



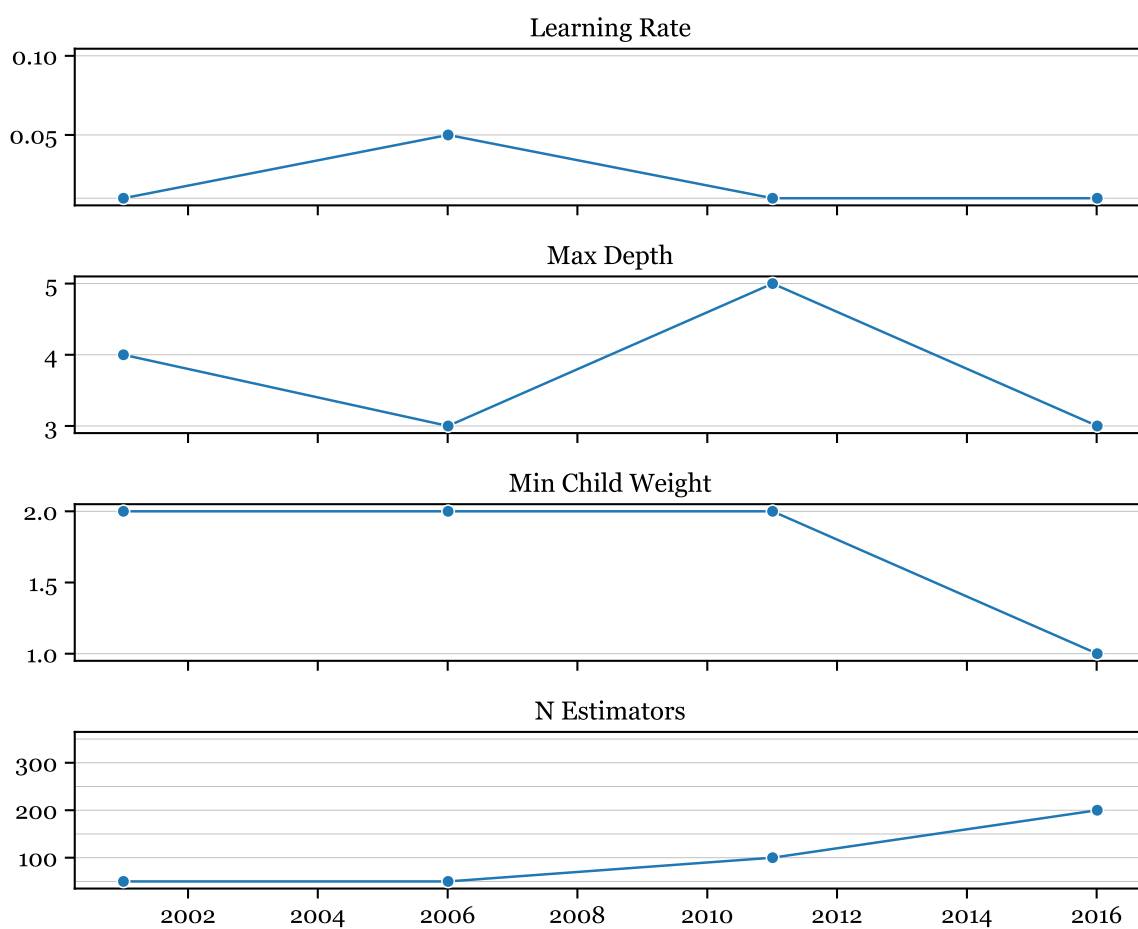
Note: The figure reports validated hyperparameters for the feed-forward neural network regression. There are multiple hyperparameters so we only report those that have a non-trivial grid. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.15: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(RF)



Note: The figure reports validated hyperparameters for the random forest regression. There are multiple hyperparameters so we only report those that have a non-trivial grid. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

Figure OA.16: VALIDATED HYPERPARAMETERS – (CTS, ALL)-(GBRT)



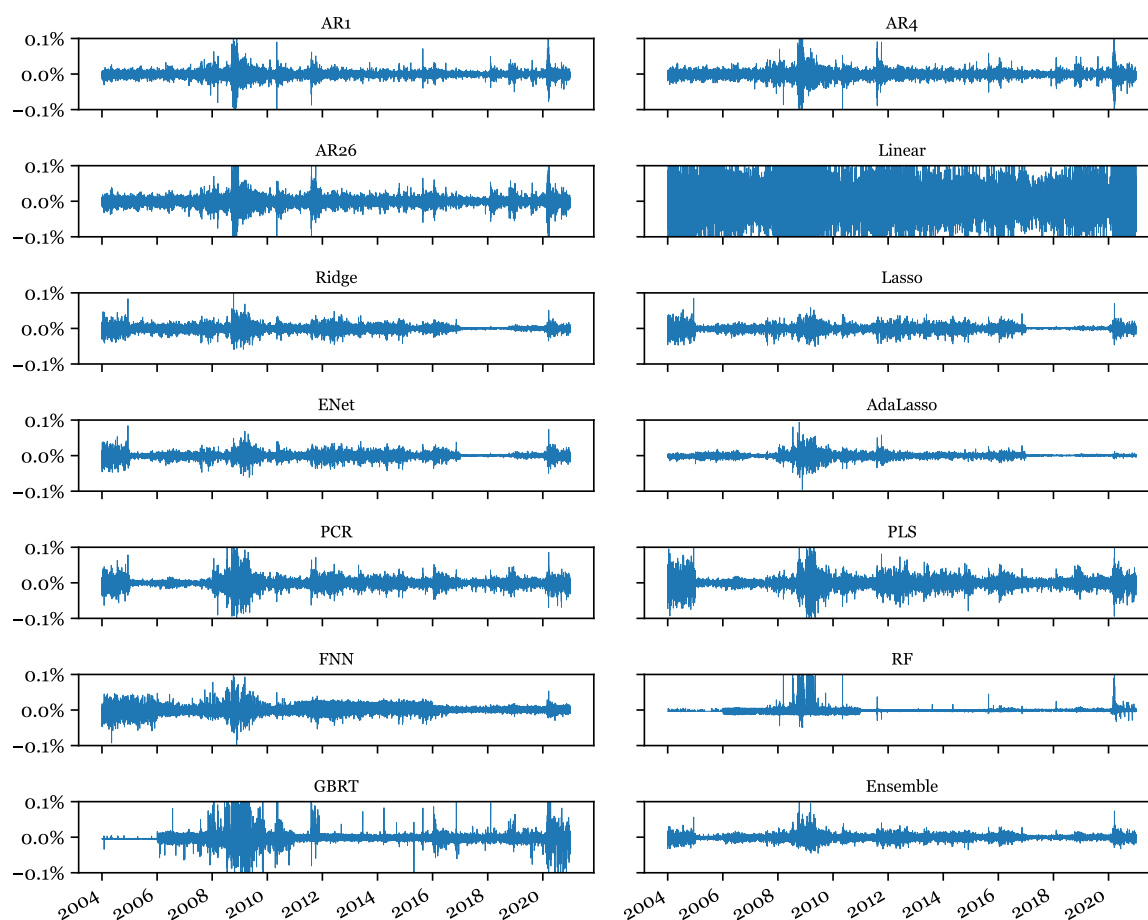
Note: The figure reports validated hyperparameters for the gradient boosted regression trees (XGBoost) regression. There are multiple hyperparameters so we only report those that have a non-trivial grid. Each point corresponds to the start date from which the hyperparameter choice is used. The gray lines denote the underlying grid.

OA.4 Forecasting the Market Portfolio

In this section, we detail some additional results pertaining to our forecasts for the Fama-French market portfolio.

OA.4.1 Forecast Summary Statistics

Figure OA.17: FORECAST TIME SERIES – (CTS, ALL) SPECIFICATION

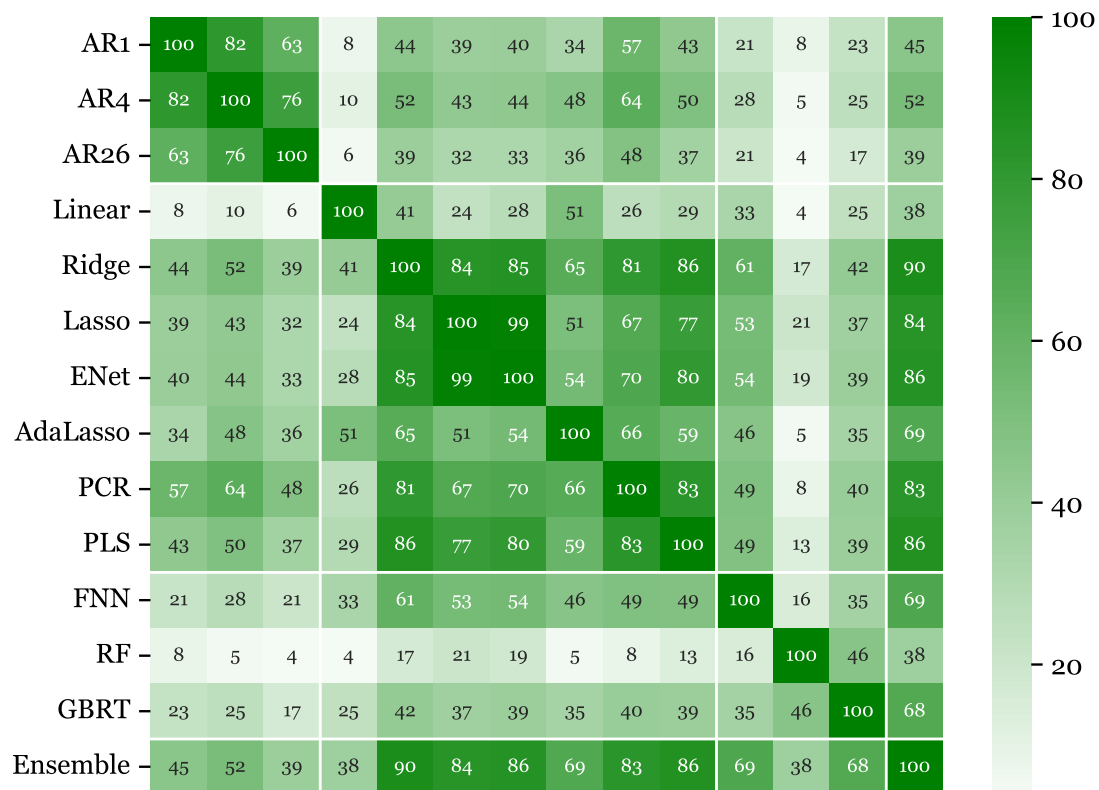


Note: We plot the intraday market return forecasts for each of our models under the (Cts, All) specification over our out-of-sample period, 2004 to 2020. The first three forecasts, from left to right and top to bottom, are the benchmark autoregressive specifications. The remaining forecasts correspond to those of the machine learning models, concluding with the Ensemble model. For visual clarity, we truncate the y-axis.

To begin, we plot the time-series of the forecasts themselves under the (Cts, All) specification in Figure OA.17. The majority of the models seem to produce similar patterns in the forecasts; the magnitude of the forecasts tend to be higher around the 2009-2010, coincident with the European Sovereign Debt Crisis, and in 2020, during the COVID pandemic. The forecasts for the linear regression model are unsurprisingly large, likely a result of its inability to deal with the inherent multicollinearity in our factors. The random forest and gradient boosting regression appear slightly different than the other

forecasts, which may be a result of their reliance on trees rather than smooth functionals as is the case for neural nets and the linear models. Of all the models, the Ensemble model appears to be the most tempered, with moderate-sized forecasts throughout the sample and a volatility pattern mimicking that of the market portfolio returns themselves.

Figure OA.18: FORECAST CORRELATIONS – (CTS, ALL) SPECIFICATION

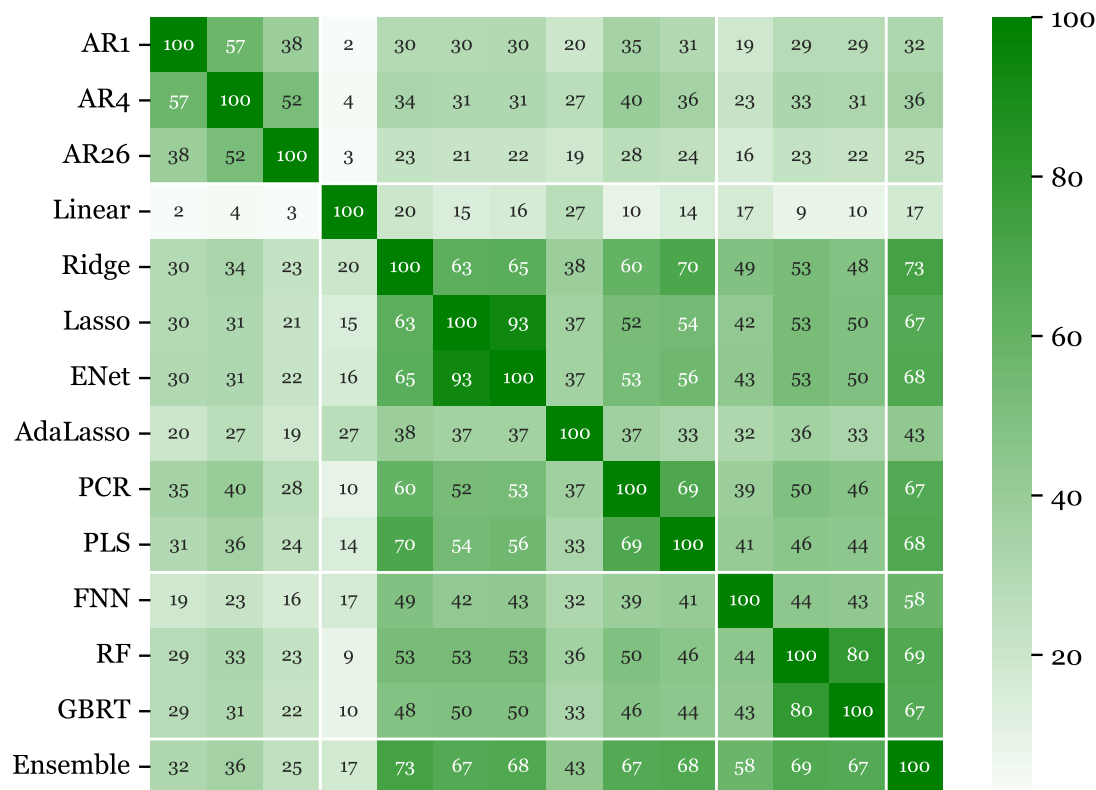


Note: This heatmap shows the correlations between the forecasted return predictions for each of our models over our out-of-sample period, 2004 to 2020. The heatmap is further segmented to highlight the benchmark models, the linear ML models, the non-linear ML models, and the ensemble model. The annotated values are percentages.

Next, we report the correlations between the forecasts in Figure OA.18. These correlations broadly summarize the joint information content between the predictions. For brevity, we focus on those forecast obtained through the *Cts*, *All* specification. Firstly, we find that the benchmark models are all highly correlated, likely because the underlying structure of these are simple AR models. The linear models, as demarcated by the next “block” in the heatmap, are also quite similar. The ENet model appears to almost exactly match the Lasso model, suggesting that the former primarily relies on its L1 penalty. The AdaLasso model also appears to differ moderately from these former two models, suggesting that the adaptive penalty does make a substantial difference in terms of the ultimate results. Of all the linear models, the conventional linear regression looks the most different, probably because it is the least able to handle the multicollinearity of

the data as noted earlier. The next block consist of the non-linear models which appear to be only weakly correlated with one another; this may be because each of the models takes a substantially different approach towards accommodating non-linearities. Finally, the Ensemble model exhibits a fairly moderate correlation with the full set of models.

Figure OA.19: FORECAST SIGN CORRELATIONS – (CTS, ALL) SPECIFICATION



Note: This heatmap shows the correlations between the signs of our forecasted return predictions for each of our models over our out-of-sample period, 2004 to 2020. The heatmap is further segmented to highlight the benchmark models, the linear ML models, the non-linear ML models, and the ensemble model. The annotated values are percentages.

To further evaluate whether these correlations are driven by outliers, we also report the same correlations but computed instead on the signs in Figure OA.19. These results are notably different in a few ways. Firstly, the linear regression appears to be far less correlated with the remaining models in this plot. Secondly, the random forest, which was seemingly only weakly correlated with the other models based on the earlier plot, exhibits far more correlation here; in other words, the RF model primarily different from the other models in terms of magnitude but not really in terms of direction. And, lastly, we can see a much stronger correlation between the RF and GBRT models – both of which rely on decision trees – compared to the previous. Otherwise, most of the takeaways from this model are similar to the earlier Figure OA.18.

OA.4.2 Intraday Explanatory Power Bounds

To help gauge the intraday R^2 's, suppose that the market price dynamics may be described by a continuous-time process with constant drift and volatility calibrated to the mean (out-of-sample, 2004-2020) intraday return and realized volatility of our high-frequency market portfolio, 1.07% and 13.35%. The uncentered population R^2 from a regression of Δ -spaced returns against a simple intercept term then⁵

$$R^2(\Delta) = 1 - \frac{(0.13345 \cdot \sqrt{\Delta})^2}{(0.01068 \cdot \Delta)^2 + (0.13345 \cdot \sqrt{\Delta})^2}. \quad (\text{OA.1})$$

For the 15-minute returns used in our empirical analyses, this translates into an R^2 of just 0.000084%. By comparison, the annual, monthly, and daily uncentered R^2 s based on these same calculations equal 0.54%, 0.046%, and 0.0022%, respectively. On the other hand, allowing the drift to change on an annual basis in accordance with the annual sample means, the R^2 for the 2004-2020 out-of-sample period obtained by combining the results from sequential year-by-year 15-minute return regressions equals 0.01049%. This latter R^2 is obviously substantially larger than the 0.000084% based on the assumption of a constant drift over the full sample. It arguably also provides a more reasonable, albeit practically infeasible, statistical benchmark against which to judge the forecasts from our ML models.

OA.4.3 Mincer-Zarnowitz Regressions

In Table OA.3, we test the rationality of our Ensemble forecasts using Mincer-Zarnowitz (1969) regressions. Each column corresponds a different specification, while rows contain the associated regression coefficients. We include 95% confidence intervals, constructed using Newey-West standard errors, for the two fitted parameters in parentheses.

For the first two specifications, we find that the intercepts, which are annualized by multiplying by the average number of observations per year (6,534), are not statistically distinguishable from zero. However, the economic significance of these coefficients is negligible given that the intercepts, when rescaled to 15-minutes, are essentially zero across all four specifications. Next, we find that the $\beta_{forecasts}$ estimates are all statistically distinguishable from one, being generally too large. This finding may be due the fact that some ML algorithms trade off variance for additional bias, resulting in forecasts that vary “too little.” To test this, we also estimated MZ regressions for the individual models; the penalized linear models generally exhibited betas above one (less variance, more bias) while the non-linear models exhibited betas below one (more variance, less bias). These

⁵ In this setting, we can write the discretized return on the market as $r_t = \Delta_t \mu + \sigma \Delta_t W$ where $\Delta_t \mu$ is the drift over the interval and $\Delta_t W$ is the innovation in the Brownian motion over the same interval. The R^2 after differencing out the mean converges to $E[(\sigma \Delta_t W)^2] / E[(\Delta_t \mu + \sigma \Delta_t W)^2]$.

Table OA.3: OUT-OF-SAMPLE FORECAST RATIONALITY FOR THE ENSEMBLE MODEL

Regressors	All		All+Jmp	
	All	Cts	All	Cts
Intercept (Annualized)	0.043 (-0.026, 0.113)	0.016 (-0.052, 0.085)	0.030 (-0.039, 0.098)	0.002 (-0.066, 0.071)
$\beta_{forecast}$	1.398 (1.048, 1.747)	1.682 (1.276, 2.089)	1.540 (1.139, 1.941)	1.692 (1.222, 2.162)
N	111087	111087	111087	111087
R^2	0.2312%	0.2459%	0.2277%	0.2132%

Note: This table reports the results from Mincer-Zarnowitz (1969) regressions for each of our forecast specifications. 95% confidence intervals for each of the coefficients are reported in parentheses. The standard errors are estimated using Newey-West procedure with 26 lags. We annualize the intercept coefficient for readability.

findings line up with the usual intuition that regularization helps cut variance, while adding more generality via non-linear functionals does the opposite. Besides this, we also note that both the specifications that use continuous regressands exhibit larger betas and intercepts than those that use the regular return as the regressand. This result is driven by the fact that identified continuous returns naturally have a smaller volatility than the standard returns; the scale of the forecasts themselves simply mirror this point. Lastly, we find that the *Cts*, *All* specification exhibits the highest R^2 but only very slightly so. Overall, our forecast rationality tests do not allow us to cleanly rule out any of the specifications.

OA.4.4 Sign Classification Accuracy

In addition to assessing the R^2 in the main text, we also study here the sign classification accuracy of our forecasts with respect to the thus studied Fama-French market portfolio. This analysis, while purely statistical, relates to our economic analysis, which heavily relies on the direction of the forecasts to devise portfolio weights for a trading performance evaluation. We begin by more formally defining this loss function:

$$SignClassAccuracy = 2 \times \left(\left(\frac{1}{\#obs} \sum_{t,i} \mathbb{I}[sign(\hat{r}_{t,i}^{mkt}) = sign(r_{t,i}^{mkt})] \right) - 50\% \right), \quad (OA.2)$$

where we are essentially normalizing the sample average of the inner indicator variable between -100% and 100%, so that 0% accuracy implies no better performance than a coin flip while 100% corresponds to perfect sign prediction. We report our results in Table OA.4. Interestingly, we find that the models targeting the continuous component of the return, *All*, *Cts* and *All*, *All+Jmp*, tend to outperform the equivalent models

Table OA.4: OUT-OF-SAMPLE SIGN CLASSIFICATION ACCURACY

Regressors	All		All+Jmp	
	All	Cts	All	Cts
AR1	-0.72	-0.50	-0.69	-0.51
AR4	-0.64	-0.47	-0.63	-0.45
AR26	-0.25	-0.11	-0.33	-0.16
Linear	0.06	0.21	0.10	0.26
Ridge	-0.12	0.42	-0.30	0.11
Lasso	0.18	0.55	-0.03	0.27
ENet	0.17	0.61	0.06	0.30
AdaLasso	-0.21	0.30	-0.43	0.44
PCR	-0.17	0.09	-0.16	0.06
PLS	-0.06	0.21	-0.13	0.00
FNN	0.42	0.98	0.34	0.47
RF	0.67	0.85	0.09	0.80
GBRT	0.62	0.72	0.60	0.67
Ensemble	0.36	0.59	0.26	0.46

Note: The table reports the out-of-sample accuracy for the signs of our forecasts for the 15-minute return on the Fama-French market portfolio. Each column corresponds to one of our four specifications while each row corresponds to a particular machine learning algorithm. The sign accuracy is measured as the usual classification accuracy metric applied to the sign of returns; 100% corresponds to perfectly accurate prediction of the signs of returns, -100% corresponds to perfectly predicting the opposite sign of returns, and 0% corresponds to the signs of the predictions and realized returns being orthogonal. All values are percentages.

targeting the entire return. This point also holds machine learning models like the Elastic Net and the Random Forest, which, as reported in the main text, had obtained a higher R^2 under the *All*, *All* specification than the *Cts*, *All* specification.

To further address the statistical significance of these differences, we run Diebold-Mariano tests comparing the out-of-sample sign classification accuracy realizations for the four specifications applied to our Ensemble model. Our results, reported in Table OA.5, are obtained by running the usual DM tests on the $\mathbb{1}[\text{sign}(\hat{r}_{t,i}^{mkt}) = \text{sign}(r_{t,i}^{mkt})]$ studied above. We find that the earlier differences between the *All*, *All* and *Cts*, *All* specifications are statistically significant, as evidenced by positive DM values for the corresponding row/column pair. The same point holds for the *All*, *All+Jmp* and *Cts*, *All+Jmp* as well. Overall, these results seem to imply that that forecasting the entire return helps get the level/scale of the forecast right, as evidenced by the higher R^2 as reported in the main text, but makes it more difficult to get the sign right. By the dropping the jump returns, it would then seem that we are less likely to overfit on this “untradeable” variation, which could then explain why our *Cts*-based strategies perform relatively better in terms of trading performance.

Table OA.5: DIEBOLD-MARIANO TESTS ON THE SIGN CLASSIFICATION ACCURACY OF THE ENSEMBLE MODEL

Regressors	All		All+Jmp	
	All	Cts	All	Cts
All, All	0.000	2.859	-1.007	1.049
All, Cts	-2.859	0.000	-3.069	-1.343
All Jmp, All	1.007	3.069	0.000	2.211
All Jmp, Cts	-1.049	1.343	-2.211	0.000

Note: The table reports the Diebold-Mariano test-statistics for our Ensemble model (reference) versus four different baseline models for forecasting the 15-minute return on the Fama-French market portfolio. The “SampleMean” model refers to setting the forecast to the (ex post) out-of-sample average return. Each row/column corresponds to one of our four specifications. Positive values indicate that the column model is *better* – statistically significantly higher out-of-sample sign classification accuracy – than the row model.

OA.5 Implementation

OA.5.1 Choice of Forecasting Target

In the main text, we argue that reusing our forecasts for the Fama-French market portfolio to trade market proxies is more reasonable than using direct forecasts of said proxies. In this section, we expand on this point and study the statistical and economic performance of forecasts that are based on either the Fama-French market portfolio (FF-MKT) or SPDR S&P 500 ETF (SPY).

To begin, we restate our argument for using forecasts of the market portfolio rather than the market proxies more explicitly. Our key point is that forecasting the market proxies runs the risk of identifying idiosyncrasies associated with proxy-specific autocorrelated market microstructure noise as predictable variation, as opposed to predictability embedded in the “true” latent market portfolio return. To further intuit this idea, let $r_{t,i}^F$ denote the actual high-frequency return on any one of the proxies. Next, recall the standard market model based on the hypothetical Fama-French market portfolio,

$$r_{t,i}^F = \beta_{t,i}^F r_{t,i}^{mkt} + \varepsilon_{i,t}^F,$$

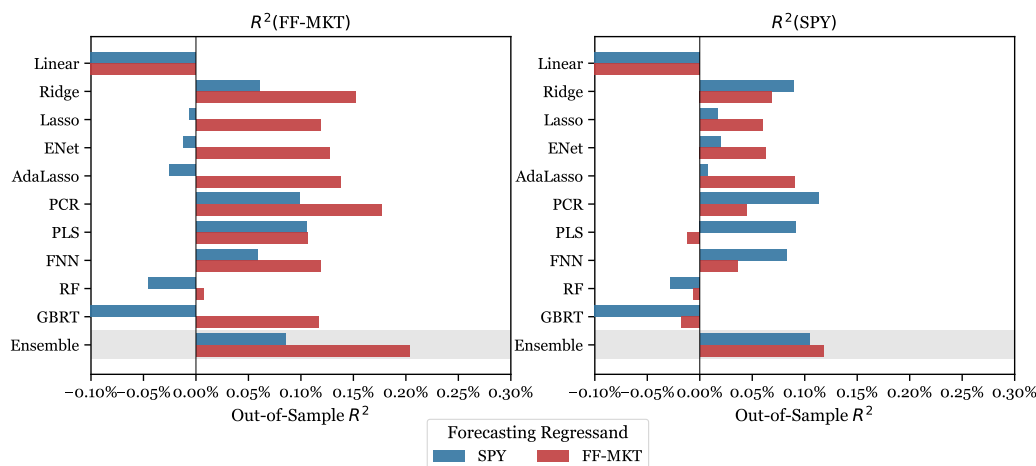
where $\beta_{t,i}^F$ denotes the market beta of the proxy and $\varepsilon_{i,t}^F$ is comprised of the proxy’s tracking error along with market microstructure noise affecting the observed high-frequency proxy returns.

Assuming that $\varepsilon_{i,t}^F$ is unpredictable, or alternatively that we do not seek to exploit any predictable variation that might exist in this combined noise term, the optimal predictions for the true market return $r_{t,i}^{mkt}$ translate proportionally into the optimal predictions for market proxy return $r_{t,i}^F$. Further assuming that $\beta_{t,i}^F$ remains approximately close to unity, as is the case empirically for the ETF-based proxies, the forecasts for the proxies may readily be inferred from the broad-based Fama-French market portfolio forecasts. In contrast, direct forecasts of the left-hand-side market proxy return risks contamination from the $\varepsilon_{i,t}^F$ term; this term may simply increase the variance of the forecasts or, worse, bias the forecasts if it happens to contain economically-irrelevant but predictable variation. To sum up, we can and should reuse the forecasts for the less-noisy market portfolio rather than its proxies.

One testable hypothesis that emerges from the above argument is that the economic performance of the forecasts trained on FF-MKT returns should be better than those trained on the market proxies. We study this below and do indeed find that it holds empirically. At the same time, how the statistical performance of the same two sets of forecasts compare is less clear. That is, accurate forecasting of the market microstructure noise of the proxy should raise the out-of-sample R^2 ’s, but this same noise and other

types of noise may impede the ability to form effective forecasts in the first place and thus reduce R^2 . We begin our empirical analysis below by first trying to understanding how the forecasting target affects the statistical performance and then study the impact on the economic performance.

Figure OA.20: STATISTICAL PERFORMANCE BY REGRESSAND – (CTS, ALL)

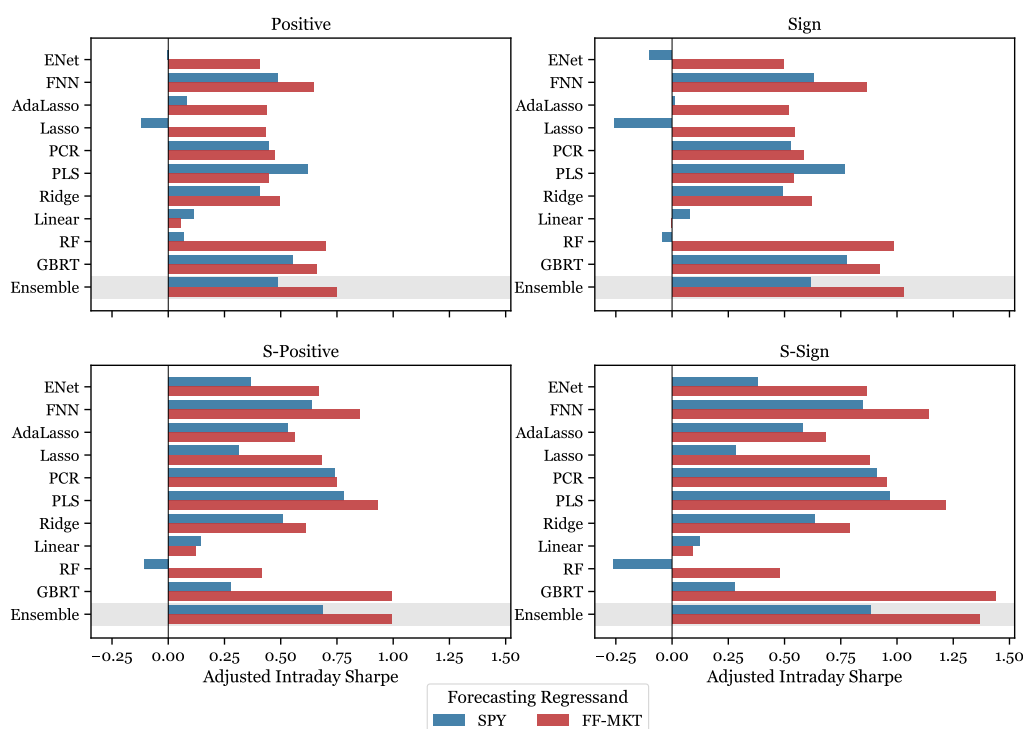


Note: The figure reports out-of-sample R^2 , computed over 2004-2020, for the 15-minute returns on the FF-MKT portfolio and the SPY. The left-subplot reports the R^2 's for the FF-MKT portfolio, while the right-subplot reports the R^2 's for the SPY ETF. The two shades of bars represent two different sets of forecasts – one set using FF-MKT as the regressand and the second set using the SPY as the regressand. All forecasts are based on the *Cts, All* specification. The x-axis is truncated for visibility since the Linear model produces strongly negative R^2 's

Correspondingly, we compute and report the the out-of-sample R^2 's for these two sets of forecasts – one set being trained on the FF-MKT portfolio and another set being trained on the SPY – in Figure OA.20. The left subplot reports the R^2 's computed on the high-frequency Fama-French market portfolio returns. Within this subplot, the blue bars report the forecasting performance of models fit on the SPY returns while the red bars report that for the FF-MKT returns; these results directly reveal how the R^2 changes based on the regressand. The main result here is unsurprising: forecasts of the FF-MKT returns generally do better than forecasts of the SPY returns when trying to predict the FF-MKT. The right subplot, arguably more interesting, is similar but reports the R^2 's computed on the returns of the SPY ETF. These results here are far more mixed. For some models, forecasts of the FF-MKT perform better while, for others, forecasts of the SPY perform better. The Ensemble model shows quite similar performance regardless of the forecast regressand. Overall, these mixed results may be a product of the two counter-acting effects described above.

Next, to empirically evaluate the main hypothesis that the economic performance of the FF-MKT based forecasts should dominate, we now turn to the intraday Sharpe ratios and alphas obtained by trading on the same forecasts studied above. We report the

Figure OA.21: ECONOMIC PERFORMANCE BY REGRESSAND – (Cts, All) – IN-TRADAY ADJUSTED SHARPE RATIOS

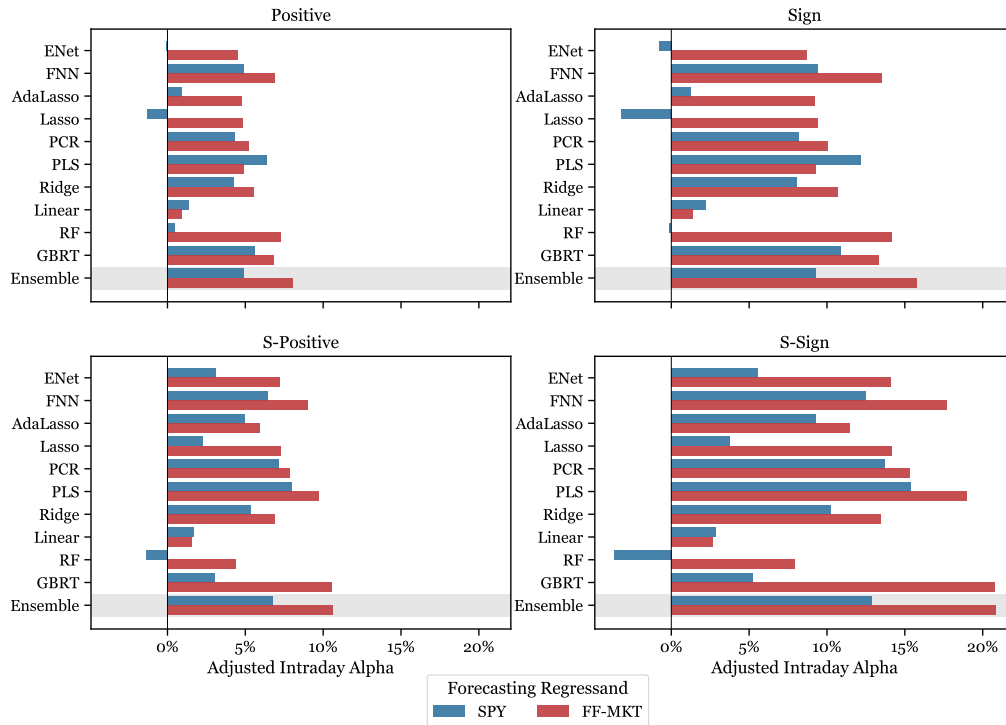


Note: The figure reports intraday adjusted Sharpe ratios, computed over our out-of-sample period 2004-2020. Each subplot corresponds to the results for one of our four trading strategies. The two shades of bars represent two different sets of forecasts – one set using FF-MKT as the regressand and the second set using the SPY as the regressand. All forecasts are based on the *Cts*, *All* specification.

results for the Sharpe ratios in Figure OA.21 and those for the alphas in Figure OA.22; again, for brevity, we focus on forecasts constructed using the *Cts*, *All* specification. The results shown in these subplots are unambiguous – trading on forecasts of the FF-MKT portfolio results in substantially higher performance than trading on forecasts of the SPY. This result holds across strategies for the Ensemble model and for many of the underlying ML models. Interestingly, this point also generally holds for models where using the SPY as the regressand for predicting the SPY itself was over-performing using FF-MKT as the regressand. In other words, such models may have been forecasting economically-irrelevant microstructure noise as conjectured earlier.

To offer a more direct comparison across strategies and specifications, we also report in Table OA.6 the differences in the intraday adjusted Sharpe ratios for the Ensemble model between using the FF-MKT portfolio as the regressand and the SPY ETF as the regressand. Positive values imply that switching to using the FF-MKT as the regressand improves the performance. So, for example, the (Cts, All)-(Ensemble)-(S-Sign) approach attains a Sharpe of 1.366 when using the FF-MKT portfolio as the regressand but this drops to 0.882 when using the SPY as the regressand instead – the difference, 1.366 –

Figure OA.22: ECONOMIC PERFORMANCE BY REGRESSAND – (CTS, ALL) – IN-TRADAY ADJUSTED ALPHAS



Note: The figure reports intraday adjusted alphas, computed over our out-of-sample period 2004-2020. Each subplot corresponds to the results for one of our four trading strategies. The two shades of bars represent two different sets of forecasts – one set using FF-MKT as the regressand and the second set using the SPY as the regressand. All forecasts are based on the *Cts, All* specification.

Table OA.6: DIFFERENCE IN ECONOMIC PERFORMANCE BY REGRESSAND – (ENSEMBLE) – INTRADAY ADJUSTED SHARPE

Regressors	All		All+Jmp	
Regressand	All	Cts	All	Cts
Positive	-0.059	0.262	-0.040	0.243
Sign	-0.054	0.414	0.018	0.405
S-Positive	0.165	0.305	0.069	0.145
S-Sign	0.267	0.484	0.237	0.285

Note: The table difference between the performance of our Ensemble model when using either the FF-MKT portfolio or the SPY ETF as the regressand when fitting the underlying forecasting models. A positive value implies a loss of performance when switching to targeting the SPY ETF. All values are the differences in the annualized adjusted intraday Sharpe ratios.

0.882 = 0.484, is reported in the last row and second column of Table OA.6. In general, the values are positive suggesting that using FF-MKT as the regressand broadly leads to better results. Note that there are negative values in the table for a few of the

specifications where the *Positive* and *Sign* trading strategies are used. In these cases, the baseline Sharpe ratios are quite small due to the incurred transactions costs and the corresponding results are therefore not as relevant to our main point; also, the differences themselves are quite small in magnitude. Thus, overall, the results reported in this table and the earlier figures suggest that reusing our FF-MKT forecasts in the trading implementation is an effective choice both in theory and in practice.

OA.5.2 Sufficient Strategy

As mentioned in our main text, our “Sufficient” strategies, *S-Positive* and *S-Sign*, can be justified through a static problem where an agent aims to maximize expected return while facing transactions costs. In what follows, we explore more general strategies based on the same underlying rationale.

To begin, we lay out the aforementioned optimization problem more formally. Suppose we have an agent who aims to solve

$$\max_w E(r) \cdot w - |w - v| \cdot (1/2) \cdot Spread, \quad (\text{OA.3})$$

where w represents the next period weight on the asset, $E(r)$ represents the return over the next period, and v represents the current weight. The term $E(r) \cdot w$ is the return from the held portfolio while the term $|w - v| \cdot (1/2) \cdot Spread$ represents the cost of paying the half-spread; the sum can also be understood in the framework of mean-variance utility maximization with risk-neutrality. In order to intuit the solution, notice that, on the margin, the agent gains $E(r)$ from raising w but loses $Spread/2$ if $w > v$. Therefore, it is always worth increasing w if the return exceeds the half-spread. If we make a symmetrical argument and impose that the portfolio weights lie in $[-100\%, 100\%]$, we obtain a well-defined solution that is precisely our *S-Sign* strategy.

This approach for deriving an optimal trading strategy is limited in two ways: firstly, it implicitly assumes risk-neutrality and, secondly, it assumes a static setting. The first flaw is more so a deliberate choice given that it allows us to derive a solution that does not depend on the variance of the asset. This characteristic is important since we would like our strategies to be sole functions of our return forecasts for simplicity. The second flaw is similarly a deliberate choice to simplify the solution. To better understand why, note that our portfolio choice today affects our position tomorrow; thus, in a multi-period setting, the optimal position depends on not only on our expectations for tomorrow but our expectations for every period following. A full treatment of this dynamic problem is outside the scope of this paper and we instead refer the reader to Gârleanu and Pedersen (2013, 2016) and Jensen et al. (2022) who derive explicit, closed-form solutions. Because the exact solutions are quite complex, Jensen et al. (2022) recommend some more practical alternatives.

One of these alternatives, adapted to our setting, is to instead consider the objective

$$\max_w E(r) \cdot w - \frac{1}{\phi} \cdot |w - v| \cdot (1/2) \cdot Spread, \quad (\text{OA.4})$$

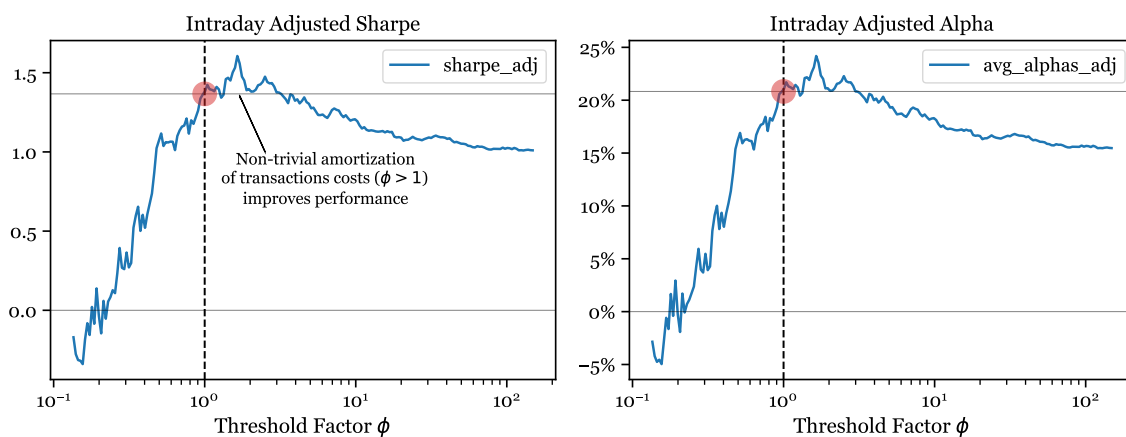
where the ϕ parameter amortizes the transaction cost from the position change over ϕ periods. This “fudge factor” converts the single-period transactions cost into a flow, akin to the utility flow from holding the asset itself, somewhat alleviating the use of a static

setup (Jensen et al., 2022). The solution to this problem, when constraining the weights to $[-100\%, 100\%]$, is quite similar to our S -Sign strategy:

$$[\phi\text{-Sign}] \quad w_{t,i+1} = \begin{cases} 1 & \text{if } \hat{r}_{t,i+1}^{mkt} > \text{Spread}_{t,i}/(2 \cdot \phi), \\ -1 & \text{if } \hat{r}_{t,i+1}^{mkt} < \text{Spread}_{t,i}/(2 \cdot \phi), \\ w_{t,i} & \text{otherwise.} \end{cases} \quad (\text{OA.5})$$

This new strategy, which we call ϕ -Sign, simply modifies the usual trading threshold and naturally corresponds to S -Sign when $\phi = 1$. The best value of ϕ ultimately depends on the data and the objective (mean-variance utility, Sharpe, or alpha). However, we should expect it to be larger than one since rebalancing does not occur in every single period, meaning transactions costs ought to be at least slightly amortized.

Figure OA.23: ECONOMIC PERFORMANCE BY ϕ PARAMETER – (CTS, ALL)-(ENSEMBLE)-(ϕ -SIGN)



Note: The figure reports intraday adjusted Sharpe and alphas, computed over our out-of-sample period 2004-2020, for the Ensemble model estimated with the *Cts, All* specification and traded with the ϕ -Sign strategy. We consider how the performance of this strategy varies with ϕ , which we select from a log grid for visual clarity. We use horizontal line to mark zero alpha and Sharpe as well as another horizontal line to mark the performance at $\phi = 1$, which corresponds to the S -Sign strategy. The performance of the S -Sign strategy is again highlighted using a red marker. The pattern in the alpha matches the pattern in the Sharpe closely, because the realized variance of the sign-based strategy does not depend on ϕ , while the returns do. Since the returns of the sign-based strategy are almost pure alpha, both the Sharpe and alpha appear highly similar.

We explore this point in Figure OA.23 where we plot the returns on the Ensemble model estimated with the *Cts, All* specification and traded with the ϕ -Sign. The marked value at $\phi = 1$ is equivalent to the performance of the S -Sign strategy. Interestingly, we see that a choice of $\phi = 1$ obtains reasonably good performance despite not amortizing the transactions costs at all. Additionally, we find that slightly larger choices around $\phi = 1.6$ appear maximize the Sharpe ratio and alpha. While these values are better, this is still an ex post analysis and such values may not be obvious ex ante; moreover,

the optimal ϕ may be changing over time as well. Thus, for the main analysis, we stick with $\phi = 1$, thereby opting for the *S-Sign* strategy. However, these results do suggest the potential for even stronger performance.

OA.5.3 Sampling Frequency

For our main analysis, we have focused on implementing our strategies at a 15-minute frequency, guided by a trade-off between sample size and microstructure noise that emerges in a high-frequency setting. In this subsection, we explore the use of a 5-minute sampling frequency on our results. We focus on the Ensemble model combined with the *Cts*, *All* specification for brevity. Additionally, to be consistent with the 15-minute results, we similarly use the 15-minute, 1-hour, and 1-day returns on the factors as our regressors; thus there is three times more data and we are now forecasting the 5-minute forward return, but the regressors are unchanged.

We begin by studying the statistical performance of our 5-minute forecasts. Like before, we compute bounds on the R^2 to better understand what magnitudes we should expect. According to Ross's bound (Ross, 2004), we should expect an R^2 of 0.00253% for 5-minute data; this number is about $(15min/5min =)3$ times smaller than the 15-minute bound in the main text because 5-minute data is even more so dominated by the second moment. Likewise, a more direct bound computed using the use approach described in a main text gives a bound of 0.000017%. In either case, the R^2 we should expect is quite small. In contrast, based on our 5-minute forecasts for the (Cts, All)-(Ensemble) model, we find an out-of-sample R^2 of 0.094%. This R^2 is only slightly smaller than the equivalent result for the 15-minute forecasts and is therefore quite substantial at a 5-minute frequency. Relative to Ross' bound, it is about 37 times larger and even more so compared to the alternative bound. Put simply, there appears to be predictability at the 5-horizon beyond that rationalizable by time-varying risk premia.

Going a step further, we report the economic performance as measured by the intraday Sharpe and alphas of our 5-minute (Cts, All)-(Ensemble) forecasts. The results, reported in Table OA.7, broadly show strong performance in terms the unadjusted measures but comparatively worse performance in terms of the the adjusted measures. This finding suggests that the statistical predictability found at a 5-minute level does not translate into transaction-cost adjusted economic performance.

There are two factors that may be driving these results. Firstly, using a 5-minute sampling frequency causes the return predictions to diminish in scale and thus fail to exceed the spread. To intuit this point, note that the magnitude of the return forecasts will be diminishing at an $O(\Delta)$ rate, since we are targeting the drift; the spread will of course be fixed and is thus $O(1)$. Thus, the likelihood that return forecast crosses the spread-based threshold in the *S-Sign* strategy is diminishing. This point is consistent

Table OA.7: PERFORMANCE OF THE (CTS, ALL)-(ENSEMBLE) FORECASTS – 5 MINUTE FREQUENCY

Strategy	Unadjusted		Adjusted	
	Sharpe	Alpha	Sharpe	Alpha
Positive	0.952	10.711	-0.369	-3.544
Sign	1.364	21.739	-0.609	-7.649
S-Positive	0.705	7.950	0.585	6.646
S-Sign	1.009	16.216	0.763	12.555

Note: The table reports the economic performance of our (Cts, All)-(Ensemble) based 5-minute forecasts of the Fama-French market factor. The table consists solely of intraday measures based on trading said forecasts using our four strategies. The asset used for the implementation is the SPY ETF, while the sample period is 2004-2020. The alphas, reported as percentages, are estimated using yearly regressions against the FF6 factors. The Sharpe ratios and alphas are both annualized. The first two columns do not account for transactions costs while the latter two columns do.

with the results – we correspondingly find that the 5-minute forecasts combined with the S-Sign strategy actually has *less* turnover than the equivalent value for the 15-minute forecasts. In contrast, the *Sign* strategy plus the 5-minute forecast has more than twice the turnover of the equivalent for the 15-minute forecasts; this opposite result is simply a result of higher sampling causing the position to be reevaluating more frequently. In short, either we trade too much because we position is re-evaluated too often or we trade too little because the return forecasts are too small to cover the spread.

The second factor driving the diminished performance has to do with signal-to-noise ratios. That is, if the drift of the process scales with the sampling interval Δ and martingale term scales with $\sqrt{\Delta}$, as does Brownian motion, then the signal-to-noise ratio for the 5-minute data is about $\sqrt{3}$ times smaller than that of the 15-minute data. Accordingly, the ML algorithms may be having a more difficult time extracting predictive content at this frequency. This point would explain why the unadjusted Sharpe ratios are lower than their equivalents for the 15-minute results.

OA.5.3.1 Multi-Horizon Forecasting

As noted above, the higher sampling frequency may reduce the precision of our forecasts as well as detrimentally affect trading decisions. The former issue is not straightforward to deal with, but it may be possible to handle the latter issue. To this end, we now construct multi-horizon forecasts of the forward return on the Fama-French market portfolio. We then employ a variant of our *S-Sign* strategy that accounts for the multiple horizons when choosing when to trade. This alternative strategy will allow for additional trading by way of implicitly amortizing the spread by accounting for how our portfolio weight will

change over time. Overall, by employing this multi-horizon approach, we can improve our trading decisions and thus improve the performance.

To begin, we first define this variant of the *S-Sign* to simultaneously help motivate the value of multi-horizon forecasting. Formally, we write this new, multi-horizon S-Sign strategy as,

$$[\text{M-S-Sign}] \quad w_{t,i+1} = \left(\arg \max_{w \in [-1,1]^h} \sum_{s=1}^h \hat{r}_{t,i+s}^{mkt} \cdot w_{t,i+s} - (1/2) \cdot \widehat{\text{Spread}}_{t,i+s-1} \cdot |\Delta w_{t,i+s}| \right)_1. \quad (\text{OA.6})$$

Note that the inner problem is maximizing the transaction-cost adjusted return in each period $i + s$ up to the final horizon of the forward forecasts ($s = h$) with respect to the corresponding weights, which are constraint to lie in $[-100\%, 100\%]$. The inner solution is defined as a vector of weights $(w_{t,i+1}, \dots, w_{t,i+h})$ and the *M-S-Sign* strategy itself is the first element of this vector.

To intuit why this strategy is useful, suppose that the spread was fixed to 3% and we forecasted returns of 1% for the next h horizons. In any singular period, the return is smaller than the half-spread, so the traditional *S-Sign* strategy would never bother trading; in other words, the *S-Sign* strategy is overly myopic, ignoring the fact that high future returns would allow us to hold our position over a longer period. In contrast, the *M-S-Sign* strategy recognizes precisely this possibility and the fact that the spread cost could be implicitly amortized by maintaining a long position. As a result, the *M-S-Sign* strategy would suggest going long in period one. More generally, in this way, the *M-S-Sign* strategy is able to better think through the longer-term costs of rebalancing. One last important note is that if we set $h = 1$ and only consider the one-period-ahead maximization problem, the M-S-Sign collapses to *S-Sign*; so, it can be viewed as a direct extension of the same intuition discussed in the previous [Section OA.5.2](#).

In order to implement this strategy for non-trivial h , we clearly need to solve the inner maximization problem. Fixing $h \geq 1$ to some finite value, we can solve this problem via backward induction. Accordingly, we first solve the final period problem; following the discussion in [Section OA.5.2](#), the solution here is in fact the *S-Sign* strategy. The key information gleaned from this is that the final period weight will either be the previous period weight or in $\{-1, 1\}$. Analogously, the solution to the second-to-final period problem will be similar – it will be in $\{-1, 1\}$ or equal to the previous period weight. Ultimately, assuming that the period-zero weight is in $\{-1, 1\}$, this tells us that the weight vector will lie on the corner of the hypercube $[-1, 1]^h$. Thus, the exact solution is rather easy to compute.

Additionally, rather than checking every corner which would take $O(2^h)$ time, dynamic programming can be used to reduce the numeric computation to $O(h)$ time, taking only

a few seconds in practice. More precisely, the solution algorithm is to simply compute the payoff for each sub-problem (period $i + s$ to period $i + h$) under two possible starting weights, -1 or 1 , and then pick the previous period weight that produces the best total payoff; one then stores this payoff and continues working backwards.⁶ We go ahead and implement this solution using our multi-horizon return forecasts up to $h = 3$, essentially exploring if it is useful to form three 5-minute return forecasts up to 15-minutes forward. We use these multi-horizon forecasts for the $\hat{r}_{t,i+s}^{mkt}$ variable. Next, for the $\widehat{Spread}_{t,i+s}$ variable, we simply use the last observed spread, $Spread_{t,i}$. With these variables defined, the solution can be computed for each interval (t, i) .

Based on our results, we find that the *M-S-Sign* strategy does improve intraday adjusted performance; it achieves a Sharpe ratio of 0.987 and an annualized alpha of 16.03%, notably beating the simpler, single-horizon-based *S-Sign* strategy. Furthermore, the turnover of this strategy is about the same level as that of our 15-minute strategy, suggesting that we have overcome the aforementioned issue with the *S-Sign* approach. Still, while the performance is better, it does not beat that of our main results. The cause may be due to the second point discussed earlier – 5-minute returns simply have a low signal-to-noise ratio making them difficult to forecast to begin with. Thus, we do not use this frequency for our main analysis.

OA.5.4 Market Impact Costs

Our analysis relies on effective spreads to account for transactions costs, ignoring the effect that trading itself has on the spread. To account for this, we now explore the market impact of trading at various capital levels when implementing our main strategy.

We begin by collecting market impact measures from WRDS TAQ. In particular, we rely on the “TSignSqrtDVol2” measure, which corresponds to λ in the regression

$$\log(M_s/M_{s-300}) = \lambda \cdot \text{SSqrtDVol}_s + \varepsilon_s \quad (\text{OA.7})$$

$$\begin{aligned} \text{SSqrtDVol}_s = & \text{Sign} \left(\sum_{s-300}^s \text{BuyDolVol} - \sum_{s-300}^s \text{SellDolVol} \right) \\ & \times \sqrt{\left| \sum_{s-300}^s \text{BuyDolVol} - \sum_{s-300}^s \text{SellDolVol} \right|} \end{aligned} \quad (\text{OA.8})$$

where the term $\log(M_s/M_{s-300})$ is the log change in the bid-ask mid-price over the last 300 seconds at second s . In essence, the parameter λ captures how much prices move given a particular amount of signed square-root dollar volume. We collect the relevant

⁶ We have also checked our work/algorithm against a brute force approach consisting of doing grid search over a mesh bounded by the hypercube $[-1, 1]^h$. The results from this alternative approach match exactly those generated by our dynamic programming solution.

data on this term for the SPY ETF for our full out-of-sample period. The resulting time-series, which is defined at a daily frequency, may dip below zero due to estimation error. To handle this, we smooth the series with a 1-day-halflife EWMA and replace any remaining negative values with zeroes; the first step estimates the true mean of the process while the second is a conservative adjustment.

We then compute the market-impact-adjusted return of our trading strategy as follows:

$$\text{ImpAdjAvgRet}_{t,i}^{F,m,k} \equiv \text{AdjAvgRet}_{t,i}^{F,m,k} - \underbrace{\lambda_{t,i}^F \times \sqrt{W|\Delta w_{t,i}^{m,k}|}}_{\text{market impact cost}} \quad (\text{OA.9})$$

where $\text{AdjAvgRet}_{t,i}^{F,m,k}$ is the same as described in the main text, accounting for both spread and shorting costs. The term $W|\Delta w_{t,i}^{m,k}|$ is the total dollar volume of our executed trade – this depends on our wealth W which we will specify later. The total market impact cost is then the change in the log price induced by our trade, which we are assuming is temporary; in other words, for a purchase, we end up buying at $\lambda_{t,i}^F \times \sqrt{W|\Delta w_{t,i}^{m,k}|}$ above the present price before it reduces back to its original value. This approach is fairly conservative for two reasons. Firstly, some of the impact may be permanent, causing our trades to push future prices towards the direction we are trading; ignoring this would understate our returns. Secondly, the price level is changing as we incrementally fill our order. Thus, the first dollar traded receives a “better deal” than the last; our specification of the impact cost assumes that the cost for the full order is equivalent to the marginal cost for the last, thereby overstating the total cost. Therefore, impact costs should actually be somewhat less than implied above.

In any case, we go ahead and compute the market-impact-adjusted returns obtained by trading on our forecasts. For brevity, we focus on our main forecasts, (Cts, All)-(Ensemble), implemented with the *S-Sign* strategy. However, note that the *S-Sign* strategy is motivated by maximizing spread-adjusted returns (see [Section OA.5.2](#)) as opposed to returns that further account for market impact. To this end, we also consider an alternative strategy – Impact Sufficient Sign or *I-S-Sign* for short – that solves the more relevant objective:

$$\begin{aligned} [\text{I-S-Sign}] \quad \arg \max_{w \in [-1,1]} \quad & \left\{ E(r) \cdot w - |w - v| \cdot (1/2) \cdot \text{Spread} \right. \\ & \left. - \mathbb{1}[w < 0] \cdot \text{ShortFee} - \lambda \times \sqrt{W|w - v|} \right\}, \end{aligned} \quad (\text{OA.10})$$

which is simply maximizing the adjusted expected return with respect to the weight. To implement the solution in practice, we use the lagged spreads on the SPY ETF and a 2% annualized rate for the short fee, as in the main analysis. Additionally, for the λ term, we use the corresponding lagged value for the same asset. Lastly and naturally, we proxy

for the next-period expected returns using our forecasts. With the parameters and data defined, the solution is computed numerically for each period in our data by selecting w from a 1000-point grid defined between -100% and 100% to maximize the objective.

Table OA.8: IMPACT-ADJUSTED OVERNIGHT-INCLUSIVE PERFORMANCE FOR (CTS,ALL)-(ENSEMBLE)

Strategy	Log10(Wealth)	Sharpe	Return	Alpha	Turnover (%)
S-Sign	7	1.106	21.798	20.827	22.230
S-Sign	8	0.536	11.192	10.220	22.230
S-Sign	9	-1.252	-22.348	-23.320	22.230
S-Sign	10	-6.315	-128.412	-129.383	22.230
S-Sign	11	-14.074	-463.813	-464.785	22.230
I-S-Sign	7	0.857	17.079	19.012	15.400
I-S-Sign	8	0.648	13.205	13.451	12.170
I-S-Sign	9	0.463	9.547	11.750	11.250
I-S-Sign	10	0.171	4.280	4.811	10.950
I-S-Sign	11	-0.402	-6.143	-5.513	10.810

Note: The table reports the impact-adjusted performance of the *S-Sign* and *I-S-Sign* strategies using our (Cts,All)-(Ensemble) model forecasts. All performance measures are overnight-inclusive and are annualized; average turnover is calculated over each interval. The impact-adjusted performance is computed using market-impact-adjusted returns for the SPY ETF, over the period 2004-2020.

We report our results for both this strategy and the *S-Sign* strategy in Table OA.8. The first column reports the strategy used, while the second reports the assumed wealth of the portfolio. For simplicity, we assume that the underlying portfolios receive cash injections/removals to keep the wealth constant, so the scale of the market impact cost term in Equation OA.9 is constant.

We find that the *S-Sign* strategy obtains positive returns up to the \$1 billion mark where the transactions costs overwhelm the performance. In contrast, the *I-S-Sign* strategy continues to perform at a wealth of \$1 billion, obtaining an annualized Sharpe of 0.463 and an annual alpha of 11.75% at this level. This finding is generally unsurprising, since the *I-S-Sign* is deliberately designed to account for the market impact costs. When further scaling up the strategy, the performance does eventually decline, reaching a negative value at the \$100 billion mark; even at \$10 billion, we can only obtain an alpha of 4.81%. Still, this strategy represents a major improvement over *S-Sign* when facing market impact costs. Lastly, even better performance may be obtained by trading the E-Mini futures which, as a result of greater liquidity,⁷ would likely have smaller market impact costs.

⁷ The dollar volume of the E-Mini futures is about ten times that of the SPY ETF according to CME (see <https://www.cmegroup.com/education/courses/understanding-the-e-mini-nasdaq-100/why-trade-futures-instead-of-etfs.html>).

OA.6 Feature Importance and Model Interpretability

OA.6.1 SHAP Analysis

In this subsection, we report some additional results related to our SHAP analysis.

OA.6.1.1 Individual Factor importance

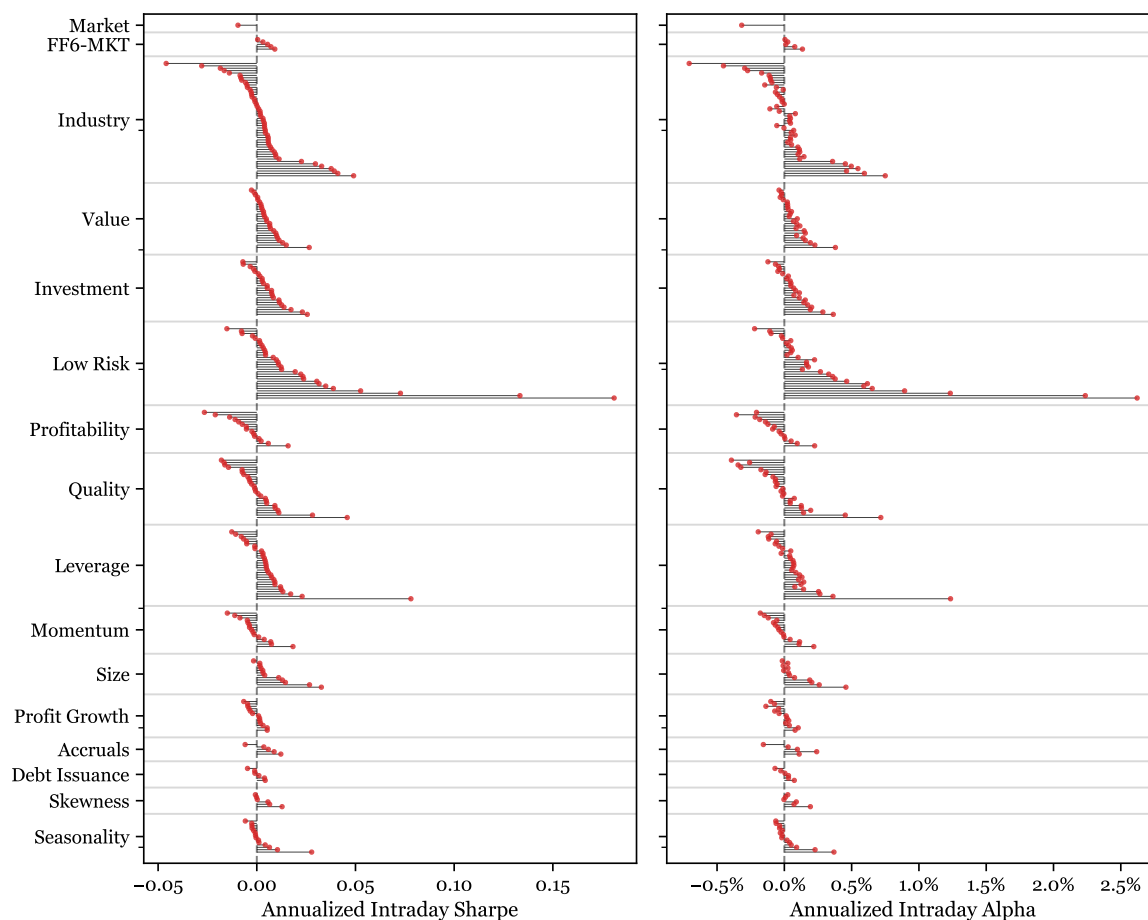
To begin, we report the SHAP values for each of our 272 portfolios in [Figure OA.24](#). This figure, which further sorts the portfolios by theme, reiterates the findings from the main text that most of our performance gains come from the Low Risk cluster. Interestingly, these gains appears to be somewhat dispersed within this cluster, with a few having outsized performance. This pattern likely stems from relying on algorithms such as Ridge and PLS; these algorithms tend to involve a large number of regressors when forming forecasts. In contrast, equivalent figures for the sparse linear models (Lasso and ENet) tend to have correspondingly sparser SHAP contributions. The Ensemble model, combining both effects, naturally exhibits both sparse and non-sparse patterns. We elaborate further on this point in the following section.

OA.6.1.2 SHAP Curves

Here, we study how the SHAP values associated with each of our ML algorithms are distributed. To this end, we begin by estimating the SHAP values for each of the 272 factors in our dataset. We then sort the SHAP values by their overnight-inclusive adjusted Sharpe and then compute the running sum. We plot our results in [Figure OA.25](#) for the *Cts*, *All* specification with the forecasts used to trade the SPY ETF with the *S-Sign* strategy. The value at zero represents the performance of each model with each regressor zeroed out, generally corresponding to simply holding the market, while the sum at 272 corresponds to the performance of the full/complete model. Red markers are used to indicate variables where the marginal contribution (first difference of the plotted running sum) is precisely zero.

As shown by the figure, the Lasso, ENet, and AdaLasso models, and to a lesser extent RF and GBRT, exhibit a sort of variable selection effect where many of the factors have a zero SHAP value due to never actually being used by the model. In contrast, the Ridge, PCR, PLS, and FNN models tend to rely on a broader set of variables for forming their predictions. The results for ridge are particularly consistent with its underlying Bayesian interpretation, which assumes that the predictive content of the variables is generally smeared across the full dataset rather than concentrated as in the Bayesian prior for Lasso. When the full set of models is averaged, we arrive at the Ensemble model which

Figure OA.24: PORTFOLIO SHAP CONTRIBUTIONS BY CLUSTER – (CTS, ALL)-(ENSEMBLE)-(S-SIGN)



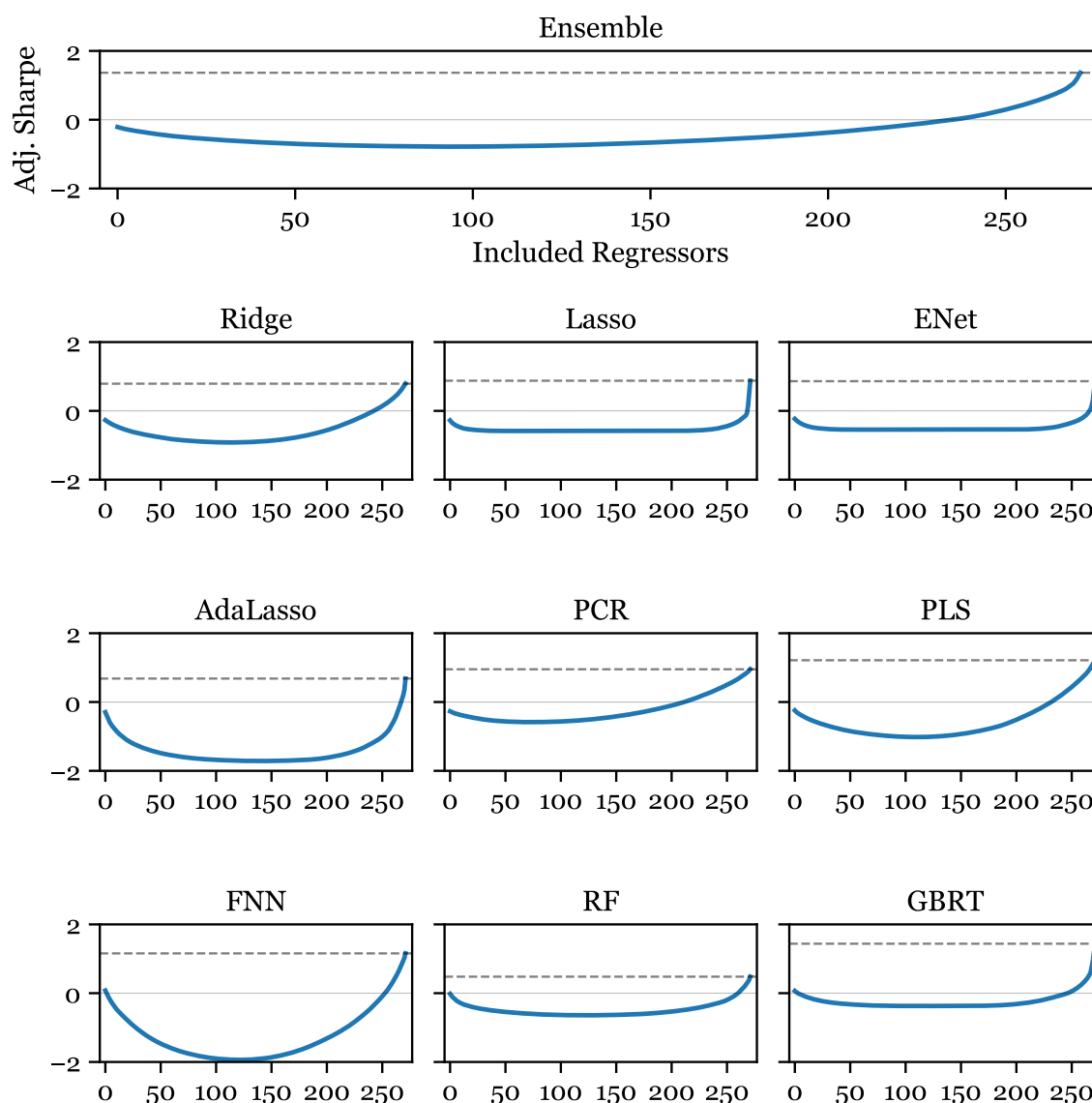
Note: The figure reports the SHAP values for the different portfolios grouped by the factor clusters – along with the market portfolio (MKT), the Fama-French six portfolios except for the market (FF6-MKT), and the 48 industry portfolios (Industry). All measures are intraday, adjusted for transactions costs, and correspond to the performance obtained from trading the SPY ETF according to our (Cts, All)-(Ensemble)-(S-Sign) trading strategy. The Sharpe ratios and alphas are both reported in annualized units. The SHAP values sum to the corresponding performance metrics of the full model minus the empty model (with all regressors zeroed out).

displays a somewhat flat center as in the sparse models, but more generally is curved like the second set of models, thereby indicating a broader reliance, however weak, on a large set of predictors. Still, the curve for the Ensemble model displays a kink at the right-hand-side, as seen in the sparse models, indicating that there are still a small number of factors with an outsized importance.

OA.6.1.3 SHAP Values by ML Algorithm

To better understand which factors drive the performance of each of our ML algorithms, we now report the SHAP values thereof. For brevity, we focus on forecasts obtained using the *Cts*, *All* specification and traded with the *S-Sign* strategy. We report our

Figure OA.25: SHAP CURVES BY ML ALGORITHM FOR OVERNIGHT-INCLUSIVE SHARPE – (CTS, ALL)-(S-SIGN)



Note: This figure reports the cumulative SHAP values for the overnight-inclusive adjusted Sharpe ratios of each ML algorithm. Each of the underlying forecasts are based on the *Cts*, *All* specification and are then implemented with the *S-Sign* strategy. The subplots are constructed for each ML algo by plotting the cumulative sum of the corresponding SHAP values. The dashed horizontal lines indicate the performance of the complete model – that which includes all the regressors. The performance of the null models, using only the averages of the fitted parameters and corresponding to left-most values, still varies by ML algo due to differences in the estimated intercept terms of each.

results, the SHAP values for the for the overnight-inclusive adjusted Sharpe, grouped by cluster and algorithm, in Figure OA.26. Consistent with the Ensemble results, all of the ML algorithms appear to rely heavily on the Low Risk cluster. The results for the alphas, reported in Figure OA.27, show a similar a pattern. We conclude that the importance of the Low Risk cluster in a predictive sense is therefore robust to the choice of ML algorithm. Furthermore, the performance of each algorithm is relatively simple to

interpret as they all exploit the variation in this particular set of features.

Figure OA.26: CLUSTER-AGGREGATED SHAP VALUES BY ML ALGORITHM FOR OVERNIGHT-INCLUSIVE SHARPE – (CTS, ALL)-(S-SIGN)



Note: This heatmap reports the SHAP values for the overnight-inclusive adjusted Sharpe ratios of each ML algorithm. Each of the underlying forecasts are based on the *Cts*, *All* specification and are then implemented with the *S-Sign* strategy. The SHAP values are aggregated within each of the different factor clusters along with the market portfolio (MKT), the Fama-French six portfolios except for the market (FF6-MKT), and the 48 industry portfolios (Industry).

Figure OA.27: CLUSTER-AGGREGATED SHAP VALUES BY ML ALGORITHM FOR OVERNIGHT-INCLUSIVE ALPHAS – (CTS, ALL)-(S-SIGN)



Note: This heatmap reports the SHAP values for the overnight-inclusive adjusted FF6 alphas of each ML algorithm. Each of the underlying forecasts are based on the *Cts*, *All* specification and are then implemented with the *S-Sign* strategy. The SHAP values are aggregated within each of the different factor clusters along with the market portfolio (MKT), the Fama-French six portfolios except for the market (FF6-MKT), and the 48 industry portfolios (Industry).

OA.6.1.4 SHAP Values for the Ensemble

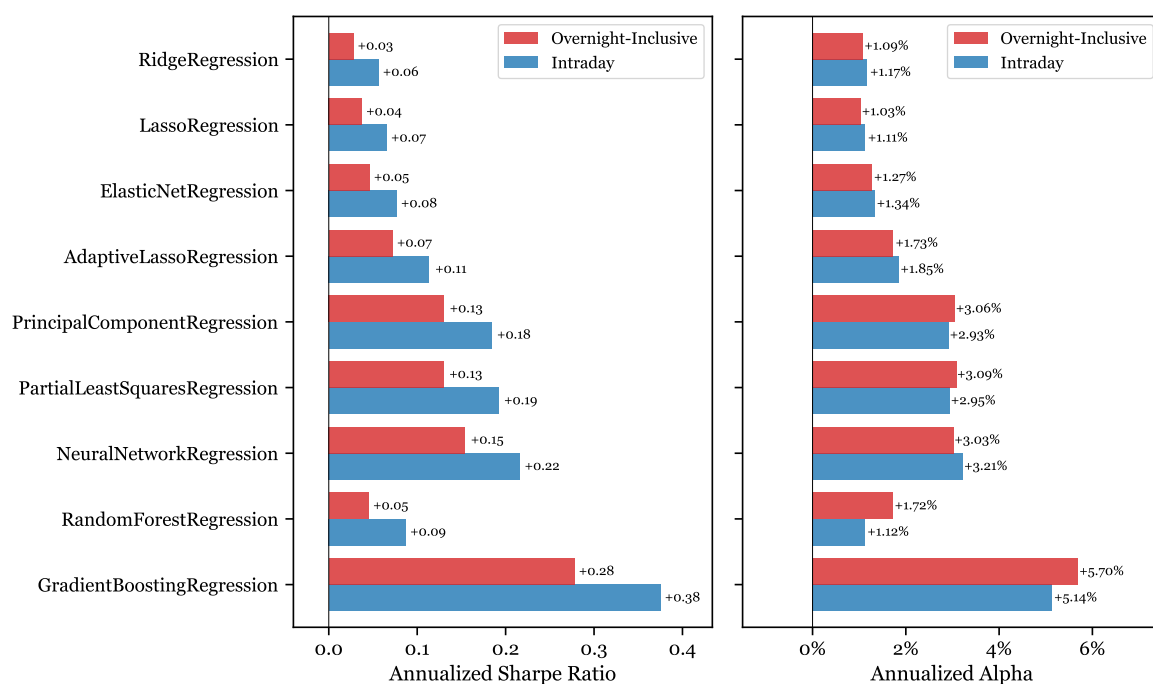
To better understand which of the machine learning algorithms drive the performance of our Ensemble model, we study here the SHAP values computed for each algorithm.

We begin by estimating the SHAP values using the same permutation-based Monte Carlo approach as described in the main text. However, rather than incrementally adding the regressors, we incrementally add each ML-based forecast to the “mean forecast,” which means the process eventually culminates to the Ensemble model – the mean of all the ML forecasts. Besides for this change, the methodology is otherwise the same. As we incrementally add each ML forecast, recompute the performance measures and eventually compute the marginal effects for each model. This constitutes one Monte Carlo trial for a given permutation. By sampling⁸ 500 different permutations, we eventually arrive at a reasonably precise estimate of each ML model’s contribution to the total performance.

We report our results for the *Cts*, *All* specification combined with the *S-Sign* strategy in Figure OA.28. To start, we note that all of the ML algorithms have positive SHAP values, so none actually hurt the Ensemble. This justifies our use of a suite of ML approaches as it does seem to payoff. Interestingly, we also find that both the linear and non-linear models contribute about equally to the overall performance. In fact, in terms of the intraday adjusted Sharpe ratios, the total contribution from the non-linear models is 0.6875 while that of the linear models is 0.6789; the equivalent numbers of the alphas are 9.47% and 11.34%, suggesting that the linear models are only slightly more helpful here. In other words, both classic and modern approaches are valuable. Lastly, and maybe unsurprisingly, we find that the SHAP values for the ML algorithms are generally consistent with the trading performance of the individual ML-based forecasts. So, for instance, trading on the forecasts from GBRT results in quite good performance – as reported in the main text – while its SHAP values are similarly relatively large; on the other hand, RF underperforms in terms of both forecast-based trading performance and SHAP. In this regard, the SHAP analysis helps reconfirm the results from the main text; there is marked heterogeneity in the performance of the models, but all do seem to perform at a reasonable level.

⁸ As in the main text, we use antithetical sampling. This is implemented by setting the permutations for the odd trials equal to the reverse of the permutation used in the preceding even trial.

Figure OA.28: ML ALGORITHM SHAP CONTRIBUTIONS TO (CTS, ALL)-(ENSEMBLE)-(S-SIGN)



Note: The figure reports the SHAP values for underlying machine learning algorithms, pertaining to the intraday and overnight-inclusive Sharpe ratios and FF6 alphas from trading the SPY ETF based on the (Cts, All)-(Ensemble)-(S-Sign) strategy. Each value correspond to the gain in performance obtained from including the corresponding ML algorithm in the ensemble. The underlying SHAP analysis uses 500 Monte Carlo trials to estimate the SHAP values. The Sharpe ratios and alphas are both reported in annualized units.

OA.6.2 Liquidity Factors and Predictability

This subsection reports additional analyses of the liquidity-related factors that appear to drive our results. We begin with some simple descriptive statistics for the factors and follow with a deeper analysis of their predictive content.

OA.6.2.1 Descriptive Statistics

Table OA.9: CORRELATIONS FOR KEY LIQUIDITY FACTORS

	JKP-Turnover126d	JKP-ZeroTrades21d	JKP-ZeroTrades126d	JKP-ZeroTrades252d	CZ-BetaTailRisk
JKP-Turnover126d	1.000	0.971	0.983	0.894	0.666
JKP-ZeroTrades21d	0.971	1.000	0.959	0.880	0.663
JKP-ZeroTrades252d	0.983	0.959	1.000	0.886	0.658
JKP-iVolCAPM252d	0.894	0.880	0.886	1.000	0.614
CZ-BetaTailRisk	0.666	0.663	0.658	0.614	1.000

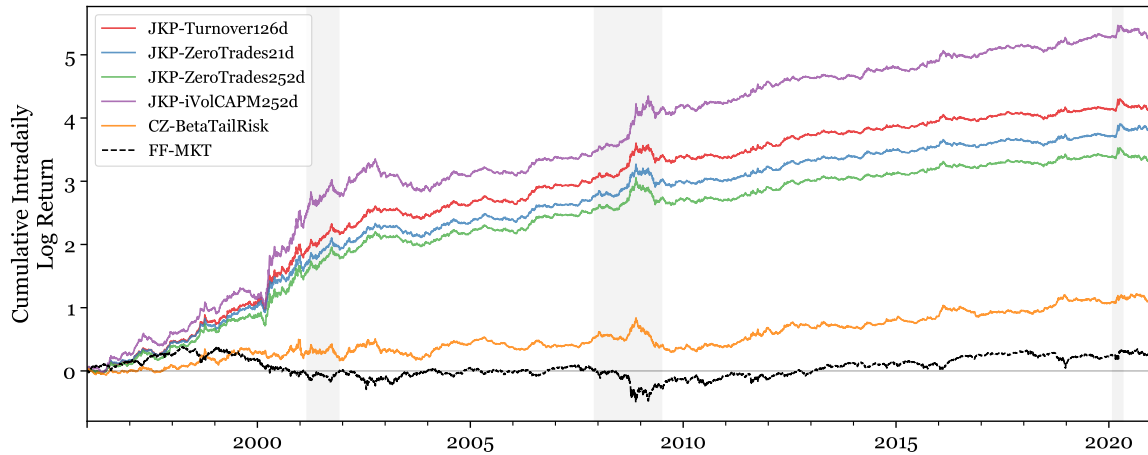
Note: The table reports the correlations between the 15-minute returns on four liquidity-related factors and the tail risk factor that we focus on in SHAP analysis of the main text. The sample period is from 1996-2020.

We begin by reporting additional descriptive statistics that were briefly mentioned in the main text. To start, we report the correlations between the five factors mentioned in the main text – JKP-Turnover126d, JKP-ZeroTrades21d, JKP-ZeroTrades252d, JKP-iVolCAPM252d, and CZ-BetaTailRisk – in [Table OA.9](#). The first three factors, all being based on measures of liquidity, are highly correlated with the average correlation lying around 95%. The fourth factor, JKP-iVolCAPM252d, sorts on idiosyncratic risk and also seems to be quite correlated with the first three, averaging about 90% in correlation. The last factor somewhat less correlated so with the others, with correlations lying around 65%. We also plot the cumulative returns for these same factors in [Figure OA.29](#). Consistent with the correlations, the first four factors tend to move together and draw about the same risk premia while the the tail risk factor somewhat differs and draws a much lower average return.

OA.6.2.2 Predictive Power

If the high turnover decile portfolio is able to absorb new information before the market, its returns should have predictive power with respect to future market returns. Likewise,

Figure OA.29: CUMULATIVE RETURNS FOR KEY LIQUIDITY FACTORS

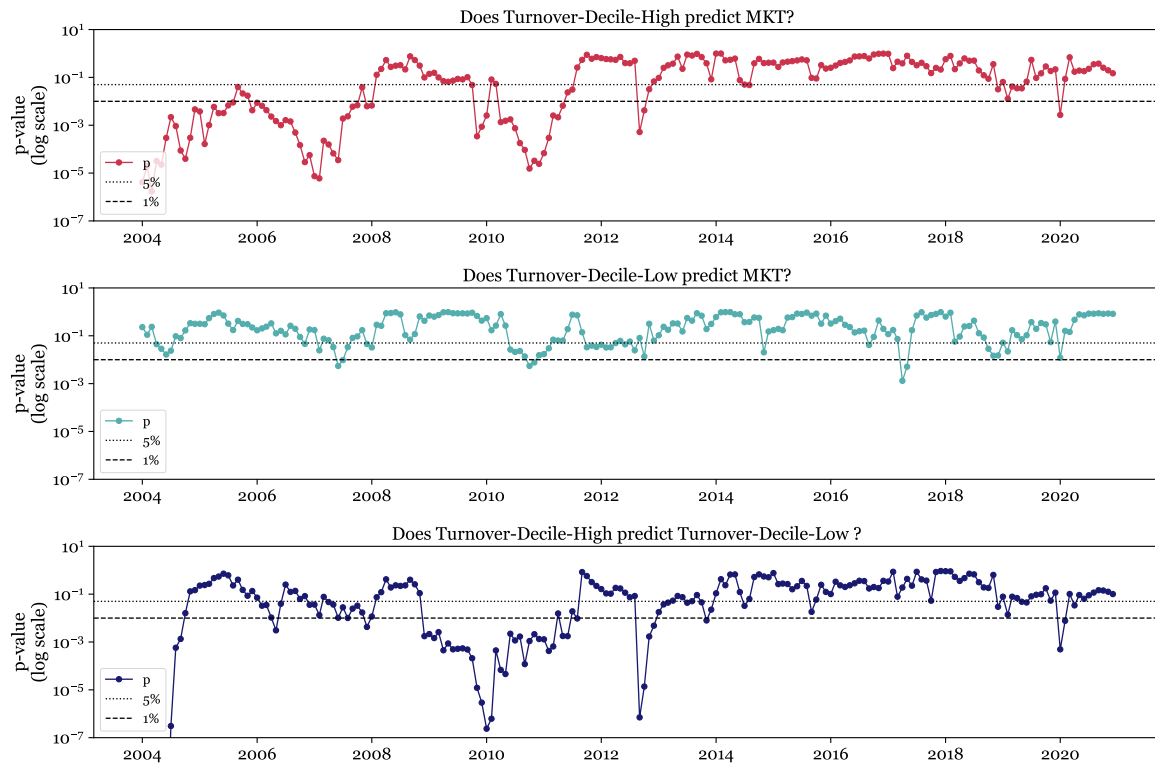


Note: We plot the cumulative log 15-minute returns for the five factors of interest: JKP-Turnover126d, JKP-ZeroTrades21d, JKP-ZeroTrades252d, JKP-iVolCAPM252d, and CZ-BetaTailRisk. Additionally, we include the cumulative return on the Fama-French market portfolio in black. NBER-defined recessions are highlighted in gray.

this portfolio should also be useful in predicting the returns on the low decile portfolio. To test this formally, we run Granger causality tests between these three portfolios. Since predictive power may be time varying, we run the tests on a monthly basis. We plot the corresponding p-values in Figure OA.30. As shown in the first subplot, the high turnover decile portfolio offers a statistically significant amount of predictive power with respect to the market return around 2007-2008, 2010-2011, and in 2020. The timing of the significance is also consistent with when we obtain alpha. Next, as shown by the second subplot, the low decile portfolio helps predict the market in 2007, 2010, 2017, and 2020. To corroborate our earlier argument that the high decile portfolio moves before the lower decile portfolio, we also run a Granger causality test between these two portfolios. The associated test results, shown in the third subplot of Figure OA.30, are similarly strong. The upper decile portfolio offers statistically significant predictive power in many months, particularly those where we obtain our alpha.

In order to determine whether the decile portfolios offer a non-trivial level of predictive power, we run a series of predictive regressions between the market portfolio and the two decile portfolios. Each regression consists of the 15-minute return on the market regressed against the lagged 15-minute return on one of the decile portfolios. We run these regressions on a yearly basis and compute out-of-sample R^2 s using the market returns in each following year; we plot our results in Figure OA.31. The predictions formed using the high decile portfolio and a “high minus low” portfolio constructed by differencing the high and low deciles appear to offer the best predictive performance. This performance peaks in 2008 in 2020, both years where we obtain large alpha with our main strategy. And, the magnitudes of the R^2 's are considerable relative to the out-of-sample

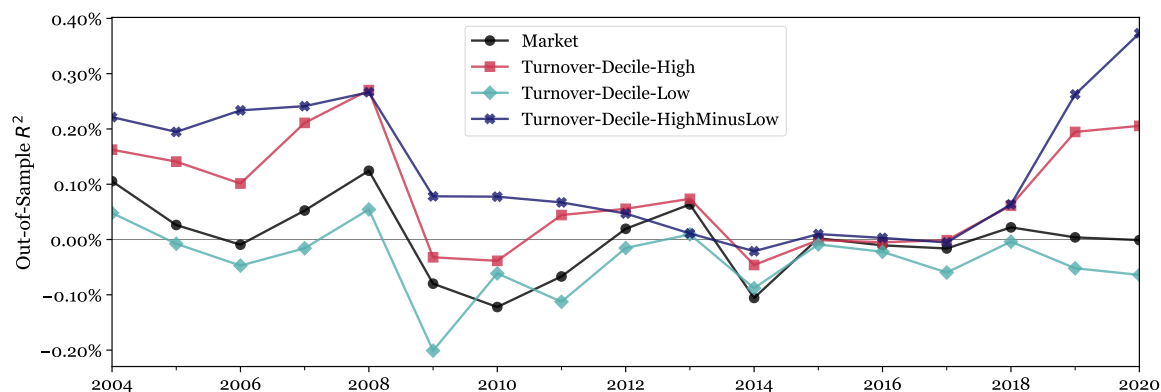
Figure OA.30: TURNOVER DECILE PORTFOLIOS – GRANGER CAUSALITY TESTS



Note: We run univariate Granger causality tests between high/low decile portfolios constructed using the turnover factor from Chordia, Subrahmanyam, and Anshuman (2001) and the Fama-French market factor. The tests are run on a monthly basis and 15-minute return data. Each subplot shows the corresponding p-values for each test and month. The p-values are derived from an F-test and computed using Newey-West standard errors with automatic lag selection; the underlying VARs are restricted to a maximum of 26 lags and use AIC for lag selection. The first subplot shows the results for tests of whether the high decile portfolio predicts the market portfolio. The second subplot shows the same but for whether the low decile portfolio predicts the market portfolio. And, the third subplot shows the results for tests of whether the high decile portfolio predict the low decile portfolio. All p-values are plotted on a log scale. The horizontal lines mark 5% and 1% significance level cutoffs.

R^2 of our main forecast specification, 0.0828%. On the other hand, the out-of-sample R^2 s obtained by simply using the lagged market return as well as the low decile portfolio return do not appear to be substantially different from zero. Overall, these results suggest that the predictive power of the high-minus-low turnover factor is solely driven by the high turnover leg.

Figure OA.31: TURNOVER DECILE PORTFOLIOS – PREDICTIVE REGRESSIONS OOS R^2



Note: For each portfolio listed in the legend, we run a predictive regression on the 15-minute intradaily returns of the Fama-French market portfolio. We run these regressions on a yearly basis, from 2003 to 2019, using one-year windows and then use the regression coefficients to form predictions for the market returns of the following year. Each line in the plot reports the R^2 measures for these predictions in each of the out-of-sample years given by the x-axis.

OA.7 Robustness

This section presents robustness checks with respect to various choices made in our statistical and economic performance analysis.

OA.7.1 Choice of Market Proxy

We begin by studying whether our trading performance results are dependent on the choice of market proxy used to implement the strategies. As mentioned in the main text, we consider four highly liquid market ETFs – SPY, QQQ, IVV, and IWM – which have relatively low spreads; in addition, we also consider the E-Mini S&P 500 Futures contract which is even more liquid than the SPY. Furthermore, as evidenced by the correlation matrix in Table OA.10, the 15-minute returns on all of these proxies fairly closely track those of the Fama-French market portfolio and therefore represent ideal stand-ins for this portfolio.

Table OA.10: CORRELATIONS FOR 15-MINUTE RETURNS ON MARKET PROXIES

	FF-MKT	SPY	QQQ	IVV	IWM	E-Mini
FF-MKT	1.000	0.980	0.912	0.963	0.882	0.978
SPY	0.980	1.000	0.899	0.964	0.853	0.983
QQQ	0.912	0.899	1.000	0.878	0.807	0.895
IVV	0.963	0.964	0.878	1.000	0.838	0.958
IWM	0.882	0.853	0.807	0.838	1.000	0.850
E-Mini	0.978	0.983	0.895	0.958	0.850	1.000

Note: The table reports the correlations between the 15-minute intraday returns on our five market proxies. The proxies (underlying targets) consist of SPY (S&P 500), QQQ (Nasdaq-100), the IVV (S&P 500), the IWM (Russell 2000), and the E-Mini S&P 500 Futures (S&P 500). The sample period is 2004-2020.

To simplify our robustness check, we focus on those forecasts obtained from our (Cts, All)-(Ensemble)-(S-Sign) approach, as in the main text. Our results, reported in Table OA.11, show reasonable performance across the ETFs, with the majority of the performance variation being driven by differences in transactions costs. Interestingly, the IWM ETF, which is about as liquid as the SPY, performs extremely well; the overnight alpha for IWM particularly stands out, averaging 34% per year. Trading the E-Mini futures also delivers strong performance rivaling that of the SPY.⁹ The QQQ and IVV ETFs perform slightly worse, likely due to the fact that they have about double the spreads of the SPY and thus incur relatively more transactions costs. We illustrate this point by comparing the adjusted alphas with the unadjusted alphas in the last row of each panel;

⁹ When computing the transactions costs on the E-Mini, we use the twice the spreads on the SPY, since we lack finer quote data on this contract. However, because the E-Mini is relatively more liquid, this is an extremely conservative assumption.

Table OA.11: ADJUSTED TRADING PERFORMANCE OF (CTS,ALL)-(ENSEMBLE)-(S-SIGN) BY MARKET PROXY

Panel A. Adjusted Intraday Measures					
	SPY	QQQ	IVV	IWM	E-Mini
Return (%)	19.889	16.090	15.871	24.783	17.258
Sharpe	1.367	0.950	1.083	1.264	1.184
Alpha (%)	20.828	16.456	17.051	26.322	17.863
t(Alpha)	7.096	6.512	7.467	6.777	7.461
Alpha/UnadjAlpha (%)	79.990	59.346	61.620	80.152	65.342
Panel B. Adjusted Overnight-Inclusive Measures					
	SPY	QQQ	IVV	IWM	E-Mini
Return (%)	26.731	25.198	22.093	35.114	20.700
Sharpe	1.439	1.209	1.168	1.479	1.132
Alpha (%)	25.758	23.891	21.103	34.268	19.395
t(Alpha)	8.002	8.095	8.361	8.043	7.702
Alpha/UnadjAlpha (%)	80.915	63.789	62.471	81.971	63.382

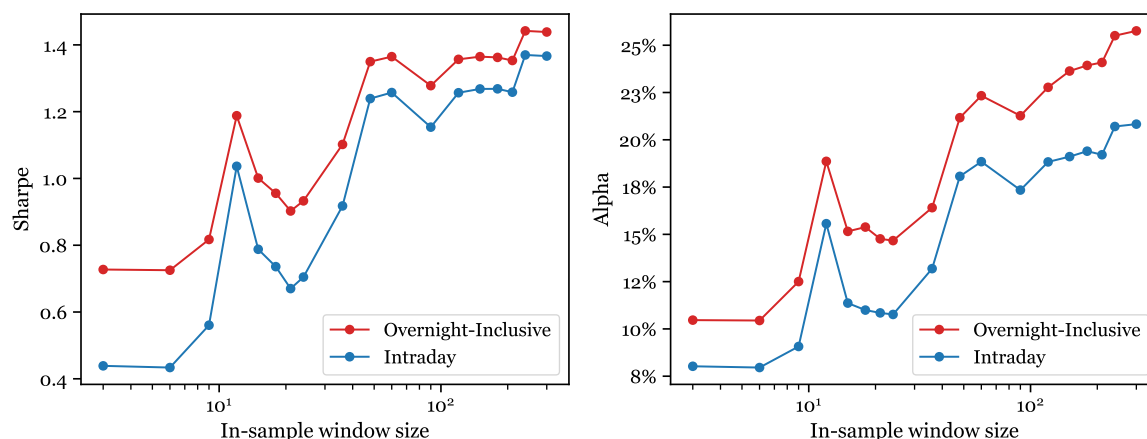
Note: The table reports the transaction-cost adjusted performance measures for the Ensemble model estimated with the *Cts*, *All* specification and implemented using the *S-Sign* trading strategy on four different market proxies. The top panel reports performance measures computed using only the intradaily returns. The bottom panel reports measures that include the overnight return. Each column reports the performance obtained by implementing on the corresponding proxy. The sample period is 2004-2020; the intraday performance measures disclude the 16:00-9:30 overnight return while the overnight-inclusive performance measures consider all returns. The alphas are estimated using yearly regressions against the FF6 factors. The returns and alphas are reported in percentage annualized form. The last row reports the adjusted alpha as a percentage of the unadjusted alpha.

for the SPY and IWM, about one-fifth of the alpha is lost to transactions costs while the same proportion for the QQQ and IVV is about two-fifths. In any case, the economic significance of our forecasts is reasonable across these alternative market proxies, albeit dependent on their liquidity. For this reason, and following convention in the literature, we focus on the SPY in our main analysis.

OA.7.2 In-sample Window Size

Next, we explore whether our results are sensitive to the choice of the in-sample window used for the rolling forecasting procedure. Recall that this window is used to define both the training and validation splits, with 80%/20% of the data within the window being devoted to each. For our main analysis, we had set the window size to 300 months, which is equivalent to using an expanding window given that our forecasting data sample, 2000 to 2020, consists of 252 months. For our robustness check, we will consider windows ranging from range from 12 months to 240 months.

Figure OA.32: ECONOMIC PERFORMANCE BY IN-SAMPLE WINDOW SIZE – (CTS,ALL)-(ENSEMBLE)-(S-SIGN)



Note: The figure reports adjusted Sharpe ratios and alphas, computed over our out-of-sample period 2004-2020, across a range of in-sample window sizes used to estimate underlying machine learning models. All results are based on the Ensemble model estimated with the *Cts*, *All* specification and implemented with the *S-Sign* strategy.

We report our main results in Figure OA.32 where we show how the performance of our Ensemble model, estimated with the *Cts*, *All* specification and traded with the *S-Sign* strategy, changes with the in-sample window size used for the underlying forecast estimation. Overall, we find that larger in-sample windows produce better performance, consistent with the idea that the quality of the estimates should improve with a larger sample size. At the same time, these results run counter to the idea that shorter windows with more recent data are more beneficial, particularly in a setting with time-varying parameters. However, based on these results, reducing variance via sample size seems to be more important for obtaining better results, possibly because our data is high-dimensional and of a low signal-to-noise ratio.

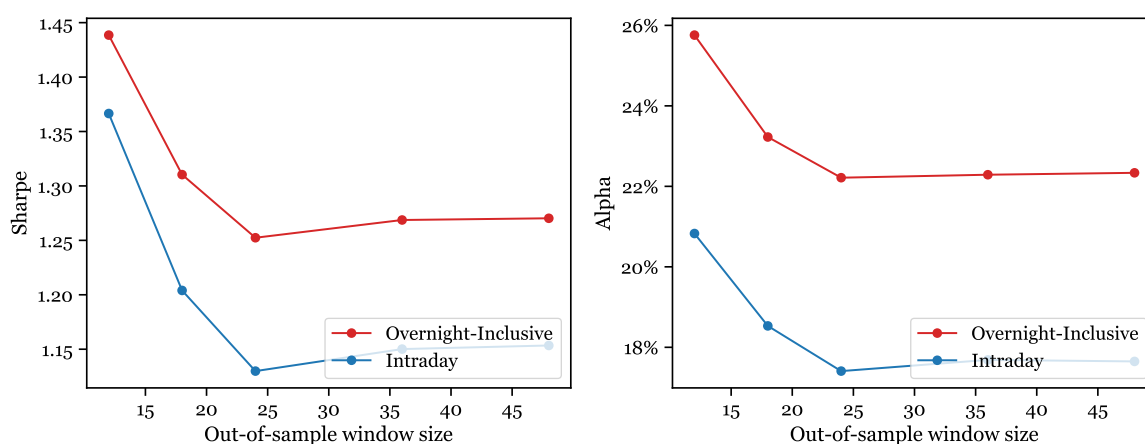
Another interesting finding is the non-monotonicity of the performance with respect to the sample size. This pattern appears in not only the plotted Sharpe ratios and alphas, but also in the out-of-sample R^2 's of the forecasts as well. This latter finding suggests that our data/models may be exhibiting “double descent,” particularly the sample-wise case explored in Nakkiran (2019). Put simply, traditional double descent describes a scenario where the MSE of a prediction declines with the dimensionality of the model/data (bias reduction dominates), then rises (variance increase dominates), then decreases again (second descent). The first descent and rise is driven by the usual bias-variance trade-off creating a U-shape in the performance with respect to the underlying complexity; the second descent is more difficult to interpret and is receiving increasing attention as the machine learning literature has trended towards tackling high-dimensional data with highly over-parametrized models. We refer to work by Bartlett et al. (2020), Nakkiran

et al. (2021), and Schaeffer et al. (2023) for possible explanations for this behavior.

OA.7.3 Out-of-sample Window Size

We also consider how our results vary with the size of the out-of-sample window. Note that this window essentially determines how frequently we re-estimate our model. For the main analysis, we use a 12-month window for the linear models and a 60-month window for the non-linear models to accommodate computational constraints. Here, we focus explicitly on the relationship between the out-of-sample window size for the linear models and our trading performance; we leave the out-of-sample window for the non-linear models fixed at 60 months again for computational reasons. We report our results in Figure OA.33; as before, we focus on the results for (Cts,All)-(Ensemble)-(S-Sign). In general, we find that shorter out-of-sample windows generally correspond to better performance, although the variation in the performance measures is quite small. While we could consider even shorter windows for the main analysis, we leave the window at 12-months to keep computational usage at a reasonable level.

Figure OA.33: ECONOMIC PERFORMANCE BY OUT-OF-SAMPLE WINDOW SIZE – (CTS,ALL)-(ENSEMBLE)-(S-SIGN)



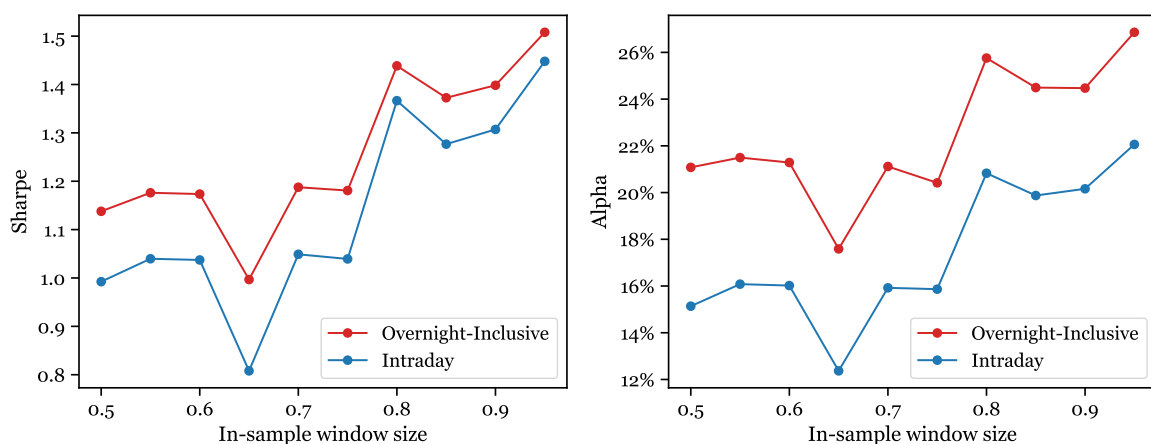
Note: The figure reports adjusted Sharpe ratios and alphas, computed over our out-of-sample period 2004-2020, across a range of out-of-sample window sizes used to estimate underlying machine learning models. All results are based on the Ensemble model estimated with the *Cts, All* specification and implemented with the *S-Sign* strategy. Only the out-of-sample windows for the linear models varies for the underlying estimates; the non-linear models have their out-of-sample windows fixed to 60 months.

OA.7.4 Train-Valid Window Split

In this subsection, we report how our trading performance depends on how we split the in-sample window into the training and validation data. As noted in the main text, we follow much of the existing literature and set aside a fixed proportion of the in-sample

window for each, opting for an 80/20 split for our main analysis. In Figure OA.34, we show how the trading performance for the (Cts,All)-(Ensemble)-(S-Sign) approach changes as we vary this proportion, with 80 on the x-axis corresponding to our default value. We find that 80/20 is a reasonably good split, although we could obtain slightly stronger results if we further increased the proportion devoted to the training window to 95%. However, based on the y-axis, the differences in performance are not particularly substantial, suggesting this parameter not particularly important. We do note, however, that much smaller values of the parameter perform noticeably worse, possibly due to the fact that reducing the amount of available training data makes the already high-dimensional and high-noise forecasting exercise substantially more difficult. Moreover, reducing the size of the validation window is not a major concern, since high-frequency data makes it relatively easy to get precise estimates of the validation objective, R^2 .

Figure OA.34: ECONOMIC PERFORMANCE BY TRAIN-VALID WINDOW PROPORTION – (CTS,ALL)-(ENSEMBLE)-(S-SIGN)



Note: The figure reports adjusted Sharpe ratios and alphas, computed over our out-of-sample period 2004-2020, across a range of train-valid split proportions used to estimate underlying machine learning models. All results are based on the Ensemble model estimated with the *Cts*, *All* specification and implemented with the *S-Sign* strategy.

OA.7.5 Overnight Strategy

Our transactions costs adjustments depend on the change in the weight on the market portfolio over each period. Correspondingly, the trading strategy used for the overnight period affects the intraday performance, since the first intraday position of each day must rebalance against the overnight position. In the main text, we had assumed that the overnight position was fixed to 100% for simplicity and did not explore alternative overnight strategies. To be comprehensive and further evaluate whether the overnight strategy is pivotal for our intraday results, we evaluate here the effects of using three alternative overnight strategies – *Zero*, *Half*, and *Maintain* – on the performance of our

main model. The first strategy, *Zero*, sets the overnight weight to 0%; the second, *Half*, to 50%; and the third, *Maintain*, to whatever weight was used in the intraday period preceding the overnight period, thereby maintaining the previous position.

Table OA.12: INTRADAY PERFORMANCE OF THE (CTS, ALL)-(ENSEMBLE) FORECASTS – ALTERNATIVE OVERNIGHT STRATEGIES

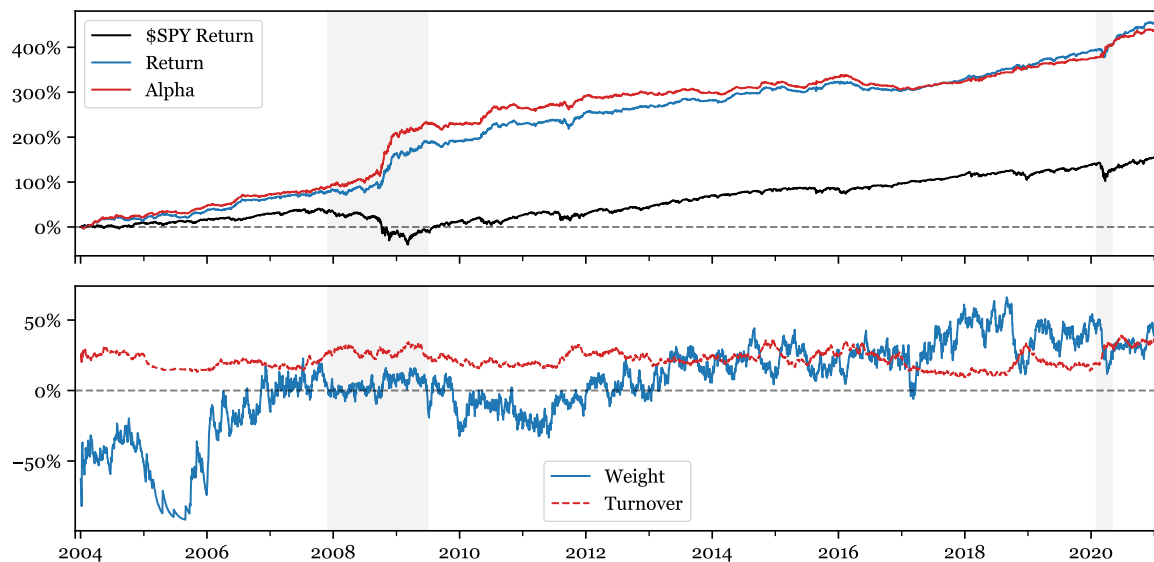
Strategy	Return (%)	Sharpe	Alpha (%)	t(Alpha)
<i>Panel A. Zero</i>				
Positive	8.57	0.74	7.12	3.69
Sign	14.92	1.02	14.79	4.24
S-Positive	11.13	0.99	9.79	5.10
S-Sign	19.89	1.37	19.98	5.75
<i>Panel B. Half</i>				
Positive	8.62	0.74	7.59	4.27
Sign	14.97	1.02	15.26	4.34
S-Positive	11.13	0.99	10.22	5.76
S-Sign	19.89	1.37	20.40	5.81
<i>Panel C. Maintain</i>				
Positive	8.78	0.76	8.12	4.56
Sign	15.25	1.04	15.84	4.54
S-Positive	11.49	1.03	10.74	6.04
S-Sign	20.62	1.42	21.02	6.04

Note: The table reports the transaction cost adjusted economic performance of our (Cts, All)-(Ensemble) based forecasts under various overnight strategies. *Zero* refers to setting the overnight weight to 0%, *Half* refers to setting it to 50%, and *Maintain* refers to maintaining the position of the weight preceeding the overnight period (15:45-16:00 weight). The asset used for the implementation is the SPY ETF. The sample period is 2004-2020; the intraday performance measures disclude the 16:00-9:30 overnight return while the overnight-inclusive performance measures consider all returns. The alphas are estimated using yearly regressions against the FF6 factors. The returns, Sharpe, and alpha are annualized.

We report the transaction cost adjusted performance for our (Cts, All)-(Ensemble) forecasts under these three alternate overnight strategies in Table OA.12. In general, we find modifying the overnight strategy has little impact on our results. Interestingly, all three alternative overnight strategies obtain the same or slightly higher Sharpe under the S-Sign strategy, although the gains are not particularly large. Overall, these results broadly suggest that the transactions costs induced by the overnight position are generally irrelevant. Furthermore, they suggest that we could further improve our our ultimate performance by separately and directly devising an optimal trading strategy for the overnight period without substantially affecting our intraday performance – we leave this for future work.

OA.8 Additional Tables and Figures

Figure OA.35: CUMULATIVE OVERNIGHT-INCLUSIVE RETURNS AND TURNOVER – (CTS, ALL)-(ENSEMBLE)-(S-SIGN)



Note: The first subplot reports the cumulative overnight-inclusive returns for the SPY ETF, our (Cts, All)-(Ensemble)-(S-Sign) trading strategy for the SPY, and the corresponding FF6 alphas. The trading strategy returns and alphas are both adjusted for transaction costs. The second subplot shows the average monthly weights for the SPY and turnover for our main trading strategy. The average weights and turnover are computed on a daily basis and further smoothed with a 21-trading-day EWMA. Shaded regions mark NBER-defined recessions.

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